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Defense Technical Information Center (DTIC), AD No.:
This TOP supersedes TOP 01-2-612, Nuclear Environment Survivability, dated 24 October 2008; TOP 01-2-618, Initial Nuclear Radiation Hardness Validation Test, dated 3 November 2008; and TOP 01-2-619, Nuclear Thermal and Blast Hardness Validation Test, dated 3 November 2008.

Marginal notations are not used in this revision to identify changes, with respect to the previous issue, due to the extent of the changes.

14. ABSTRACT
This TOP provides a general outline and guidelines of the test and analysis procedures required to determine the effects of specified tactical nuclear weapons effects environments on Army material. The purpose of these test and analysis procedures is to ascertain the degree to which the Initial Capabilities Document (ICD), Capabilities Development Document (CDD), System Evaluation Plan (SEP), and Army Nuclear Hardening Criteria (NHC) are met.

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U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

Test Operations Procedure 01-2-612A 28 October 2014
DTIC AD No.

NUCLEAR ENVIRONMENT SURVIVABILITY

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1. SCOPE.

a. This Test Operations Procedure (TOP) is a general outline on test and analysis procedures required to determine the effects of a specified nuclear environment on Army materiel. The purpose of these test and analysis procedures is to ascertain the degree to which the Initial Capabilities Document (ICD), Capabilities Development Document (CDD), Army Regulation (AR) 70-75[1], System Evaluation Plan (SEP)/Independent Assessment Plan (IAP) criteria, and Army Nuclear Hardening Criteria (NHC) are met. Army materiel can consist of complete end items, subsystems, Line Replaceable Units (LRUs), components, or piece-parts of major systems. All materiel shall be tested and analyzed to its NHC with respect to the performance of all its mission essential functions. Realistic hardware, and practical test

*Superscript numbers correspond to Appendix K, References.

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configurations and scenarios, shall be tested and analyzed in order to achieve an accurate and complete Nuclear Test and Survivability Assessment (NTSA). All NTSAs shall include a three phase approach, a) the electronic microcircuit (piece part) test phase; b) the system/subsystem level test phase; and c) the system assessment phase, in order to meet the requirements of Department of Defense Directive (DODD) 5000.01[2], AR 70-75 and its NHC. Pertinent data, results, and hardware information is archived into the system's Life-cycle Nuclear Survivability program database for Hardness Assurance (production), Sustainment Assurance (post-production), and Surveillance Tests.

b. This TOP adheres to an integrated set of test principles and procedures that will result in timely, reliable, and consistent data for nuclear survivability analysis. This document is encouraged for use by all nuclear survivability testers (government and contractor) for test planning, for test conduct, and for acquiring and analyzing data in technical and customer tests.

2. FACILITIES AND INSTRUMENTATION.**

2.1 Nuclear Airblast Facilities and Instrumentation.

2.1.1 Nuclear Airblast Criteria Parameters.

a. These system level criteria parameters shall be thoroughly analyzed to ensure that acceptable facilities and appropriate instrumentation are utilized.

Airblast Parameter	Unit
Peak Static Overpressure (P)	kilo Pascal (kPa)
Overpressure Duration (t_p)	second (sec)
Overpressure Impulse (I_p)	kPa-sec
Peak Dynamic Pressure (q)	kPa
Positive Duration (t_q)	sec
Dynamic Pressure Impulse (I_q)	kPa-sec
Peak Underpressure (P_{neg})	kPa
Arrival Time (t_a)	sec

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b. Performance criteria requirements of the test system include allowable downtime, recovery procedures, operate through, acceptable damage and degradation, and the availability of replacement parts and time required to implement repairs.

2.1.2 Nuclear Airblast Facilities.

a. Acceptable test facilities can be categorized as either large-scale High-Explosive (HE) field test, or threat relatable shock tubes. Mission critical military systems as large as a Blackhawk Helicopter (19.5 meters (m) X 6.4 m X 3.75 m) should utilize a large-scale HE field test because of the system's size, mass, and response. However, HE tests of large systems like the Blackhawk Helicopter are no longer performed; rather, Large Blast/Thermal Simulator (LB/TS) shock tube facilities are used. When the test item incident surface area exceeds 20% blockage of the shock tube, use a rarefaction wave eliminator and place the item outside of the tube end to obtain a more realistic airblast environment. In general, items that can translate can be tested at the exit end of LB/TS to a less than desired environment

Facility	Type	Location	Comments
Large Blast / Thermal Simulator (LB/TS)	Shock Tube / Thermochemical Reaction	White Sands Missile Range (WSMR), NM	Tube Width - 20.0 m. Max Overpressure - 241 kPa. Max Duration - 3300 millisecond (ms). System Level - 0.1 kilo-ton (kt) to 3 Mega-Ton (MT).

b. Additional Nuclear Airblast facilities are listed in "A Complete Guide to Nuclear Weapons Effects (NWE) Simulator Facilities and Applications", (2013 Edition)[3]. It shall be noted that this edition was the last published NWE Simulator Facilities and many of these facilities are not currently operational. The Test Officer (TO) shall ensure that the airblast test facility utilized is the foremost facility available for the desired criteria, test system configuration, and the anticipated responses. The TO or project engineer is also responsible for developing the cost estimates, initial planning, experimental design/development, the U.S Army Test and Evaluation (ATEC) Decision Support System (ADSS) development (if required), project/program safety, environmental, security information, scheduling resources, testing documentation , test execution and test recovery.

c. It is important that a pre-test analysis be performed to determine potential response areas and the number, type, and location of gages and high speed photography. It is emphasized that available facilities will provide only a simulated nuclear blast environment. Therefore, in addition to test data, adequate analysis shall be performed to account for the facility deficiencies which shall be known, quantified, and documented.

2.1.3 Nuclear Airblast Instrumentation.

Measurement Parameter	Preferred Device	Measurement Accuracy
Pressure	Pressure Transducer	± 10%
Strain	Strain Gages	± 10%

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Measurement Parameter	Preferred Device	Measurement Accuracy
Acceleration	Accelerometers	± 10%
Translation/Overturn/Flexing	High Speed Camera	250-400 frames per second (fps)
Damage	Digital Camera	> 2 megapixels

These measuring devices should be positioned at locations on the test item based upon the pre-test response analysis. Data transmissions are normally through twisted pair cable. Transmitted data are normally input to adjustable gain instrumentation amplifiers and transient data recorders with an operating bandwidth of 200 kilohertz (kHz).

2.2 Nuclear Thermal Radiation Facilities and Instrumentation.

2.2.1 Nuclear Thermal Radiation Criteria Parameters.

a. These system level criteria parameters shall be thoroughly analyzed to ensure that acceptable facilities and appropriate instrumentation are utilized.

Airblast Parameter	Unit
Pulse Width	sec
Thermal Flux (Q_{dot})	kilo-Joule (kJ)/m²-sec
Thermal Fluence (Q)	kJ/m²
Time to Maximum Irradiance (t_{max})	sec

b. Performance criteria requirements of the test system include allowable downtime, recovery procedures, operate through, acceptable damage and degradation, and the availability of replacement parts and time required to implement repairs.

2.2.2 Nuclear Thermal Radiation Facilities.

a. Test facilities can be categorized as solar collectors, electrical resistance heaters, or thermo-chemical reactions. The determination and experimental solution for samples evaluation, analysis, and modeling depends on the best experimental design based upon cost, schedule and assumed risk. It shall be noted that the complete process shall be adequately defined to verify and validate the analytical methods. For system level response, usually thermo-chemical reaction facilities are used because they are the only facilities that can irradiate these large systems. However, these simulators can only produce an unusable test environment for Man-Machine Categories I (man and equipment in Open) and V (fixed wind and rotary aircraft). Solar collectors and electrical resistance heaters are preferred and should be utilized on small systems, material samples, and when spectrum fidelity is a concern. If optical paths are being tested, then a non-occurring source is required. An example of a test facility is:

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Facility	Type	Location	Comments
White Sands Solar Furnace	Solar Collector	WSMR, NM	Excellent Spectrum / Subsonic Wind Tunnel / Shaped Nuclear Thermal. Pulse / No limit on Fluence. Peak flux - 100 cal/cm^2-sec. Area - 15 cm diameter.

b. Additional Nuclear Thermal facilities are listed in "A Complete Guide to Nuclear Weapons Effects (NWE) Simulator Facilities and Applications", (2013 Edition). It shall be noted that this edition was the last published NWE Simulator Facilities and many of these facilities are not currently operational. Ensure that the thermal radiation test facility utilized accurately simulates the desired test environment criteria and test item response in order to adequately test the system configuration. It is emphasized that available facilities will provide only a simulated nuclear thermal radiation environment. Therefore, in addition to good test data, adequate analysis shall be performed to account for the facility deficiencies which shall be known, quantified, and documented. For some test items there are no facilities which can produce the required levels; in these cases modeling shall be performed and the facility test is to determine model acceptability.

2.2.3 Nuclear Thermal Radiation Instrumentation.

Measurement Parameter	Preferred Device	Measurement Accuracy
Energy	Calorimeters	± 5% Cal
Temperature	Thermocouples	± 5 °Celsius (C)
Damage	Digital camera	> 2 megapixels

The nuclear thermal radiation pulse will be monitored and recorded generally through the use of a calorimeter. The waveform generated by the calorimeter will be utilized to determine the simulated thermal radiation environment against the thermal NHC specified for the system. Thermocouples should be attached to the test item to monitor thermal response and recorded utilizing analog and digital recording instruments.

2.3 High Altitude Electromagnetic Pulse (HEMP) / Source Region Electromagnetic Pulse (SREMP) Facilities and Instrumentation.

2.3.1 HEMP / SREMP Criteria Parameters.

a. These system level criteria parameters shall be thoroughly analyzed to ensure that acceptable facilities and appropriate instrumentation are utilized.

HEMP Parameter	SREMP Parameter	Unit
Electric Field (E-Field)	E-Field	volts/meter
Magnetic Field (H-Field)	H-Field	amp-turns/meter

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HEMP Parameter	SREMP Parameter	Unit
Risetime	Risetime	nanoseconds
	Gamma Dose Rate	centiGray (cGy) silicon (Si)/sec
Pulse Width	Pulse Width	nanoseconds

b. Performance criteria requirements of the test system include allowable downtime, recovery procedures, operate through, acceptable damage and degradation, re-boot, and the availability of replacement parts and time required to implement repair.

2.3.2 HEMP / SREMP Facilities.

a. HEMP test facilities can be categorized as radiating HEMP, hybrid HEMP, or bounded wave HEMP. Simulators in these categories can be vertically or horizontally polarized. Vertically polarized simulators should be utilized on systems which respond vertically such as missiles or those possessing large vertical antennas. Horizontally polarized simulators should be utilized on all military land systems, distributed systems, and aircraft. In addition to being the largest, gamma dose rate facility, the High Energy Radiation Megavolt Electron Source (HERMES) III facility located at Kirtland Air Force Base (KAFB), NM, is a SREMP facility. Examples of test facilities are the following:

Facility	Type	Location	Comments
United States Army (USA) Horizontal Polarized Dipole (HPD)-II	Radiating/Horizontal	WSMR, NM	Max E-Field - 65 kilovolt per meter (kV/m). Area - 76 m diameter. System level testing. Quadripartite Standardization Agreement (QSTAG) 244 (Edition 4)[4] and Military Standard (MIL-STD)-2169B[5] compliant.
USA Advanced Fast Electromagnetic Pulse Simulator (AFEMP)	Radiating/Horizontal	WSMR, NM	Max E-Field - 75 kV/m. Area - 200 m diameter. System level testing. QSTAG 244 (Edition 4) and MIL-STD-2169B compliant.
HERMES-III	SREMP	KAFB, NM	System level testing.

b. Additional HEMP / SREMP facilities are listed in "A Complete Guide to Nuclear Weapons Effects (NWE) Simulator Facilities and Applications", (2013 Edition). It shall be noted that this edition was the last published NWE Simulator Facilities Guide and many of these facilities are not currently operational. Ensure that the HEMP test facility utilized accurately simulate desired criteria and test system response in order to adequately test the system configuration. More than one facility may be required to adequately test a system to account for

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horizontal and vertical responses as well as SREMP effects. It is emphasized that available facilities will provide only a simulated HEMP /SREMP environment. Therefore, in addition to good test data, adequate analysis shall be performed to account for the facility deficiencies which shall be known, quantified, and documented.

2.3.3 HEMP / SREMP Instrumentation Dosimetry.

Measurement Parameter	Preferred Device	Measurement Accuracy
Current	Current Probes	± 5%
E-Field	D-Dot Probe	± 5%
H-Field	B-Dot Probe	± 5%
Gamma Dose	CaF_2(Mn)	± 10%
Gamma Dose Rate	Compton Diode	± 10%
Test Setup	Digital Camera	> 2 megapixels

a. The data acquisition system for the free-field tests should consist of transient digitizers with an operating bandwidth of 250 MHz and 500 MHz (small test items), with a 1 Giga-sample per second sampling rate. A fiber optic data transmission system that is capable of transmitting data equal to the operating digitizer bandwidth is also required. All utilized probes shall be responsive to at least 1 GHz.

b. Measurements of each illumination shall be monitored by a B-dot probe (measures the time rate of change in the H-Field) or D-dot probe (measures the time rate of change in the E-Field) so that the magnitude of the E-Field and pulse shape information is obtained. This information should be digitized, analyzed, and stored for a later detailed analysis.

c. In the case of SREMP testing, the dose rate at selected locations based upon critical electronics / cabling paths on the test system will be measured using Compton diodes. The measured gamma dose values will be expressed in cGy(Si) and cGy(tissue) by these general ratios, however the American Society for Testing and Materials (ASTM) Standards E666-14[6] and E668-13[7] must be referenced for each test:

$$cGy(Si)/cGy(CaF_2) = 1.02 \text{ and } cGy(tissue)/cGy(CaF_2) = 1.13, \text{ respectively}$$

2.4 Gamma Dose Rate (GDR) Facilities and Instrumentation.

2.4.1 Gamma Dose Rate Criteria Parameters.

a. These systems, subsystem, piece part level criteria parameters shall be thoroughly analyzed to ensure that acceptable facilities and appropriate instrumentation are utilized.

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b. Prompt gamma radiation pulses generate the production of charge carriers and subsequent photocurrents. These damaging photocurrents, which flow across device junctions, induce transient upset, latch-up, and/or burnout in the semiconductor devices.

Gamma Dose Rate Parameter	Unit
Peak Gamma Dose Rate	cGy(Si)/sec
Pulse Width	nanoseconds
Energy	Mega-electron Volt (Mev)

c. Performance criteria requirements of the test system include allowable downtime, recovery procedures, operate through, acceptable damage and degradation, and the availability of replacement parts and time required to implement repairs.

2.4.2 Gamma Dose Rate Facilities.

a. Test facilities can be categorized as electron Linear Accelerators (LINAC), or Flash X-Ray simulators. Major military systems and subsystems should utilize flash x ray simulators because they can irradiate large test systems or LRUs; while LINACs should be utilized on electronic piece-parts, components, and circuit card assemblies because of cost effectiveness, pulse width variability, and quick turn-around times. Examples of test facilities are:

Facility	Type	Location	Comments
USA WSMR LINAC	LINAC	WSMR, NM	Max dose rate - 2E12 cGy(Si)/sec. Pulse width - 10 nanosecond (ns) to 10 microsecond (µs). Piece part and component level testing.
USA WSMR PI538	Flash x-ray	WSMR, NM	Max dose rate - 2.6E11 cGy(Si)/sec. Pulse width - 70 to 90 ns. Up to system LRU level testing.
Department of Energy (DOE) Sandia National Laboratories (SNL) HERMES III	Flash x-ray	KAFB, NM	Max dose rate - > 5E12 cGy(Si)/sec. Pulse width - 20 ns. Large system level testing.

b. Additional GDR facilities are listed in "A Complete Guide to Nuclear Weapons Effects (NWE) Simulator Facilities and Applications", (2013 Edition). It shall be noted that this edition was the last published NWE Simulator Facilities Guide and many of these facilities are not currently operational. Ensure that the GDR test facility utilized accurately simulate the test criteria over an adequate exposure area and test item responses in order to adequately test the system or piece part configuration. It is emphasized that available facilities will provide only a simulated GDR environment. Therefore, in addition to good test data, adequate analysis shall be performed to account for the facility deficiencies which shall be known, quantified, and documented.

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2.4.3 Gamma Dose Rate Instrumentation / Dosimetry.

Measurement Parameter	Preferred Device	Measurement Accuracy
Photocurrent	Photocurrent Probes	± 5%
Gamma Dose	$CaF_2(Mn)$[a]	± 10%
Gamma Radiation Pulse	P-Type/Intrinsic/N-Type (PIN) Diode	± 10%
	Compton Diode	± 10%
Current	Multimeter/Digitizer	± 10%
	Oscilloscope	± 10%
Voltage	Multimeter/Digitizer	± 5%
	Oscilloscope	± 5%

[a] Other materials may be utilized instead of $CaF_2(Mn)$ to determine gamma dose. However, the material's calibration and detection shall conform to the procedures outlined in the Annual Book of ASTM Standards[8], Section 12.

a. The gamma dose is generally measured using $CaF_2(Mn)$ Thermoluminescent Dosimeters (TLDs). The measured gamma dose values will be expressed in cGy(Si) and cGy(tissue) by these general ratios, but ASTM Standards E666 and E668 shall be referenced for each test:

$$cGy(Si)/cGy(CaF_2) = 1.02 \text{ and } cGy(tissue)/cGy(CaF_2) = 1.13, \text{ respectively}$$

b. Each radiation pulse will be measured using a PIN or Compton diode and digitized on a transient digitizing system. The pulse width, Full Width Half Maximum (FWHM), of each radiation pulse will be obtained from this digitized signal. The gamma dose rate for each pulse will then be determined from the dose recorded on the TLDs divided by the pulse width obtained from the digitizers.

2.5 Neutron Fluence Facilities and Instrumentation.

2.5.1 Neutron Fluence Criteria Parameter.

a. This system, subsystem, piece part level criteria parameter shall be thoroughly analyzed to ensure that acceptable facilities and appropriate instrumentation are utilized.

b. Fast neutrons interact with semiconductor material in electronic piece-parts by elastic collisions with lattice atoms which decrease minority carrier lifetimes and increase device

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resistivity. This resulting damage alters electrical parameters of the device which can cause failure in the semiconductor devices or circuit applications.

Neutron Fluence Parameter	Unit
Neutron Fluence	1 Mev(Si) n/cm^2

c. Performance criteria requirements of the test system include allowable downtime, recovery procedures, operate through, acceptable damage and degradation, and the availability of replacement parts and time required to implement repairs.

2.5.2 Neutron Fluence Facilities.

a. Test facilities can be categorized as a Fast Burst Reactor (FBR) or a Training, Research, Isotopes, General Atomics (TRIGA) reactor. (Sometimes, Californium-252 is utilized for piece-part testing.) These reactors generally can be utilized in the pulse or steady-state mode of operation. In the pulse mode of operation, the FBR can generate neutron fluence up to 5E14 n/cm^2 with energies > 10 keV and gamma dose rate up to 1E9 cGy(Si)/sec with a pulse width of approximately 50 µsec. However, when total neutron fluence is the primary concern, the steady-state mode of operation is typically used. An example of a test facility is:

Facility	Type	Location	Comments
USA WSMR Fast Burst Reactor	Fast Burst Reactor	WSMR, NM	Peak Pulse Power - 6.5E4 MW. Neutron Fluence - 7E13 n/cm^2. FWHM Pulse Width - 40 to 3000 µs. Up to system level testing.

b. Additional Neutron fluence facilities are listed in "A Complete Guide to Nuclear Weapons Effects Simulator Facilities and Applications", (2013 Edition). It shall be noted that this edition was the last published NWE Simulator Facilities Guide and essentially all of these facilities are not currently operational. Ensure that the neutron fluence test facility utilized to accurately simulate desired test environment criteria and test item responses in order to adequately test the system, subsystem, or piece part. It is emphasized that available facilities will provide only a simulated neutron fluence environment. Therefore, in addition to good test data, adequate analysis shall be performed to account for the facility deficiencies which shall be known, quantified, and documented.

2.5.3 Neutron Fluence Instrumentation / Dosimetry.

Measurement Parameter	Preferred Device	Measurement Accuracy
Neutron Fluence	Sulfur Activation Foil[a]	± 10%
Gamma Dose	$CaF_2(Mn)$[a]	± 10%
Current	Multimeter/Digitizer	± 5%
	Oscilloscope	± 5%

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Measurement Parameter	Preferred Device	Measurement Accuracy
Voltage	Multimeter/Digitizer	± 5%
	Oscilloscope	± 5%

[a] Other materials or techniques may be utilized instead of sulfur and CaF_2(Mn) to determine neutron fluence and gamma dose, respectively. However, the material's calibration and detection shall conform to the procedures outlined in the Annual Book of ASTM Standards, Section 12.

a. The neutron fluence at each test location will be measured using sulfur activation foils which measure neutrons with energies greater than 3 MeV; but, the current versions of the ASTM Standards, E720[9], E721[10], and E722[11] shall be referenced. The measured fluence will be converted to 1 MeV(Si) equivalent damage fluence by the following relationship:

1 MeV Equiv Fluence = Fluence > 3MeV * (Total Fluence/Fluence > 3 MeV) * Hardness Parameter

(1) The Hardness Parameter is the total neutron spectrum folded with the Silicon displacement damage cross section. This hardness parameter is quantified by employing the procedures and methods detailed in all three of the ASTM Standards respectively; E720 – one determines the selection and use of Neutron-Activation Foils for determining Neutron Spectra employed in radiation-hardness testing of electronics, E721- one determines the Neutron Energy Spectra from Neutron Sensors for radiation-hardness testing of electronics and E722 – one characterizes the Neutron Energy Fluence Spectra in Terms of an Equivalent Monoenergetic Neutron Fluence for radiation-hardness testing of electronics.

(2) The Total Fluence is the fluence provided by the neutron source throughout its energy spectrum ranging from eV to 14 MeV utilizing the methods specified in E720 for Neutron-Activation Foil selection and utilization to determine this neutron spectra.

b. The gamma dose will be measured using CaF_2(Mn) TLDs. The measured gamma dose values will be expressed in cGy(Si) and cGy(tissue) by these typical ratios, but the ASTM Standards E666-14 and E668-13 shall be referenced for each test:

$$cGy(Si)/cGy(CaF_2) = 1.02 \text{ and } cGy(tissue)/cGy(CaF_2) = 1.13, \text{ respectively}$$

2.6 Gamma Total Dose Facilities and Instrumentation.

2.6.1 Gamma Total Dose Criteria Parameter.

a. This system, subsystem, piece part level criteria parameter shall be thoroughly analyzed to ensure that acceptable facilities and appropriate instrumentation are utilized.

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b. Gamma Total Dose (GTD) generates hole-electron pairs through the process of ionization in the semiconductor material resulting in trapped charges. These total dose effects are exhibited either as a change in electrical parameters or as a catastrophic failure in semiconductor devices.

Gamma Total Dose Parameter	Unit
Gamma Total Dose	cGy(Si)

c. Obtaining the proper gamma total dose test criterion can be difficult. First obtain the NHC and identify the subheading: "Silicon Absorption / Displacement Damage". Under this subheading, is the title "Max Combined Neutron and Gamma Ionizing Dose, (cGy(Si))" referred to as D_i. In addition, the correct utilized value should have in quotes (single pulse duration of 60 seconds). This acquired value is the actual Vertical Plane Center (VPC) GTD to be received by the test item.

d. Performance criteria requirements of the test system include allowable downtime, recovery procedures, operate through, recovery time, degradation and/or acceptable damage, and the availability of replacement parts and time required to implement repairs.

2.6.2 Gamma Total Dose Facilities.

a. Test facilities typically utilize a Cobalt-60 source or multiple Cobalt-60 sources. Large systems are extremely difficult to test adequately because no large scale Department of Defense (DOD) / DOE gamma dose facility is available. Therefore, most testing should be accomplished at the electronic component and LRU level. Whole body irradiations are typically limited to surfaces < 1.5 meters on a side for gradient and disposition rate reasons. Examples of test facilities are:

Facility	Location	Comments
USA WSMR Gamma Radiation Facility (GRF)	WSMR, NM	Max Dose Rate - 1700 cGy(Si)/sec. Exposure Area - 13x6x4 (height) m-4 E sources. Piece part, LRU subsystem and system level testing.
SNL Gamma Facility	KAFB, NM	Max Dose Rate - 600 cGy(Si)/sec. Exposure Area – 15.2 cm diameter x 20.3 (height) cm. Piece part, small LRU / subsystem level testing.

b. Additional gamma total dose facilities are listed in "A Complete Guide to Nuclear Weapons Effects Simulator Facilities and Applications", (2013 Edition). It shall be noted that this edition was the last published NWE Simulator Facilities Guide and many of these facilities are not currently operational. Ensure that the GTD test facility utilized accurately simulate desired test criteria over an adequate exposure area and test item responses in order to adequately test the piece part, subsystem, or system configuration. It is emphasized that available facilities will provide only a simulated GTD environment. Therefore, in addition to good test data, adequate analysis shall be performed to account for the facility deficiencies which shall be known, quantified, and documented.

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2.6.3 Gamma Total Dose Instrumentation / Dosimetry.

Measurement Parameter	Preferred Device	Measurement Accuracy
Gamma Dose	CaF_2(Mn) TLDs[a]	± 10%
Current	Multimeter/Digitizer	± 5%
	Oscilloscope	± 5%
Voltage	Multimeter/Digitizer	± 5%
	Oscilloscope	± 5%

[a] Other materials or techniques may be utilized instead of CaF_2(Mn) to determine gamma dose. However, the material's calibration and detection shall conform to the procedures outlined in the Annual Book of ASTM Standards, Section 12.

The gamma dose will be measured using CaF_2(Mn) TLDs. The measured gamma dose values will be expressed in cGy(Si) and cGy(tissue) by these typical ratios for Cobalt 60, but ASTM Standards E666-14 and E668-13 shall be referenced for each test:

$$cGy(Si)/cGy(CaF_2) = 1.02 \text{ and } cGy(tissue)/cGy(CaF_2) = 1.13, \text{ respectively}$$

2.7 Radiation Protection Factor Facilities and Instrumentation.

2.7.1 Radiation Protection Factor Criteria Parameter.

This procedure determines the value of the radiation protection or transmission factor of the item's structure or shielding, as such there is no fixed limit or criterion for this value. As with all radiation exposure, the goal is as low as reasonably achievable. Guidelines for materiel selection and composition are provided in Allied Engineering Publication 14 (AEP-14)[12]. The definitions for the protection factors are provided below. Please note that transmission factor is just the inverse of protection factor.

a. Protection Factor (PF).

(1) $PF = D_o/D_i$

(2) D_o = Free-field Radiation Dose

(3) D_i = Dose inside due to radiation penetrating the shield

b. Neutron Protection Factor (NPF).

(1) $NPF = D_{no}/(D_{ni} + D_{gn})$

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(2) D_{no} = Free Field (1m height) Neutron Dose

(3) D_{ni} = Shielded Neutron Dose

(4) D_{gn} = Secondary Gamma Dose

c. Neutron Reduction Factor (NRF).

(1) $NRF = D_{no}/D_{ni}$

(2) D_{no} = Free Field (1m height) Neutron Dose

(3) D_{ni} = Shielded Neutron Dose

(4) D_{gn} = Secondary Gamma Dose

d. Gamma Protection Factor (GPF).

(1) $GPF = D_{go}/D_{gg}$

(2) D_{go} = Free Field (1m height) Gamma Dose

(3) D_{gg} = Dose due to primary gammas penetrating shield

2.7.2 Radiation Protection Factor Facilities.

Test facilities are required to produce a measurable (intensity or duration) radiation environment (neutron and or gamma) at a distance to minimize gradients. The gamma measurements typically utilize a Cobalt-60 source or multiple Cobalt-60 sources. The neutron measurements typically utilize a fission reactor, which can be operated in the steady state mode in an open area. The radiation field shall be uniform within 20 percent over the item surface being irradiated and within 40 percent throughout the air volume occupied by the item. Examples of test facilities are:

Facility	Location	Comments
USA WSMR GRF	WSMR, NM	Max Dose Rate - 1700 cGy(Si)/sec at exposure head.
USA WSMR FBR	WSMR, NM	Outdoor capable steady state at 8 kilowatts (kW).

2.7.3 Radiation Protection Factor Instrumentation / Dosimetry.

Measurement Parameter	Preferred Device	Measurement Accuracy
Neutron Fluence	Sulfur Activation Foil	± 10 %
Neutron Dose	ROSPEC, Rotating Spectrometer - 4	± 10 %

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	Proton recoil Chambers.	
	Proton Recoil System - Determines the neutron spectrum from 50 keV to approximately 4 MeV.	± 10 %
	Neutron rem detector (NRD) remmeter or neutron Ball	± 10 %

Measurement Parameter	Preferred Device	Measurement Accuracy
Gamma Dose	TLD $CaF_2(Mn)$	± 10 %
	Far West Tissue Equivalent Chambers - 3 instruments available (1 cubic centimeter (cc), 80 cc, and 1000 cc), operating one at a time with the calibrated electrometer.	± 10 %
Source for ION chambers	Keithly 6514 electrometer or equivalent	± 5 %
Calculation Verification	Standard North Atlantic Treaty Organization (NATO) 2 meter box	± 10 %

The instrumentation shall be calibrated to National Institute of Standards and Technology (NIST) or ASTM standards, and the same instrument must be used for free field and internal measurements.

3. REQUIRED TEST CONDITIONS.

3.1 Test Preparation.

3.1.1 Scope of Testing.

Once a test program is initiated, the first concern is the establishment of the objectives and the scope of the program. In essence, these questions shall be addressed: What equipment and support items are required, how shall the equipment be tested in order to maximize determination of its performance, what test environments and at what assembly levels shall testing occur, what data are required and how it will be collected, how shall the information be processed and analyzed in order to obtain an accurate and complete survivability analysis of the test system, and ultimately the system configuration to the criteria environment. For the piece part, subsystem, or system there shall be a thorough understanding of the operation of all mission essential functions, test criteria, test facility limitations, test objectives, operational and maintenance procedures, performance and operational checkouts, material composition,

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instrumentation, dosimetry, system integration, environmental considerations, Nuclear Weapons Effects (NWE), Transient Radiation Effects on Electronics (TREE), statistical processes, and safety considerations to adequately devise a realistic test scenario, test schedule, and performance analysis program.

3.1.2 Cost Estimates.

Upon devising an appropriate test scenario, an ATEC cost estimate shall be prepared In Accordance With (IAW) ATEC Pamphlet (PAM) 73-1, Volume I[13], System Test and Evaluation Procedures. The cost estimate shall adequately account for all reasonable expenditures of the proposed nuclear test and analysis program. These direct expenditures are for man-hours, material and supplies, travel, contractual service, equipment, minor construction, facilities, repair, and replacement of test related consumables. IAW the National Defense Authorization Act 2003, only direct expenditures can be charged to DOD or DOD sponsored customers. Additionally, these cost estimates shall be posted in the ADSS and Versatile Information Systems Integrated On-Line (VISION) Digital Library System (VDLS).

3.1.3 Test Coordination.

a. From the initiation to the completion of the test program, test coordination is a constant and essential task that shall be coordinated effectively with a multitude of various personnel in order to properly plan, resource, execute, and determine the nuclear survivability of a test piece part, subsystem, or system. Without proper and effective test coordination, a NTSA program will experience cost overruns, unnecessary test delays, inadequate test data, improper determinations, and improper usage of manpower. In conclusion, test coordination is one of the most important aspects and is essential to the conduct of a successful NTSA program.

b. If military personnel are required, ensure a Test Schedule and Review Committee (TSARC) request is submitted within one year from the start of testing or as early as possible. A Safety Release (SR) shall be obtained from the U.S. Army Evaluation Center (AEC) prior to using military personnel as test participants.

3.1.4 Environmental Impact.

a. An important pre-test requirement IAW AR 200-1[14], and AR 200-2[15], is an environmental analysis. This analysis will help alleviate environmental problems that could interfere with the test schedule and completion of the NTSA program. The proper documents, such as an environmental record IAW local policy, shall be completed and submitted to the environmental office and/or personnel who regulate and control environmental practices at the test execution site prior to start-of-test.

b. The actual time requirement for document submission before test execution is dependent on the level of preparation required, type of system, and required documentation as well as the work load of the environmental office. Most of the required information can be obtained from the Program Manager (PM) office.

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3.1.5 Safety Analysis.

Another important pre-test coordination task is the safety analysis which shall be prepared IAW U.S. Army Materiel Command Regulation (AMC-R) 385-100[16]. Like the environmental analysis, it should be prepared, submitted, and approved As Soon As Possible (ASAP) to alleviate safety problems which could affect the completion of the NTSA program. The safety analysis can usually be obtained from the PM's office or system's contractor. If a complete initial safety analysis is needed, extra time and funds shall be allotted to identify the necessary safety procedures and prepare the documentation.

3.1.6 Preferred Nuclear Environment Test Methodology.

a. Sufficient analysis shall be performed to account for deficiencies in the simulated nuclear test environment versus the United States Army Nuclear and Combating Weapons of Mass Destruction Agency (USANCA) issued environments, variations between the test and production configuration, and the corresponding variations in hardware response. One shall initially assume that neither the test issued environment nor hardware are accurate representations of the NHC and system configuration, respectively. There will be differences which shall be identified and quantified in order that a survivability analysis can be successfully performed. To accurately achieve compliance with the test objective, one shall accomplish the following:

(1) First, perform the pre-test analysis to identify:

(a) Instrumentation and dosimetry for required response and environment data.

(b) Test hardware for each environment.

(c) Location of instrumentation and dosimetry.

(d) Test facilities and limitations.

(e) Test levels per environment.

(f) Test piece part, subsystem or system's performance and operational checkouts to adequately analyze all mission essential functions.

(g) Required test data per environment.

(h) Electromagnetic (EM) energy paths and port-of-entries.

(i) Potential test system's responses.

(j) Potential susceptibilities and hardness levels in all nuclear environments with respect to the USANCA criteria.

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(k) Test piece part, subsystem, or system's configuration with respect to each environment.

(l) Test piece part, subsystem ,or system's configuration baseline.

(m) Differences between test and production configuration.

(2) Second, thoroughly document and analyze the test hardware which is to be utilized during the NTSA. This documentation and determination includes the test's material composition, shape, size, mass, fastening schemes, shielding and attenuation characteristics, nuclear hardening concepts and devices, mission essential functions, and Hardness Critical Items (HCIs) and circuits. Then, analyze the test hardware relative to fielded or proposed fielded system hardware and identify all of the relevant differences. With all this information, identify and establish the test system and proposed system baseline configuration. This baseline will be utilized for the survivability analysis as well as a basis for analysis of all product improvements, Engineering Change Proposals (ECPs), and configuration changes to ensure that the test system remains nuclear survivable during production, maintenance, and deployment.

(3) Third, identify the environmental tests that will best meet the requirements identified in the pre-test analysis. The nuclear test environments are Nuclear Airblast (system level), Nuclear Thermal Irradiation (component or coupon level), electromagnetic pulse (EMP) (system level) (Endo-atmospheric (SREMP) and Exo-atmospheric (HEMP)) and Initial Nuclear Radiation (INR) (gamma dose rate, gamma total dose, and neutron fluence) (piece part, subsystem, system level). All INR testing should be conducted in the following sequence: gamma dose rate, total gamma dose, and neutron fluence. This sequence is based on actual occurrence and the fact that some semiconductor devices may generate inaccurate failure thresholds if this sequence is not preserved. If time constraints or test facility scheduling forces deviation from this INR test sequence, analysis shall be performed to insure that any out-of-sequence related effects on the test system are identified and accounted for. All effects produced by a nuclear weapon are dependent upon weapon yield, weapon type, height of burst (HOB), and distance. Figure A-2 (in Appendix A) presents the time history of environments produced by a 27 kT fission weapon detonated at a HOB of 180 m at a distance of 1 kilometer (km) (note that Figure A-2 is a chronological representation of the environments only and is not intended as a reference). As is indicated in this figure (y-axis values are normalized per environment and do not represent any relationship to magnitudes), the first environment to arrive at the location of interest is the gamma environment (this is the GDR and Prompt Gamma Total Dose (PGTD)). This is followed by a continuing Gamma Total Dose (GTD) environment (at a much lower deposition rate through 1 minute) and then a short time later by the Neutron Fluence (NF) environment. This environment delivery scheme provides a simple test chronology. The GDR is always tested as the first environment of the NTSA followed by the GTD and finally the NF environment. In addition, the NF activates the test items; therefore, to minimize personnel radiation exposure, it is tested last. As can be seen in the figure, approximately 50% of the GTD is provided prior to and 50% of the GTD is provided after the arrival of the neutrons. The physics of damage in common electronic microcircuits to the GTD and NF environments is such that the environments could be performed in either order; however, NF last is preferred. Additionally, based upon the experimental design, these environments can be alternated to maximize test hardware allocation. However, the physics of

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damage in common electronic microcircuits when exposed to the GDR environment is such that this test can never follow the NF environment. This is due to the fact that most GDR microcircuit damage is incurred as a result of induced latch-up and burnout. This latch-up is characterized by the microcircuit entering a non-functional state and consuming excessive power supply and/or input current. The latch-up path is part of the microcircuit physical layout and is the unintentional creation of parasitic secondary four-layer devices which resemble Silicon Controlled Rectifiers (SCRs) whose performance is degraded by the NF environment. Therefore, NF should never be performed before GDR. Testing conducted in the EMP, nuclear air blast, and nuclear thermal radiation environments can usually be performed independently with disregard to the test sequence because of the nature of their effects. However, HEMP is preferred after airblast and thermal because: HEMP is produced by an exo-atmospheric detonation and can occur after a surface/near-surface event, and the response of the damaged test item is likely to be more adverse. Synergistic effects on the test system, particularly for thermal and airblast, shall be considered because it is real and may significantly enhance damage, especially to light armored systems.

(4) Lastly, analyze and determine the test system's performance with a detailed post-test analysis. This post-test analysis includes test environments and results of the pre-test analysis, documentation and detailed determination of the test system's performance, determination of all shortcomings and failures and determination of obtained environmental data compared against the USANCA criteria. To effectively determine criteria compliance, a thorough understanding of the simulation fidelity of each test facility is required. All test facilities have one or more parameter deficiencies; therefore, these deficiencies shall be well understood and analysis performed to establish the effects of these parameter deficiencies on the results of the test. This analysis can adequately determine the environmental test parameters against the desired USANCA criteria. In order to effectively analyze survivability of the system configuration requires a thorough understanding of the differences between the test system's and the system's configuration, and corresponding effects on the analyzed system.

b. Combining the piece part, subsystem and system test data, pre-test analysis or other analytical data, the system can then be analyzed for its survivability to the USANCA requirements.

3.1.7 Test Plan.

Incorporate all the factors and ideas presented in paragraphs 3.1.1 through 3.1.6 into a test plan that shall be written IAW ATEC PAM 73-1. It is critical that all required steps in ATEC PAM 73-1 are followed. The test plan shall be developed and submitted to ATEC approximately sixty days prior to, and approved by ATEC approximately thirty days prior to, test execution. Test plans should contain the following information:

a. Section I: Introduction.

(1) 1.1 System Description.

(2) 1.2 Summary.

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(3) 1.5 Unique Test Personnel Requirements.

b. Section II: Subtests (for each environment).

(1) 2.1 Name of Subtest.

(2) 2.11 Objectives.

(3) 2.1.2 Criteria and Data Analysis/Procedure

(4) 2.1.3 Test Procedures and Data Required.

c. Section III: Appendices.

(1) A. Test Criteria.

(2) B. Test Schedule.

(3) C. Informal Coordination.

(4) D. References.

(5) E. Abbreviations.

(6) F. Distribution List.

d. Events are likely to occur during the test execution that causes the utilization of sound engineering judgment to deviate from the original test plan. Major deviations shall be approved by ATEC before implemented. All deviations shall be documented in the detailed test report.

3.2 Test Execution.

3.2.1 Pre-Test Analysis / Modeling.

Before the execution of any nuclear test program (piece part, subsystem, or system), a pre-test analysis shall be performed. During the pre-test analysis, thoroughly examine the test system and manipulate engineering principals and nuclear effects responses to estimate where potential nuclear survivability problems exist. Evaluate test facilities to ensure that the best facility is scheduled, sufficient data acquisition is available and scheduled, and required test configurations/orientations can be tested. In order to perform an adequate pre-test analysis, accurate schematics, part lists, details of deliberate hardening methods/hardware, previous test and/or analytical data, material composition, wiring diagrams, cable shielding details, and piece-part specifications shall be available. Based on the pre-test analysis, establish functional models where significant data can be obtained on the expected performance of the test system through all the different nuclear environments.

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3.2.2 Piece-Part / Circuit Analysis Program.

One of the major limitations in NTSA programs is the difficulty of establishing survivability confidences on systems with extremely small sample sizes. To effectively establish confidence levels and the survivability of the baseline system, it is expected to implement an analysis program. For INR, this program will utilize piece-part test data, circuit analysis, modeling methods, and statistical procedures to determine design margins and confidence levels. The piece-part / circuit analysis program will identify all potential nuclear survivability deficiencies by accounting for response variances due to different manufacturing processes. For EMP, this program will identify and analyze grounding schemes, cabling, cable shielding, transient, and terminal protection devices. For airblast and thermal analysis, the material composition, shape, size, mass, and fastening schemes are analyzed. Only by having adequate design margins in all nuclear survivability environments, can an acceptable airblast, thermal, HEMP, and/or INR survivability analysis be performed on the system's baseline configuration.

3.2.3 Test Organization and Documentation.

The formulation of a detailed test plan and effective test coordination prior to the test execution is critical to test organization and execution, and cost effectiveness. Test organization consists of a set of preset procedures for accomplishing specific test execution tasks. Proper test organization will result in superior test execution. The test procedure shall assign and explain to each test support personnel their specific tasks and schedules. Examples include test system placement, dosimetry placement, probe placement, test documentation, data acquisition, performance checkouts, maintenance procedures, etc. The most important of these specific tasks is test documentation. Ensure that all aspects of the nuclear test program are carefully, completely, and correctly documented. To achieve effective documentation, test specific control forms should be generated. Improper documentation can lead to an inaccurate and incomplete NTSA. In conclusion, careful organization and adequate documentation of the test is essential.

3.2.4 Sound Engineering Judgement.

During the entire execution of the test, utilize sound engineering judgment to effectively test and analyze the test item and maintain schedules and costs. Sound engineering judgment becomes extremely critical when schedule impacts occur such as facility downtime, inclement weather, failures and/or re-prioritization. Under such conditions, determine the problem, deviate from the original test plan, and devise an alternate plan or set of procedures. Any deviations from the test plan shall be recorded and reported in the test report.

3.3 Test Reporting and Life-Cycle.

3.3.1 Data Reduction and Analysis.

After the completion of all survivability testing, conduct data reduction and analysis on the raw data. The raw test data are manipulated into an understandable format and documented in Appendix B and summarized in the Test Results section of the test report. The actual data

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reduction procedures selected is dependent upon the performance parameters, the test environment, and the criteria parameters. All data reduction procedures shall be standardized for each individual test and documented. Clear and concise data reduction and analysis will enhance and enrich the final product, the survivability analysis.

3.3.2 Statistical and Error Analysis.

Other forms of analysis that should be performed on the test data are statistical and error analysis. Use statistical analysis to obtain the nuclear survivability probability of electronic piece-parts based on test data, circuit analysis, and safety margins. The preferred confidence window is the 99/90 tolerance level (0.99 probability of being survivable with a 90 percent confidence). Also, statistical analysis should be utilized to obtain the criteria compliance between actual environment parameters and desired criteria. An error analysis should be performed to account for and eliminate sources of error present in the raw test data. Possible sources of error are: instrumentation and data acquisition, human, test setup, probe, dosimetry, and round off. Utilize this error analysis to help predict how accurate the simulated test environment was to the specified USANCA environment and to ensure that the test system received its nuclear survivability criteria taking into account the predicted error.

3.3.3 Survivability Analysis of the System Configuration versus USANCA Criteria.

Based on data processed (system and environmental), analyze the nuclear survivability of the test system to each test environment. Then analyze the nuclear survivability of the system's deployed configuration to each of the USANCA environments. To accomplish this, first identify and define the test system's configuration, test environments, and safety margins. Use this information to establish nuclear survivability of the test system configuration to the test environments. The test results and environments are then corrected to represent the USANCA environments by accounting for differences and deficiencies. Finally, analyze the baseline system configuration performance against the corrected or USANCA environments. This is the NTSA of the baseline configuration and is the information for the Technical Analysis section of the test report. Also, the system's baseline configuration and analysis is the basis for analyzing effects of product improvements, other system configuration changes or repairs on the hardness level and survivability of the system during its lifetime. These future analyses will involve additional piece-part / circuit analysis and piece-part and or subsystem level testing where data are not available.

3.3.4 Test Record / Report.

a. After the completion of all test execution, data analyses, and survivability analysis, a detailed test report or test record shall be written IAW ATEC PAM 73-1. It is critical that all required steps in ATEC PAM 73-1 are followed. The test record / report shall be completed and submitted to ATEC NLT the time frames specified in Table 6.2 (ATEC PAM 73-1) after test completion and approved by ATEC. Test Record / Reports should contain the following information:

(1) Foreword.

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(2) Section 1: Executive Digest.

(a) 1.1 System Description.

(b) 1.2 Summary.

(c) 1.3 Conclusions.

(d) 1.4 Recommendations.

(3) Section 2: Subtests (for each test environment).

(a) 2.1 Name of Subtest.

(b) 2.1.1 Objectives.

(c) 2.1.2 Criteria and Analysis.

(d) 2.1.3 Test Procedure and Findings.

(4) Section 3: Appendices.

(a) A. Test Criteria.

(b) B. Test Data.

(c) C. Recommendation for Classification of Risk.

(5) If no additional appendices are required to adequately quantify the test results and findings, the following appendixes are required to close the test report and will be lettered consecutively:

(a) D. References.

(b) E. Abbreviations.

(c) F. Distribution List.

b. The highlighted portions of the previous list (Summary, Test Findings, & Analysis) are the most significant sections of the test report. Special consideration shall be given to ensure these sections are detailed, accurate, concise, complete, and comprehensible.

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3.3.5 Life-Cycle Nuclear Survivability Program.

The production, operation, maturity, storage, maintenance, repairs, modification, and ambient environments shall not introduce any form of susceptibilities or unacceptable levels of degradation into a nuclear survivable system. To ensure continued nuclear survivability, a Life-Cycle Nuclear Survivability (LCNS) program shall be established IAW the NHC, AR 70-75, and the Department of Defense Instruction (DODI) 3150.09[17]. The basic purpose of the LCNS program is to control all changes to the baseline configuration during production and product improvements, ensure that an acceptable hardness level is preserved during maintenance by using certified spare parts and procedures, and verifying that the hardness level is not degraded to an unacceptable level during fielding, storage, and operation in the ambient environments.

3.4 Nuclear Airblast Pre-Test Analysis.

During the pre-test analysis, analyze and identify the high risk susceptibilities and test conditions of the test system to the airblast environment. To do this effectively:

a. Identify potentially susceptible subsystems and/or system components based upon exposure conditions, materiel composition, shape, size, mass, and fastening schemes.

b. Identify test system's configuration.

c. Define data acquisition requirements.

d. Analyze contractor's program documentation.

e. Analyze hardening and analysis performed by contractor.

f. Define baseline performance checks for test hardware.

g. Determine which set of test environment(s) from the NHC is required.

h. Identify the most realistic and severe test configuration and orientation with respect to ground zero (GZ) or the source.

i. Identify the type, number, and location of gages to measure the response of areas of concern.

j. Analyze potential test facilities to determine the one best for test system or test item(s).

k. Analyze selected test facility's response producing parameters and calculate the expected system response utilizing engineering principals.

l. Perform structural analysis of mechanical and structural response of the test system and of mission critical external fixtures/appendages to the transient loads induced by the airblast wave utilizing finite element methods or similar analytical techniques.

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m. If appropriate, perform analyses using a Verification, Validation and Accreditation (VV&A) approved code.

n. Identify detailed photography scheme.

3.5 Nuclear Thermal Radiation Pre-Test Analysis.

During the pre-test analysis, analyze and identify the high risk susceptibilities and test conditions of the test system to the thermal radiation environment. To do this effectively:

a. Identify potentially susceptible subsystems and/or system components based upon exposure conditions, materiel composition, shape, size, and mass.

b. Identify test item's configuration.

c. Define data acquisition requirements.

d. Analyze contractor's program documentation.

e. Analyze hardening and analysis performed by contractor.

f. Define baseline performance checks for test hardware.

g. Identify the most realistic and severe test configuration and orientation with respect to the thermal radiation source.

h. Identify the type, number, and location of gages to measure the response of areas of concern.

i. Determine which set(s) of thermal radiation environment from NHC for each test item.

j. Analyze potential test facilities to determine the one best for test system or test item(s). More than one may be required, for Man-Machine Categories II, III and IV.

k. Analyze selected test facility's response producing parameters and calculate the expected system's thermal response utilizing engineering principals.

l. If appropriate, perform analyses using a VV & A approved code.

m. Define and document all pre-test visual inspections and quantified performance check baselines.

n. Identify detailed photography scheme on all exposure areas.

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3.6 HEMP / SREMP Pre-Test Analysis.

During the pre-test analysis, analyze and identify the high risk susceptibilities and test conditions of the test system to the HEMP / SREMP environments. To do this effectively:

a. Analyze drawings and circuits to determine potentially harmful energy paths. The analysis should be concentrated on external unshielded cables of significant length and the interface of these cables into subsystems of the test system.

b. Identify test system's configuration.

c. Identify and determine all points-of-entry.

d. Analyze grounding scheme and shielded cables to include backshells and connectors for shielding effectiveness.

e. Identify and analyze Hardness Critical Circuits (HCC) and Terminal Protection Devices (TPDs).

f. Determine inherent hardness afforded by the system to its mission critical electronics.

g. Define data acquisition requirements.

h. Analyze contractor's program documentation.

i. Analyze hardening and analysis performed by contractor.

j. Define baseline performance checks for test hardware.

k. Utilize peak pulse power data to analyze the HCIs, HCCs, and TPDs that are inputs to these large energy paths, if available.

l. Fabricate Breakout Boxes (BOBs), as required, for all cables of concern to enable actual pin measurements to be performed during testing and current injection.

m. Analyze potential test facilities to determine the one best for the test system or its test item(s).

n. Identify the type and location of current and differential voltage probes to be utilized to measure predicted cables and pins of concern.

o. Identify all test orientations.

p. Identify test levels based on results of hardening determination.

q. Identify all test configurations and operating modes.

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r. Identify the location for all dosimetry (SREMP).

3.7 Gamma Dose Rate Pre-Test Analysis.

During the pre-test analysis, analyze and identify high risk HCIs and test conditions of the test piece part, subsystem, or system to the gamma dose rate environment. To accomplish this effectively:

a. Identify all HCIs based upon technology and topology, and utilize Appendix E (Nuclear Radiation Effects) for component family generic response levels analysis. Consider existing circuit hardening in this screening.

b. Identify test system's configuration.

c. Define data acquisition requirements.

d. Analyze contractor's program documentation.

e. Analyze hardening and analysis performed by contractor.

f. Define baseline performance checks for test hardware HCIs, subsystem, system).

g. Analyze potential test facilities to determine the one best for the test system and/or its test item(s). At least two different facilities will likely be required - one for HCIs and one for LRUs and/or systems.

h. Identify the most realistic and severe test setup/circuit with respect to the radiation exposure.

i. Identify all current limiting, power removal, and/or gamma dose rate hardening applications.

j. Identify the type, number, and location for all dosimetry and data acquisition required to collect real time response data.

k. Establish the baseline configuration of the test system.

l. Identify test orientations.

m. Identify test levels based on results of hardening determination.

n. Identify test configurations and operating modes.

o. Acquire all available test data for all priority HCIs.

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p. Analyze all HCI circuit performance characteristics.

q. Determine HCI safety margins based upon test data, circuit analysis, and statistical techniques.

3.8 Neutron Fluence Pre-Test Analysis.

During the pre-test analysis, analyze and identify high risk HCIs and test conditions of the test piece part, subsystem, or system to the neutron fluence environment. To accomplish this effectively:

a. Identify all HCIs based upon technology and topology, and utilize Appendix E (Nuclear Radiation Effects) for component family generic response levels analysis. Consider existing circuit hardening in this screening.

b. Identify test system's configuration.

c. Define data acquisition requirements.

d. Analyze contractor's program documentation.

e. Analyze hardening and analysis performed by contractor.

f. Define baseline performance checks for test hardware (HCIs, subsystem, system).

g. Analyze potential test facilities to determine the one best for the test system and/or its test item(s).

h. Identify the most realistic and severe test setup with respect to the neutron source.

i. Identify the type, number, and location for all dosimetry and data acquisition required to collect real time response data.

j. Establish the baseline configuration of the test system.

k. Acquire all test data for all high priority HCIs.

l. Analyze all available HCI circuit performance characteristics.

m. Identify correction factor to establish 1 MeV Eq. damage in silicon.

n. Determine HCI safety margins based upon test data, circuit analysis and statistical techniques.

o. Identify test orientations with respect to the neutron source.

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p. Identify test levels based on results of the hardening determination.

q. Identify test configurations and operating modes.

3.9 Gamma Total Dose Pre-Test Analysis.

During the pre-test analysis, analyze and identify high risk HCIs and test conditions of the test piece part, subsystem, or system to the gamma total dose environment. To accomplish this effectively:

a. Identify all HCIs based upon technology and utilize Appendix E (Nuclear Radiation Effects) for component family generic response levels analysis.

b. Identify test system's configuration.

c. Define data acquisition requirements.

d. Analyze contractor's program documentation.

e. Analyze hardening and analysis performed by contractor.

f. Define baseline performance checks for test hardware (HCI, subsystem, system).

g. Analyze potential test facilities to determine the one best for the test system and/or its test item(s).

h. Identify the most realistic and severe test setup with respect to the radiation source.

i. Identify circumvention and/or gamma total dose hardening techniques.

j. Identify the type, number, and location for all dosimetry and data acquisition required to collect real time response data.

k. Establish the baseline configuration of the test system.

l. Identify test orientations for exposure.

m. Identify test configurations and operating modes.

n. Identify test levels based on results of hardening determination.

o. Acquire all available test data for all high priority HCIs.

p. Analyze all HCI circuit performance characteristics.

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q. Determine HCI safety margins based upon test data, circuit analysis and statistical techniques.

r. Analyze potential for Dose Enhancement Effects.

3.10 Radiation Protection Factor Pre-Test Analysis.

During the pre-test analysis, analyze and identify risks and test conditions of the test item to the gamma and neutron environments. To accomplish this effectively:

a. Identify test system's configuration.

b. Define data acquisition requirements.

c. Analyze shielding and analysis performed by contractor.

d. Analyze potential test facilities to determine the one best for the test system.

e. Identify the most realistic and severe test setup with respect to the radiation source.

f. Identify the type, number, and location for all dosimetry and data acquisition required to collect real time response data.

g. Establish the baseline configuration of the test system.

h. Identify test orientations for exposure.

i. Identify facility test configurations and operating modes.

j. Identify test levels based on facility and instrumentation limitations.

k. Identify personnel and security requirements.

l. Identify radiation exposure limits and control area contours.

m. Collect data from standard NATO 2 meter box

4. TEST PROCEDURES.

4.1 Nuclear Airblast.

4.1.1 General.

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4.1.1.1 Test System.

Survivability of the test hardware when exposed to the airblast test environment will be analyzed by:

a. Performing a detailed pre-test analysis (this supplements planning, improves experimental, designs and enhance successful testing).

b. Ensuring that the test hardware is properly deployed and in a realistic operational state and configuration as established by the pre-test analysis.

c. Establishing the performance baseline of the test hardware prior to the event.

d. Determining effects by visual inspections, performance of the baseline checks, and detailed failure diagnosis.

e. Determining displacement translation, rotation, tipping overturn.

f. Determining performance/operational data of the test hardware.

g. Analyzing both still and high speed motion photography of the test hardware taken before, during, and after the airblast event.

h. Analyzing response measurements from instruments such as acceleration, pressure, and strain gages.

i. Determining damage and/or degradation in regards to impacts on mission accomplishment.

j. Analyzing test environment data.

4.1.1.2 Baseline System.

The survivability of the baseline system configuration when exposed to the USANCA issued NHC airblast environments will be analyzed by:

a. Analyzing the differences between the test and USANCA environments.

b. Analyzing the differences between the test and baseline system configurations.

c. Determining the response and survivability of the baseline system configuration to the USANCA environment.

4.1.2 Test Setup.

The test setup is based upon results obtained from the pre-test analysis and should consist of:

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a. Based on pre-test determination of primary response mechanism, peak static, or dynamic pressure, position the test hardware at the desired test location and in the correct test configuration.

b. Instrument the test hardware IAW the pre-test analysis to acquire critical response data of the test hardware. Instrumentation includes pressure transducers, accelerometers, and strain gages. Ensure instrumentation is properly mounted/bonded and calibrated.

c. Prior to the event, the complete test hardware shall be examined to ensure proper operation and establish the performance baseline. All problems identified will be documented and corrected if detrimental to the airblast test program.

d. Photograph the pre-test setup to include instrumentation locations.

e. Setup free-field pressure transducers.

f. Setup, check, and calibrate the complete data acquisition system. Check the cables to ensure adequate attachment to the response measuring instruments, junction box (j-box), data recorders, and amplifiers; and that they are sufficiently protected against the airblast wave. Set-up amplifiers and transient digitizers IAW predictions of the pre-test analysis. All data acquisition calibration should be accomplished at the test location to ensure accuracy.

g. Setup, checkout, and calibrate backup data channels for critical areas and responses.

h. Setup, checkout, and calibrate the pressure transducers and accelerometers to measure the principal free-field environmental parameters.

i. Setup the high speed motion cameras to record the response of the test hardware during the event. Typical speeds are 250 and 400 fps.

j. Ensure the test hardware is in the proper operational mode for the test.

4.1.3 Test.

a. After the airblast environment has been produced and the test area is considered safe, a comprehensive damage analysis shall be performed on the test hardware. This damage analysis will consist of post-event photography, a detailed visual inspection, displacement and rotation measurements, and complete post-event performance/operational checks. Response data obtained from pressure transducers, accelerometers, and strain gages will be processed and thoroughly analyzed and determined. All pertinent pre-event and post-event information shall be clearly and accurately documented and analyzed. Diagnostics of all failed and mission degraded areas of the test system shall be performed and mission impacts determined.

b. The test environment data will be processed, analyzed, and determined. The eight test environment parameters will be analyzed against the USANCA parameters to determine criteria

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compliance. These criteria compliances shall be utilized in correcting induced and projected responses in the test system and baseline configuration, respectively. All pertinent test data and results, photographs, test system and baseline configuration information and assessments will be archived into the system's life-cycle nuclear survivability program database.

4.2 Nuclear Thermal Radiation.

4.2.1 General.

4.2.1.1 Test System.

Survivability of the test hardware when exposed to the test thermal radiation environment will be analyzed by:

a. Performing a detailed pre-test analysis to identify thermal radiation sensitive elements of the system.

b. Establishing the performance baseline of the test hardware prior to testing to include pre-test photography, performance and operational checks, and visual inspections.

c. Properly position/mount calibrated instruments to measure environment and response.

d. Determining effects by repeating all performance baseline procedures and visual checks on each test hardware after exposure to thermal radiation.

e. Exposing mission critical hardware items to at least 1.3X, with 1.5X desired to establish hardness margins and confidence levels.

f. Analyzing photographs of the test hardware.

g. Analyzing all instrumentation data (thermocouples and calorimeters).

h. Analyzing all environmental data.

i. Determining damage and/or degradation in regards to impacts on mission accomplishment.

4.2.1.2 Baseline System.

The survivability of the baseline system configuration when exposed to the USANCA issued thermal radiation environments will be analyzed by:

a. Analyzing the differences between the test and USANCA environments.

b. Analyzing the differences between the test and baseline configurations.

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c. Determining the response of the baseline configuration to the USANCA environment.

4.2.2 Test Setup.

Before testing, the performance of each test item identified in the pre-test analysis will be baselined and documented utilizing photographs, visual inspections, and performance and operational checks. All problems identified will be documented and corrected if detrimental to the thermal radiation test program. The thermal radiation environment will then be monitored using calorimeters and adjusted until the specified thermal pulse is generated. Upon verifying the content of the generated pulse, the calorimeter will be removed and the instrumented test item will be properly positioned in the test chamber. Test setup photographs will be taken. The above setup procedures will be repeated for each set of criteria.

4.2.3 Test.

a. After the thermal exposure, the pre-test baseline check procedures will be performed. The test item will be repositioned to expose another thermal sensitive area to 1.0X, if required. If the test item survives, a second test sample will be positioned for testing to 1.3X to confirm the previous results and to provide confidence. If this test item fails, a third sample will be exposed to 1.0X to confirm the result of the first test and to provide additional confidence. If the sample survives 1.3X, then, the third sample will be exposed to 1.5X. This procedure will be repeated until all thermal sensitive areas and/or samples have been exposed and all sets of criteria have been addressed. It is essential that mission critical test hardware items or test samples be exposed to 1.3X to establish a safety margin to ensure nuclear thermal radiation survivability. All exposed areas will be photographed. Failures will be diagnosed and analyzed as to the cause and effects on mission performance. Response data will be processed, analyzed, and determined. All pertinent data will be documented. Where necessary, diagnostics of all failed areas shall be performed and the results determined.

b. The test environment data will be processed, analyzed, and evaluated against the criteria system performance. The four critical test environment parameters will be analyzed against the USANCA parameters to determine criteria compliance. These criteria compliances shall be utilized in correcting induced and projected responses in the test system and baseline configuration, respectively.

c. Pertinent data, results, and hardware information is achieved into the system's Life-cycle Nuclear Survivability program database for Hardness Assurance (production), Sustainment Assurance (post-production), and Surveillance Tests.

4.3 HEMP / SREMP.

4.3.1 General.

4.3.1.1 Test System.

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a. Survivability of the test system when exposed to the test HEMP and/or SREMP environment will be analyzed by:

(1) Performing the detailed pre-test analysis.

(2) Calibrating required Data Acquisition Systems (DAS) elements.

(3) Establishing the performance and operational baseline for the test system prior to testing.

(4) Determining effects by repeating the performance and operational baseline checks or abbreviated checks after each illumination.

(5) Illuminating the test system in the pre-selected orientations, configurations, and modes at 0.5, 0.75, 1.0, and 1.5 times (if possible) the E-field criterion level as defined in the pre-test analysis phase. Determining all upsets, failures, downtimes, mission performance impacts, and corrective actions.

(6) Analyzing response and environmental data.

(7) Current injecting at 1X, 3X, and 5X based upon simulator signals and/or the damped sinusoidal waveforms obtained from Conducted Susceptibility (CS)115 and CS116 in MIL-STD-461F[18] and MIL-STD-464C[19].

(8) Acquiring Shield Cable Test (SCT) measurements for baselining the test system and for the LCNS database.

(9) Analyzing system response in both the time and frequency domains.

b. Ensure that accurate, consistent and documented operational checks are utilized. Many of the problems induced by the illumination will be transient upsets and will be correctable by crew intervention such as re-cycling power.

4.3.1.2 Baseline System.

The survivability of the baseline system configuration when exposed to the HEMP/SREMP USANCA environments will be analyzed by:

a. Analyzing the differences between the test and USANCA environments.

b. Analyzing the differences between the test and baseline configurations.

c. Determining the response of the baseline configuration to the USANCA environment.

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4.3.2 Test Setup.

Prior to testing, the complete test system will be analyzed to ensure proper operation and establish the performance baseline. All problems identified will be documented and corrected if detrimental to the HEMP and/or SREMP test program. The test facility will perform calibration and noise measurements on the DAS to ensure accurate data acquisition. The DAS utilized shall account for all introduced error and be adequately protected against EM interference. The test system will be positioned in its first orientation in the facility's test volume based upon facility mapping data. BOBs will be installed, dosimetry positioned (SREMP), and current and/or voltage probes will be positioned based on information obtained from the pre-test analysis. The baseline or abbreviated baseline checks will be performed. Test setup photographs will be taken. These procedures will be repeated for each test orientation and configuration at each test level.

4.3.3.1 Test.

a. The test system will be illuminated by simulated HEMP and/or SREMP waveforms. After each illumination or set of illuminations, the test system will be analyzed to identify and quantify effects by using the pre-test baseline or abbreviated checks and diagnostic checks. Test probes and new dosimetry will be repositioned, if required, and the test system will be illuminated again. This procedure will be implemented until sufficient data are obtained for all functional modes and system configurations on all cables identified in the pre-test analysis. At the completion of the first successful test system orientation, the system's orientation will be altered IAW the pre-test analysis unless the test results dictate differently. Once adequate data are obtained for the initial test level, the test level will be incremented as specified in paragraph 4.3.1. The levels determined in paragraph 4.3.1.1.a(5) can be altered based on engineering judgment of the results/effects of the on-going test. Multiple illuminations or a substantial test sample size (seven test items are preferred based upon Normal Probability Distribution Statistics) shall be utilized to provide statistical confidence in the HEMP and/or SREMP survivability of the test system. Failures and significant upsets will be diagnosed as to cause and impacts on mission accomplishment. Response data will be processed, analyzed, and assessed to determine impact(s) on system performance. All pertinent data will be documented.

b. The test environment data will be processed, analyzed, and evaluated against the criteria. The four critical test environment parameters will be analyzed against the USANCA parameters to determine criteria compliance. These criteria compliances shall be utilized in correcting induced and projected responses in the test system and baseline configuration, respectively.

c. Pertinent data, results, and hardware information is achieved into the system’s Life-cycle Nuclear Survivability program database for Hardness Assurance (production), Sustainment Assurance (post-production), and Surveillance Tests.

4.3.3.2 Current Injection and Shielded Cable Test (SCT).

a. Current injection techniques are essential to HEMP evaluation of large distributed systems and should be utilized as an integral part of the EMP test. Current injection is greatly

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beneficial in the context of determining safety margins and enhancing and verifying HEMP simulator results. Current injection should also be an integral part of life-cycle HEMP Survivability program. But, current injection cannot be the primary means of obtaining accurate HEMP data.

b. Based upon the actual response measurements and cable/pins identified in the pre-test analysis, current can be injected, direct or inductive, at the maximum measured current level up to a level recommended for the subassemblies by CS115 and CS116 of MIL-STD-461 or MIL-STD-464. Also, dominant coupling frequencies obtained from the simulator tests will be primaries for current injected. Current and/or voltage probes will be positioned on the injected cables and response measurements will be obtained. Inductive types of current probes are preferred. The baseline or abbreviated baseline checks will be performed after each set of injections.

c. Circuit response measurements will be made IAW CS115, CS116, and dominant coupling frequencies. These data will be digitized and stored so that a detailed analysis can be performed, and to update and compare against the LCNS database. These data will be utilized to provide preferred safety margins such as 3X and 5X to account for variations in hardening features/devices and between systems, and input electronic piece-parts. A thorough baseline performance check will be performed at the completion of current injection testing.

d. The SCT should be performed on shielded cables of concern identified in the pre-test analysis or during simulator testing. These data will be obtained from a spectrum analyzer and stored so that a detailed analysis can be performed. The SCT results will be utilized to baseline the performance of shielded cables for model verification and for comparison during future LCNS tests to assess margin degradation and hardness maintenance.

4.4 Gamma Dose Rate.

4.4.1 General.

4.4.1.1 HCI.

Survivability of the test electronic device when exposed to the gamma dose rate test environment will be analyzed by:

a. Testing of a minimum of 11 samples of each high risk HCI vendor-parts (based upon Normal Probability Distribution Statistics) that are identified in the pre-test analysis and for which inadequate test data exist. One shall adequately conduct the cost trade between test device costs (i.e., parts < $50), to determine the best solution between 11 versus 24 samples.

b. Detailed characterization of all critical electrical performance parameters of each high risk HCI requiring test.

c. Calibrating the DAS.

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d. Establishing HCI performance characteristics by monitoring transient response and repeating the pre-test performance baseline checks after each exposure.

e. Performing a detailed circuit analysis for each unprotected HCI or hardened critical process analysis for protected HCIs.

f. Irradiating the test samples while energized.

g. Establishing radiation design margins (99/90) utilizing the results of the circuit analysis and the characterization/test program conducted at 1X, 2X, 3X, and 5X, unless consistent HCI failures dictates otherwise.

h. If the probability of a nuclear event occurring on a LRU while powered is small (<10%), then the LRU HCIs can be eliminated from gamma dose rate testing.

i. Accepting/rejecting high risk unprotected HCIs based on Radiation Design Margins (RDMs), which are defined as:

$$RDM = \frac{Effects\ Threshold\ Level}{Criteria\ Dose\ Rate\ Level}$$

4.4.1.2 Test System.

Survivability of the test system when exposed to the gamma dose rate test environment will be analyzed by:

a. Performing a detailed pre-test analysis.

b. Analyzing deliberate hardened non-Radiation Tolerance (RT) HCIs and/or techniques for adequacy.

c. Establishing performance and configuration baseline of the test system prior to testing.

d. Irradiating the system while energized and operating.

e. Establishing the system operational status by identifying and quantifying effects on performance and identifying performance differences after each gamma dose rate exposure by repeating the baseline and diagnostic checks.

f. Instrumenting the test system, installing BOBs, and calibrating the DAS.

g. Irradiating the test system in different configurations, orientations, and modes.

h. Determining and documenting all upsets and/or damage, downtime, mission performance impacts, and necessary corrective action procedures.

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i. Stressing the system by utilizing multiple exposures at the criterion level.

j. Exposing mission critical items to levels that can be utilized to verify a specified RDM of at least 2X with 5X desired.

k. Determining survivability of the system to the test environment. Upsets and/or latchups are expected. The first corrective action attempt will be to cycle power.

4.4.1.3 Baseline System.

The survivability of the baseline system configuration when exposed to the gamma dose rate USANCA environment will be analyzed by:

a. Analyzing the differences between the test and USANCA environments.

b. Analyzing the differences between the test and baseline configurations.

c. Determining the response of the baseline configuration to the USANCA environment.

4.4.2 Test Setup.

4.4.2.1 HCI.

Prior to testing, a detailed pre-test electrical performance characterization of each of the high risk HCIs in its specified test circuit shall be performed. Characterization data will be collected on eleven samples of each HCI utilizing digital, analog, and/or mixed signal testers. The characterization program should include all major manufacturers' parameter specifications. Before testing, the desired Source-To-Target (STT) distances and pulse widths required for each test level and configuration shall be mapped and calibrated utilizing CaF_2(Mn) TLDs and a PIN Diode. Each of the Device-Under-Test (DUT) will then be centered upon the facility's output beam at the determined STT distance to receive the first required test level. A PIN diode will be positioned next to the DUT in the beam to measure each irradiation pulsewidth. Calibrated probes to measure the response are positioned on individual pins of the piece part to monitor currents, voltages, and induced photocurrents during irradiation. These probes provide input to a DAS. The DAS should utilize shielded data cables to transmit signals to the transient digitizers and waveform processors. Protection of the DUT and DAS against Radio Frequency (RF) fields shall be provided for, otherwise, instrumentation may be damaged or data corrupted by spurious signals created during dose rate generation. Finally, the DUT's test circuit is energized and exposure configuration performance baseline is accurately established.

4.4.2.2 Test Unit/System (Item).

a. Prior to testing, a performance baseline for the test item will be established. Problems identified will be documented and corrected if detrimental to the gamma dose rate test program. The various STT distances and exposure areas will be determined from previous test data, and then refined by measurements. An area equal to the area of active electronics will be mapped

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using CaF_2(Mn) TLDs. The selection of each STT distance will also include the requirement that the gamma dose rate exposure across the target is adequate to stimulate the targeted area. The test item will be powered during each irradiation. The system will be positioned at the first STT distance to receive a specified percent of the criterion gamma dose rate level based upon the pre-test analysis. In place and prior to irradiation, the test system's baseline and operational status will be re-verified. The gamma dose rate level should be applied to the VPC of the test volume of the target area.

b. Diodes are positioned at selected locations on the test item to measure the received gamma dose and pulsewidth, respectively. The current, voltage, and/or transient photocurrent probes are positioned IAW the pre-test analysis. The DAS should be setup using double shielded data cables or fiber optics, transient digitizers, and waveform processors. Protection of the DAS against RF fields shall be provided for, otherwise, instrumentation may be damaged or data corrupted by spurious signals created during gamma dose rate pulse simulation.

4.4.3 Test.

4.4.3.1 HCI.

Eleven samples of each HCI will be characterized, and irradiated. Ten samples will be at 1X the criterion level while biased, and characterized again. This procedure will be repeated at 2X, 3X, and 5X the criterion level unless a valid failure occurs such as the HCI failing to meet manufacturer's specifications, upset, latchup or burnout. The eleventh sample will be characterized and kept as a control device to demonstrate characterization equipment consistency. If a valid failure occurs at or below the criterion level, the HCI fails qualification. If a valid failure occurs above the criterion level, then this information along with the circuit analysis will be utilized to establish the RDM, which in turn, will be utilized to determine acceptance/rejection for usage in the system. After the HCIs have been irradiated by a gamma dose rate pulse, all circuit operational checks should be initiated within three minutes IAW MIL-STD-883J[20], Methods 1020, 1021, and 1023. The facility's access time shall be taken into consideration in order to provide an accurate analysis of determining whether special action should be initiated. Analyze the test environment recorded by the facility PIN diode against the USANCA environment and establish criteria compliance. This criterion compliance will be utilized in determining the survivability of the HCI against the USANCA criteria.

4.4.3.2 Test System Test Unit/System (Item).

a. After the energized test item system has been irradiated at 1X the criterion level by a pulse of bremsstrahlung photons, all operational checks should be initiated within 5 minutes or within the allowable downtime of the test after the irradiation. The time duration after irradiation to initiate the baseline checks is dependent on the safety procedures of the utilized test facility. The facility access time shall be taken into consideration in order to provide an accurate analysis of determining whether special action should be initiated. If determined to be operational to an acceptable level, the test item’s dosimetry will be replaced and the test item will be re-positioned and irradiated again. Operational checks between gamma dose rate pulses should be thorough, but as abbreviated as possible to achieve an efficient test program. This process will be repeated

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until all configurations, modes, orientations, and levels identified by the pre-test analysis or on-going test results have been accurately tested, analyzed, and documented. The preferred test levels for system level testing are 1X, 1.5X, 2X, and 3X the criterion level with a preferred sample size of seven. If an upset or latchup occurs, the problem will be documented and diagnosed. Testing will not be continued until the problem is completely understood and its effects on the system have been analyzed. If the problem is upset or latchup, the affected subsystem(s) will be identified and testing will be repeated to ensure that the problem was environmentally induced. Test/diagnostic circuits will be employed to collect information required in determining the cause and impacts on mission critical performance parameters. Usually, resolution can be made at the HCI level employing the methods described in paragraph 4.4.3.1. Work arounds, as necessary, will be implemented to complete the testing. A follow-up investigation will be performed to identify the failure to the HCI level. The exposure environment recorded by the TLDs and facility COMPTON or PIN diode will be analyzed against the USANCA environment and establish criteria compliance.

b. A final operational baseline check will be performed on the test item at the end of the test. If possible, sufficient number of test items should be tested, analyzed, and documented as specified above to achieve extremely important statistical confidence in the gamma dose rate survivability of the test item. Usually only one test sample is available, therefore emphasizing the need to perform test and analysis of all power removal circuit and hardening critical process employed.

c. The test environment data will be processed, analyzed, and evaluated. The critical test environment parameters will be analyzed against the USANCA parameters to determine criteria compliance. This criterion compliance shall be utilized in correcting induced and projected responses in the test system and baseline configuration, respectively.

d. Pertinent data, results, and hardware information is achieved into the system's Life-cycle Nuclear Survivability program database for Hardness Assurance (production), Sustainment Assurance (post-production), and Surveillance Tests.

4.5 Neutron Fluence Test Procedures.

4.5.1 General.

4.5.1.1 HCI.

Survivability of the test system's HCIs when exposed to the neutron fluence test environment will be analyzed by:

a. Requiring testing of 11 samples of each high risk HCI that are identified in the pre-test analysis and for which inadequate test data exists. One shall adequately conduct the cost trade between part costs (i.e., parts < $50), to determine the best solution between 11 versus 24 samples.

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b. Detailed characterization of all critical performance parameters of each high risk HCI requiring test.

c. Establishing HCI performance characteristics by repeating the pre-test performance baseline checks after each exposure.

d. Performing a detailed circuit analysis for each HCI application.

e. Irradiating the DUTs while not energized unless annealing or operate through is a critical parameter.

f. Establishing RDMs (99/90) utilizing the results of the circuit analysis and the characterization/test program conducted at 1X, 2X, 3X, and 5X, unless consistent HCI failures dictates other reasonable RDMs.

g. Accepting/rejecting high risk HCIs based on RDM, which is defined as:

$$RDM(1\ Mev\ Equivalent)\ Damage\ in\ Si = \frac{Fluence\ at\ which\ Circuit/Device\ Fails}{Criteria\ Neutron\ Fluence\ Level}$$

4.5.1.2 Test Unit/System (Item).

Survivability of the test item when exposed to the neutron fluence test environment will be analyzed by:

a. Performing a detailed pre-test analysis.

b. Analyzing deliberate hardening devices non-RT HCIs and/or hardening techniques for adequacy.

c. Establishing performance and configuration baseline of the test item prior to testing.

d. Irradiating the item while not energized. Unless annealing or operate through is as critical.

e. Establishing the test item's operational performance status by identifying and quantifying effects on performance and performance differences after each neutron fluence exposure by repeating the baseline and diagnostic checks as necessary.

f. Irradiating the test item in different orientations.

g. Determining and documenting all damage, downtime, mission performance impacts, and necessary corrective action procedures.

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h. Exposing mission critical items to levels that can be utilized to verify a specified RDM of at least 2X with 3X highly desired.

i. Determining survivability of the test item to the test environment.

4.5.1.3 Baseline System.

The survivability of the baseline system configuration when exposed to the USANCA neutron fluence environment will be analyzed by:

a. Analyzing the differences between the test and USANCA environments.

b. Analyzing the differences between the test and baseline configurations.

c. Determining the response of the baseline configuration to the USANCA environment.

4.5.2 Test Setup.

4.5.2.1 HCI.

Prior to testing, a detailed pre-test electrical performance characterization of each of the high risk HCIs in its specified test circuit shall be performed. Characterization data will be collected on eleven samples of each vendor-part using digital, analog, and/or mixed signal testers. The characterization program should consist of all major manufacturers' specifications. Before testing, the desired STT distances and reactor modes required for each test level shall be obtained. The various STT distances will be determined from previous test data, and then refined by measurements, if necessary, utilizing sulfur detectors. Each of the HCIs will then be positioned at the determined STT distance to receive the first required exposure level. Pairs of TLDs and sulfur detectors will be placed VPC beside each DUT to measure the received gamma dose and neutron fluence, respectively.

4.5.2.2 Test Unit/System (Item).

Prior to testing, a performance baseline for the test item will be established. Problems identified will be documented and corrected if detrimental to the neutron fluence test program. The various STT distances will be determined from previous test data and then refined by measurements. An area equal to the area of neutron sensitive HCI will be mapped using CaF_2(Mn) TLDs and sulfur detectors. The selection of each STT distance will also include the requirement that the neutron fluence gradient across the target area is less than 10%. The test item will be deployed and tested at the first STT distance. The specified neutron fluence level should be applied to the VPC of the test item's exposure volume of concern. The TLDs and sulfur detectors will be positioned at selected locations on the test unit/system. Sulfur detectors and TLDs should also be positioned at locations at each of the STT distances in a relatively free-field environment to measure the received gamma dose and neutron fluence, respectively. If required to adequately determine the neutron fluence response of an "operate through system", the data acquisition should be accomplished utilizing photocurrent probes, shielded data cables, transient digitizers,

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and waveform processors. If the gradient requirement of 10% cannot be met and the test item does not possess an operate through requirement, the test item should be disassembled, exposed in the neutron fluence environment, reassembled, and analyzed to determine the effects on the mission essential functions of the test item.

4.5.3 Test.

4.5.3.1 HCI Level.

Ten samples of each HCI will be characterized, irradiated at 1X the criterion level, and characterized again. This procedure will be repeated at 2X, 5X, and 10X the criterion level unless a valid failure occurs, such as the HCI failing to meet its circuit's specifications. An eleventh sample will be characterized and kept as a control device to insure characterization equipment consistency. If a failure occurs, at or below the criterion level, the HCI fails qualification. If a failure occurs above the criterion level, then this information along with the circuit analysis will be utilized to establish the RDM which, in turn, will be utilized to determine acceptance/rejection. After the HCIs have been irradiated by neutrons, all post characterization should be initiated within 24 hours IAW MIL-STD-883J, Method 1017. The test environment will be analyzed against the USANCA environment and criteria compliance obtained. This criterion compliance will be utilized in determining the survivability of the HCI- against the USANCA criteria.

4.5.3.2 Test Unit/System (Item) Level.

a. After the test item has been irradiated at 1X the criterion level, all operational checks should be initiated ASAP. For non-powered experiments, usually two hours is the typical time interval between exposure and checkout. The actual access time after irradiation to perform operational checks is dependent on the safety procedures of the utilized test facility. If determined to be operational to an acceptable level, the test item’s dosimetry will be replaced and the test hardware will be re-positioned and irradiated again. This process will be repeated until all test levels and conditions identified by the pre-test analysis or on-going test results, have been accurately tested, analyzed, and documented. The preferred test levels for system level testing are 1X, 1.5X, 2X, and 3X the criterion level and the preferred sample size is seven. If a failure occurs, the problem will be documented and diagnosed. Testing will not be continued until the problem is completely understood and its effects on the system’s mission performance have been analyzed. The affected subsystem(s) will be identified and testing may be repeated on a second sample to determine whether the problem was environment induced. Test/diagnostic circuits may be employed to collect information required in determining the cause and impacts on missions. Usually, resolution can be made at the HCI level employing the methods described in paragraph 4.5.3.1 above. A follow-up investigation will be performed to identify the failure to the HCI level. A final operational baseline check will be performed on the test system at the end of the test. If possible, sufficient number of test items should be tested, analyzed, and documented as specified above to achieve extremely important statistical confidence in the neutron fluence survivability of the test unit/system.

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b. The test environment data will be processed, analyzed, and determined. The critical test environment parameters will be analyzed against the USANCA parameters to determine criterion compliance. This criterion compliance shall be utilized in correcting induced and projected responses in the test system and baseline configuration, respectively.

c. Pertinent data, results, and hardware information is achieved into the system's Life-cycle Nuclear Survivability program database for Hardness Assurance (production), Sustainment Assurance (post-production), and Surveillance Tests.

4.6 Gamma Total Dose Test Procedures.

4.6.1 General.

4.6.1.1 HCI.

Survivability of the test system's electronic HCIs when exposed to the gamma total dose test environment will be analyzed by:

a. Requiring testing of 11 samples of high risk HCIs that are identified in the pre-test analysis and for which inadequate test data exist. One shall adequately conduct the cost trade between HCI costs (i.e., parts < $50), to determine the best solution between 11 versus 22 versus 33 samples (i.e., availability, cost, and customer requirements).

b. Detailed characterization of all critical performance parameters of each HCI requiring testing.

c. Establishing HCI performance characteristics by repeating the pre-test performance baseline checks after each exposure.

d. Irradiating the test samples in one continuous pulse of less than one minute duration while energized and operating.

e. Establishing RDMs (99/90) utilizing the results of the circuit analysis and the characterization/test program conducted at 1X, 1.5X, 2X, and 3X, unless consistent HCI failures dictates otherwise.

f. If the probability of a nuclear event occurring on a LRU while powered is small, then the LRU(s) can possibly be eliminated from gamma total dose testing.

g. Accepting/rejecting HCIs based on DM which is defined as:

$$RDM = \frac{Failure\ Absorbed\ Dose}{Criteria\ Total\ Dose\ \ Level}$$

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4.6.1.2 Unit Test/System (Item).

Survivability of the test item to the gamma total dose test environment will be analyzed by:

a. Performing a detailed pre-test analysis.

b. Analyzing deliberate hardened non-RT HCIs and/or techniques for adequacy.

c. Establishing performance and configuration baseline of the test item prior to testing.

d. Irradiating the test item while energized and operating.

e. Establishing the item's operational status by identifying and quantifying effects on performance and performance differences after each gamma total dose exposure by repeating the baseline and diagnostic checks as necessary.

f. Instrumenting the test item, installing BOBs as needed, and calibrating the DAS.

g. Irradiating the test item in different configurations, orientations, and modes, as required.

h. Determining and documenting all upsets and/or damage, downtime, mission performance impacts, and necessary corrective action procedures.

i. Exposing mission critical areas to levels that can be utilized to verify a specified RDM of at least 2X with 3X highly desired.

j. Determining survivability of the test item to the test environment.

4.6.1.3 Baseline System.

The survivability of the baseline system configuration when exposed to the USANCA gamma total dose environment will be analyzed by:

a. Analyzing the differences between the test and USANCA environments.

b. Analyzing the differences between the test and baseline configurations.

c. Determining the response of the baseline configuration to the USANCA environment.

4.6.2. Test Setup.

4.6.2.1 HCI.

Prior to testing, a detailed pre-test electrical performance characterization of each of the high risk HCIs in its specified test circuit shall be performed. Characterization data will be collected on

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eleven samples of each HCI using digital, analog, and mixed signal testers. The characterization program should consist of all major manufacturers' specifications. Before testing, the desired STT distances and exposure times required for each test level and configuration shall be mapped and calibrated utilizing CaF_2(Mn) TLDs. TLDs will be placed next to each of the DUTs to measure the received gamma dose. Each of the DUTs or set of DUTs will then be positioned at the STT distance to receive the first required test level. Next, the DUT test circuit will be energized and exposure circuit baseline performance accurately established. Calibrated probes to measure the response of the DUT are positioned on individual pins to monitor currents and voltages during irradiation. These probes are the input to the DAS. The DAS should utilize shielded data cables to transmit signals to the multimeters, transient digitizers, and waveform processors.

4.6.2.2 Test Unit/Systems (Item).

Prior to testing, a performance baseline for the test item will be established. Problems identified will be documented and corrected if detrimental to the gamma total dose analysis program. The various STT distances and exposure areas will be determined from previous test data, and/or pre-test analysis, and then refined by measurements. The area equal to the area of gamma dose sensitive electronics will be mapped using CaF_2(Mn) TLDs. The selection of each STT distance will also include the requirement that the GTD gradient across the target area is less than 10%. The test item will be deployed at the first STT distance and tested in realistic configurations based on the pre-test analysis. The test item will be powered during each irradiation. In place and prior to irradiation, the test item baseline and operational status will be re-verified. The GTD level should be applied to the VPC of the test system exposure volume of concern. The TLDs will be positioned at selected locations on the test item to measure the received gamma dose. If required to adequately determine the GTDs response of the test item, a DAS will be setup and using voltage and/or current probes along with shielded data cables, multimeters, transient digitizers, and waveform processors. If at all possible, the gamma total dose exposure should be deposited on the test item within one minute.

4.6.3 Test.

4.6.3.1 HCI Level.

Ten samples of each piece part will be characterized, irradiated at 1X the criterion level while biased and operating, and characterized again. The GTD shall be delivered to the DUT in one continuous pulse within one minute. This procedure will be repeated at 1.5X, 2X, and 3X the criterion level unless a valid failure occurs such as the HCI failing to meet manufacturer's specifications or the circuit requirements. An eleventh sample will be characterized and kept as a control device to insure characterization equipment consistency. If a valid failure occurs, at or below the criterion level, the HCI fails qualification. If a failure occurs above the criterion level, then this information along with the circuit analysis will be utilized to establish the RDM which, in turn, will be utilized to determine acceptance/rejection. After the DUT has been irradiated by gammas, all circuit operational checks should be initiated within one hour IAW MIL-STD-883J, Method 1019. The test environment will be analyzed against the USANCA environment and

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criteria compliance will be determined. This criterion compliance will be utilized in determining the survivability of the part against the USANCA criteria.

4.6.3.2 Test Unit/System (Item).

a. After the energized and operating test item has been irradiated at 1X the criterion level by gamma photons, all operational checks should be initiated within one hour after the irradiation. The time duration after irradiation to initiate the baseline and operational checks is dependent on the safety procedures of the utilized test facility. If determined to be operational to an acceptable level, the test item's dosimetry will be replaced and the test item will be re-positioned and irradiated again. All necessary operation checks between GTD exposures should be thorough, but as abbreviated as possible to achieve an efficient test program. This process will be repeated until all configurations, modes, orientations, and levels identified by the pre-test analysis or on-going test results have been accurately tested, analyzed, and documented. The preferred test levels for system level testing are 1X, 1.5X, and 2X the criterion level and the preferred sample size is seven based upon availability, cost and schedule. If a failure occurs, the problem will documented and diagnosed. Testing will not be continued until the problem is completely understood and the effect on the test item has been analyzed.

b. The affected subsystem(s) will be identified and testing may be repeated on a replacement subsystem to ensure that the problem was environmentally induced. Test/diagnostic circuits will be employed to collect information required in determining the cause and impacts on mission critical parameters. Usually, resolution can be made at the HCI level employing the methods described in paragraph 4.6.3.1. A follow-up investigation will be performed to identify the failure to the HCI level. A final operational baseline check will be performed on the test item at the end of the test. If possible, sufficient number of test systems should be tested, analyzed, and documented as specified above to achieve extremely important statistical confidence in the GDR survivability of the test items.

c. The test environment data will be processed, analyzed, and determined. The critical test environment parameters will be analyzed against the USANCA parameters to determine criterion compliance. This criterion compliance shall be utilized in correcting induced and projected responses in the test items and baseline configuration, respectively.

d. Pertinent data, results, and hardware information is achieved into the system's Life-cycle Nuclear Survivability program database for Hardness Assurance (production), Sustainment Assurance (post-production), and Surveillance Tests.

4.7 Radiation Protection Factor Test Procedures.

4.7.1 General.

4.7.1.1 Test System.

Protection factor measurement of the test system to the gamma and neutron environment will be performed by:

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a. Performing a detailed pre-test analysis.

b. Establishing safety and security procedures.

c. Establishing performance and configuration baseline of the test system prior to testing.

d. Perform measurements and calculation on Standard NATO 2-meter box.

e. Instrumenting the test system and calibrating the DAS.

f. Irradiating the test system in different configurations and orientations as required.

g. Exposing the system to levels that are adequate to insure instrumentation response.

h. Establish received dose based on instrumentation, facility spectrum folding, and Monte Carlo N-Particle (MCNP) analysis.

i. Determine the protection factor of the test system to the test environment.

4.7.1.2 Baseline System.

The protection factors of the baseline system configuration will be analyzed by:

a. Analyzing the differences between the test and baseline configurations.

b. Determining the response of the baseline configuration.

4.7.2 Test Setup.

4.7.2.1 System.

Prior to testing, all area safety and security shall be coordinated. Please note, this is an extensive effort, since it requires protection of the facility, test personnel and the general public. The test system will be unpowered during each irradiation and is not required to be operational; however, the system shall, at least, be mobile and have all equipment installed. The various STT distances and exposure areas will be determined from previous test data or pre-test analysis. The selection of each STT distance will include the requirement that the dose gradient across the target surface is less than 20% and across the target volume is less than 40%. In place and prior to irradiation, the dose measurement instrumentation will be positioned at the selected personnel location and verified as operational.

4.7.3 Test.

4.7.3.1 Free Field Measurements.

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The dose measuring instrumentation will be positioned at the measurement location (for a typical system (8 x 4 meters) this will be 180 to 400 meters from the source) with the system removed. The facility (neutron or gamma) will be activated and dose measurements taken for the period of time required for data acquisition.

4.7.3.2 System Level Measurements.

a. The system will be emplaced in the first configuration and orientation at the free field measurement point and the dose measuring instrumentation will be installed at the selected personnel location within the system and verified as operational. The facility (neutron or gamma) will be activated and dose measurements taken for the period of time required for data acquisition. This measurement is performed independently three times. The process is repeated for each personnel position of interest and each system configuration/orientation.

b. Pertinent data, results, and hardware information is achieved into the system's Life-cycle Nuclear Survivability program database for Hardness Assurance (production), Sustainment Assurance (post-production), and Surveillance Tests.

5. DATA REQUIRED.

See Appendix H for examples of system test data documentation.

5.1 Nuclear Airblast.

a. Detailed description of the method of producing the airblast environment to include photographs displaying test system configuration with respect to the origin of the airblast environment.

b. Detailed description of all operational and performance baseline checks for all test items comprising the test system.

c. Results from the pre-test analysis to include data from the contractor's test/analysis and other airblast test/analysis programs performed on similar military systems.

d. Detailed description (to include composition of the exposed material and hardening hardware) serial numbers, and dimensions of all test items in the test system.

e. Results of the airblast environment measurements with the peak static overpressure expressed in kPa (±10%), the overpressure duration expressed in ms (±10%), the overpressure impulse expressed in kPa-ms (±10%), the peak dynamic pressure expressed in kPa(±10%), the dynamic pressure positive duration expressed in ms (±4%), the dynamic pressure impulse expressed in kPa-ms (±10%), the peak static under pressure expressed in kPa (±10%), and the arrival time expressed in seconds (±10).

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f. Detailed description, serial numbers, and location to include photographs of placement of all response measuring gages such as pressure transducers, strain gages, load cells, and accelerometers.

g. Detailed description of the DAS hardware/software.

h. Calibration and percent error data for all data acquisition equipment.

i. Detailed photographs of the test system before and after exposure to airblast environment.

j. Results of pre- and post-exposure visual inspections and pre- and post-exposure performance/operational checks.

k. Results of all failure diagnostics.

l. Measured system acceleration, pressure, or strain responses.

m. Baseline configuration of the test system and proposed production system.

n. List and description of all expected test system support equipment.

o. Detailed description of mission essential functions.

p. Manikins' response measurements, including acceleration (g ±4%), pressure, (kPa ±4%), pressure duration (ms ±4%), and peak force (newtons ±4%).

q. Complete set of pre-test calculated peak static overpressure levels versus radial distance of test system position from source.

r. High-speed (250 and/or 400 fps) motion picture camera photographs of the test system during the event.

s. Test Incident Reports (TIRs).

5.2 Nuclear Thermal Radiation.

a. Detailed description of the test facility and method of producing the thermal radiation environment to include photographs displaying test item or system configuration with respect to the thermal radiation environment source.

b. Results of pre-test thermal radiation analysis on the test item and/or system, identifying potentially high risk test items and/or areas.

c. Results from the pre-test analysis to include data from the contractor's test/analysis and other thermal radiation test/analysis programs performed on similar military systems.

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d. Detailed description of all performance and operational baseline checks for all items in the test system to be tested.

e. Detailed description (to include composition of the material), serial numbers, and dimensions of all test items in the test system.

f. Material Safety Data Sheets (MSDS) for all irradiated test items.

g. Results of the thermal radiation environment measurements with the total energy expressed in cal/cm² (±7%), maximum irradiance expressed in cal/cm²-sec (±7%), time to maximum irradiance expressed in seconds (±7%), and pulsewidth (FWHM) (±7%) expressed in seconds.

h. Detailed description of the DAS hardware/software.

i. Calibration and percent error data for all data acquisition equipment.

j. Photographs of the test item, before and after exposure to the thermal radiation environment.

k. Results of pre-and post-exposure visual inspections and pre-and post-exposure performance and operational checks.

l. Results of failure diagnostics.

m. Detailed description of mission essential functions.

n. Detailed description of test item configurations.

o. Baseline configuration of the test system and proposed production system.

p. Detailed description of all expected test item support equipment.

q. TIRs.

5.3 HEMP / SREMP.

a. Detailed description of the method and facility producing the HEMP and SREMP environment to include photographs of the test facility setup showing test system location relative to the HEMP and SREMP source.

b. Complete set of pre-test mapping data of the facility with the E-field expressed in volts/meter (±5%), risetime and pulsewidth expressed in nanoseconds (±5%), frequency expressed in Hertz (±5%), and H-field amplitude expressed in amp-turns/meter (±5%), gamma

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dose rate expressed in cGy(Si)/sec (±10%), and duration of each gamma radiation pulse expressed in seconds (±5%).

c. Results from the pre-test analysis to include data from the contractor's HEMP and SREMP test/analysis programs, as well as other such programs performed on similar military systems.

d. Detailed description of system performance and operational checks utilized to baseline the system and determine its post-illumination operational status.

e. Complete list of all active electronic piece parts utilized in the test system.

f. Complete set of electrical schematics and interconnect diagrams.

g. Detailed description, serial numbers, and dimensions of each subsystem of the test system.

h. Detailed description of all system cables to include type, composition, and dimensions.

i. Detailed description of all backshells and connectors to include attachment methodology, type, and composition.

j. Detailed description of the grounding scheme utilized on the test system.

k. Complete list of safety and environmental concerns.

l. Detailed description of all mission essential functions.

m. Detailed description of all deliberate EM hardening techniques/hardware to include manufacturer's specifications.

n. Detailed description of pre-test selected system configurations, orientations, and modes utilized during the test.

o. Detailed description and documentation of all inspections, downtime (sec) (±10%), performance and operational checks, and maintenance procedures.

p. Detailed description of the facility's data acquisition system to include probe calibration data, noise measurements, hardware, and software.

q. Detailed description of utilized current and voltage probes, and BOBs and probe locations employed on the test system.

r. Results of all HEMP and SREMP environment and test points measurements to include real time response and Fast Fourier Transforms (FFTs).

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s. Results obtained from the pre-test analysis, SCTs, and Current Injection (CI) tests.

t. Detailed description of the method and facility producing the SCTs and CIs.

u. Detailed description of recovery procedures and time.

v. Detailed description of the method and facility producing the gamma dose rate test environment including photographs of the test facility setup showing test system location relative to the gamma radiation source.

w. Results and locations of dosimetry utilized (SREMP).

x. Complete set of pre-test mapping data in radiation absorbed dose (cGy) in silicon (cGy(Si))(±10%) and cGy(tissue) (±10%) for each expected test location.

y. Results of all energy coupling and protection hardware analysis to include DMs.

5.4 Gamma Dose Rate.

a. Detailed description of the method and facility producing the gamma dose rate test environment including photographs of the test facility setup showing test system location relative to the gamma radiation source.

b. Complete set of pre-test mapping data in radiation absorbed dose (cGy) in silicon (cGy(Si))(±10%) and cGy(tissue) (±10%) for each expected test location.

c. Pulsewidth (FWHM) of each gamma pulse.

d. Results from the pre-test analysis to include data from other gamma dose rate test/analysis programs performed on similar military systems.

e. Test data and/or analytical data on the HCIs and the test system from the contractor such as Design Parameter Reports (DPRs).

f. List of all active electronic piece parts utilized in the test system.

g. Piece part characterization and test data on HCIs from available databases.

h. Detailed description, serial numbers, and dimensions of each subsystem of the test system.

i. Results of applicable piece part tests and circuit analysis to include DMs.

j. Description of statistical method(s) used to determine DMs.

k. Detailed description and electrical schematics of test circuits utilized.

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l. Detailed description of all mission essential functions.

m. Duration of each gamma radiation pulse (sec) (±10%).

n. Total gamma dose expressed in cGy(Si) and cGy(tissue) (±10%).

o. Detailed description of all utilized data acquisition procedures and hardware/software.

p. Detailed description of expected system configurations, orientations, and modes.

q. Detailed description and documentation of all inspections, downtime (sec) (±10%), operational checks, and maintenance procedures.

r. Type and location of dosimeters on the test system for each test exposure.

s. Conversion factors (±10%) used to convert cGy(CaF_2) to cGy(Si) and cGy(tissue).

t. Complete list of safety and environmental concerns.

u. TIRs.

v. Diagnostic data on all failure(s) or unacceptable degradation(s).

5.5 Neutron Fluence.

a. Detailed description of the method and facility producing the neutron fluence test environment including photographs of the test facility setup showing test system location relative to the neutron fluence source.

b. Time duration (sec) (±1 sec) and nominal power level (watts) (±5%) for each steady-state operation.

c. Documentation of each radiation pulse signature to include the shape, width at FWHM (sec) (±10), and burst size (Delta T °C) (±10%).

d. Results of the pre-test analysis and data from other neutron fluence test/analysis programs performed on similar military systems.

e. Test data and/or analytical data and analysis on HCIs and the test system from contractor such as DPRs.

f. List of all active electronic piece parts utilized in the test system.

g. Piece part characterization and test data on HCIs from available databases.

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h. Detailed description to include material composition, serial numbers, and dimensions of each test item of the test system.

i. Complete set of electrical schematics.

j. Results of applicable piece part test and circuit analysis to include DMs.

k. Description of statistical method(s) used to determine DMs.

l. Detailed description of all mission essential functions.

m. Detailed description of utilized data acquisition procedures, hardware/software.

n. Detailed description of test item or system configurations, orientations, and modes.

o. Detailed description and documentation of all inspections, downtime (sec) (±10%), operational checks, and maintenance procedures.

p. Type and location of dosimeters on the test system for each test exposure.

q. Conversion factors (±10%) used to convert cGy(CaF_2) to cGy(Si) and cGy(tissue).

r. Complete list of safety and environmental concerns.

s. TIRs.

t. Diagnostic data on all failure(s) or unacceptable degradation(s).

u. Results of all neutron environment measurements with neutron fluence expressed in terms of 1 Mev (Si) equivalent damage fluence (n/cm^2) ,and gamma dose expressed cGy(Si) and cGy(tissue) (±10%).

v. Complete list of possible expected radioactive isotopes.

5.6 Gamma Total Dose.

a. Detailed description of the method and facility producing the gamma total dose test environment including photographs of the test facility setup showing test system location relative to the gamma radiation source.

b. Complete set of pre-test mapping data in radiation absorbed dose (cGy) in silicon (cGy(Si))(±10%) and cGy(tissue) (±10%) for each expected test location.

c. Results of the pre-test analysis and data from other gamma total dose test/analysis programs performed on similar military systems.

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d. List of all active electronic piece parts utilized in the test system.

e. Test data and/or analytical data and analysis on HCIs and the test system from the contractor such as DPRs.

f. Piece part characterization and test data on HCIs from available databases.

g. Detailed description, serial numbers, and dimensions of each subsystem of the test system.

h. Detailed description and electrical schematics of test circuits utilized.

i. Results of piece-part tests and circuit analysis to include DMs.

j. Description of statistical method(s) used to determine DMs.

k. Detailed description of all mission essential functions.

l. Duration of each gamma radiation pulse (sec) (±5%).

m. Gamma total dose expressed in cGy(Si) and cGy(tissue) (±10%).

n. Detailed description of all utilized data acquisition procedures.

o. Detailed description and documentation of all inspections, downtime (sec) (±10%), operational checks, and maintenance procedures.

p. Type and location of dosimeters on the test system for each test exposure.

q. Conversion factors (±10%) used to convert cGy(CaF_2) to cGy(Si) and cGy(tissue).

r. Complete list of safety and environmental concerns.

s. TIRs.

t. Diagnostic data on all failure(s) or unacceptable degradation(s).

5.7 Radiation Protection Factor.

a. Detailed description of the method and facility producing the radiation protection factor test environment including photographs of the test facility(s) setup showing test system location relative to the radiation source.

b. Complete set of pre-test mapping data in radiation absorbed dose cGy(tissue) (±10%), and neutron spectrum for the test area and each expected test location.

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c. Results of the pre-test analysis and data from other radiation protection factor test/analysis programs performed on similar military systems.

d. Test data and/or analytical data and analysis on shielding of the test system from the contractor.

e. Detailed description, serial numbers, and dimensions of each subsystem of the test system.

f. Detailed description of all radiation protection factor instrumentation, instrument capabilities and limitations; and calibration sources.

g. Detailed description of testing orientations/configurations and measurement instrumentation locations.

h. Survey locations of test item and source locations to minimize placement / distance error.

i. Normalized conversion factors for each instrument based upon mRads/kW-hr or other consistent correlation.

j. Duration and level of each radiation exposure (neutron and gamma) (kW-Minutes), (kW level) and (sec) (±1%).

k. Detailed description of all utilized data acquisition procedures.

l. Type and location of dosimeters / foils and/or instrument on the radiation source to capture neutron spectrum at defined and test locations.

m. Conversion factors (±10%) if used to convert measurements to dose to cGy(tissue).

n. Complete list of safety, security, and environmental concerns and mitigations.

o. Defined test chain of command and reporting structure.

p. TIRs.

q. Diagnostic data on all instrumentation failure(s) or unacceptable degradation(s).

6. PRESENTATION OF DATA.

6.1 Data Compliance.

a. Results from the pre-test analysis, and all other applicable nuclear survivability programs will be analyzed and, whenever possible, incorporated into all facets of the NTSA on the test system. The incorporation of all available analytical and test data will be used to enhance and reduce the overall scope of the test program.

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b. Data from free-field environment measurements will be utilized to define the test environment and quantify the differences between the test and criterion environments. Differences greater than fifteen percent between the primary parameter values will be analyzed to determine the effect on the test results. Procedures and analysis utilized will be clearly documented.

c. Results from the pre-test analysis, system test and post-test determination/analysis, and environment compliance will be integrated into an analysis of the survivability of the test system's configuration and the USANCA environments. The final analysis of the test system may show different damage and mission impacts than the test system's results due to extrapolation and correction of environmental and test results to account for variances and differences.

d. The USANCA NTSA requirements are usually derived from the following documents:

(1) QSTAG 244, Edition 4: Nuclear Survivability Criteria for Military Equipment.

(2) QSTAG 620, Edition 2: Nuclear Survivability Criteria for Communications-Electronics Equipment[21].

(3) MIL-STD-2169B/C: High Altitude Electromagnetic Pulse (HEMP) Environment.

e. The final survivability analysis of the baseline system configuration to the USANCA requirements will utilize, incorporate, and integrate data and results of the test system survivability determination and analysis of the configuration differences. This final survivability analysis of the baseline configuration may show results different than the test system analysis due to extrapolations and/or corrections for configuration differences.

6.2 Data Reduction.

a. All raw data collected during nuclear survivability testing shall be processed to remove data acquisition and dosimetry error and to refine simulation deficiencies. All analytical procedures and methods utilized to process these raw data shall be documented along with example calculations in "Appendix B: Test Data" of a detailed test report. The entire collection of raw data should not be presented in the test report because of its excessive bulk. Reduced data that are pertinent to the analysis and support the determinations should be included in tabular form in the main body.

b. Quantitative and analytical techniques along with adequate response measurements shall be utilized during all nuclear survivability testing. A simple GO / NO-GO test is not acceptable and will not enable the survivability of the system to be determined.

c. The data shall demonstrate that the test hardware was adequately tested to its specified criteria in each nuclear test environment. The test environments will then be processed and combined with the pre-test results, along with the body of data analyzed, so that the survivability

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of the test configuration can be determined. Analytical techniques such as PSPICE, Messenger-Spratt and curve fitting shall be discussed with constraints and inputs to enable the reader to determine adequacy. All analytical data reduction methods shall be identified and presented in Appendix B of the test report.

d. Testing in the HEMP / SREMP and nuclear airblast environment shall include data in both the frequency and time domains as well as pertinent processed data in Appendix B of the test report.

e. Statistical analysis such as computing the mean, standard deviation, 99/90 tolerance limit, minimum data, DMs, and criteria compliance percentages should be performed on all nuclear survivability system test data. Type and quality of data will determine the statistical methods to be employed.

f. Whenever an electronic piece part possesses sample size of eleven or more and the data can be assumed to come from a normally distributed population, the 99/90 tolerance limits will be calculated. This statistical figure, calculated from the mean, standard deviation, and sample size is the limit below (or above, depending on the specific parameters of interest), which we expect (with 90% confidence) 99% of the population to survive. In cases where the underlying distribution of the data is not known and cannot be assumed to be normal, non-parametric statistics should be used. In these cases, larger sample sizes will be required to provide the same confidence of attaining the DM. It should be noted that some adjustment to the desired confidence level and/or population proportion may be necessary for non-parametric techniques (e.g., the 90/90 non-parametric tolerance limit requires a sample size of 22, the 95/90 non-parametric tolerance limit requires a sample size of 45, and the 99/90 non-parametric tolerance limit requires a sample size of 230). Any changes in population proportions or confidence levels should be coordinated with the ATEC independent evaluator or ATEC independent assessor.

g. In electronic piece part testing, the minimum preferred sample size is eleven because one device is utilized as a control device; these devices are required for the GDR, GTD, and NF phases. If possible, a minimum of four electronic devices should be tested to each required test environment.

h. In system level testing, the preferred sample size of seven is desired to provide an acceptable level of statistical confidence. However, this sample size is extremely difficult to obtain for system level testing. Therefore, stress testing is used to provide additional confidence in the results. Typical stress levels are 2X and 4X for INR, 1.3X for airblast and thermal radiation, and 25 illuminations at HEMP/SREMP criteria or 10 illuminations at 1.5X HEMP/SREMP criteria.

i. Additional data reduction and analytical techniques can be found in TOP 01-2-620[22].

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6.3 Data Presentation.

Data shall be presented in a clear and concise manner, so they are easy to understand and support the conclusions regarding the nuclear survivability of the test item/system hardware as depicted in Appendix H. To accomplish this, a combination of charts, graphs, drawings, tables, and photographs should be utilized.

a. Tables should be utilized to present the following data:

(1) Irradiation / Illumination Test Results Summary.

(2) Equipment Test Matrix.

(3) Criteria Compliance.

(4) Test Point Data and Results.

(5) Statistical Analysis.

(6) Criteria and Test Standards.

(7) Test Comparisons.

b. Photographs should be utilized to present the following data:

(1) Dosimetry Locations.

(2) Test Configurations, Orientations, and Set-ups.

(3) Test Facility's Data Acquisition Set-up.

(4) Locations of Other Utilized Measuring Devices.

(5) Real Time Response (airblast).

(6) Test Facility Layout.

(7) Visible Damage.

c. Drawings should be utilized when photography is not available or inadequate to display critical data supporting the results and/or conclusions. Drawings may also be utilized to illustrate airblast and/or thermal damage and/or effects and test orientations / configurations.

d. Charts and Graphs should be utilized to present the following data:

(1) Test Schedules.

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(2) Criteria Compliance.

(3) Previous Test Comparisons.

(4) Comparisons of Test Point Data with the Test Item in Different Configurations, Orientations, or Modes.

(5) Test Program Status.

e. Circuit analysis and DM determination for each high risk HCI shall be provided in Appendix B of the Test Report. As a minimum, the data shall include:

(1) Test Circuit Layout.

(2) Utilized Analytical Techniques.

(3) Application of Utilized Data.

(4) Design Margins.

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APPENDIX A. GENERAL NUCLEAR WEAPON EFFECTS.

The detonation of a nuclear weapon generates the following four primary effects or energy distributions: blast, thermal radiation, INR, and residual nuclear radiation. A fifth effect is generated by the interaction of the INR with the atmosphere and is designated electromagnetic effects. These distributions illustrated in Figure A-1 are for a generic tactical event at or near the surface. Height-Of-Burst, weapon type, and weapon configuration do have an effect on the shown energy distributions.

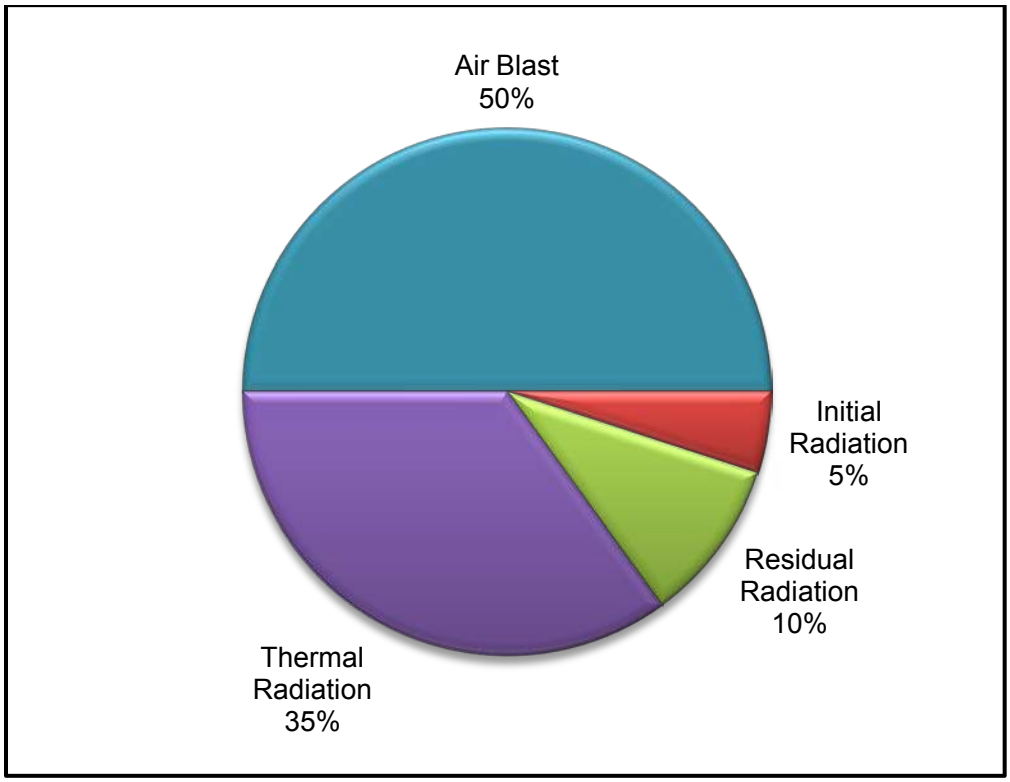


Figure A-1. Nuclear weapon energy distribution.

A.1. BLAST EFFECTS.

The blast effects of a nuclear weapon is extremely similar to those caused by a conventional weapon explosion, but possessing a much larger magnitude and longer duration. The blast wave travels radically in all directions from GZ, initially at a speed greater than sound, but decreases as a function of radial distance to subsonic and finally to zero. Blast effects are considered the most damaging to exposed military equipment except armor. Blast effects are typically the greatest cause of collateral damage, but is not typically the greatest personnel casualty producer.

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APPENDIX A. GENERAL NUCLEAR WEAPON EFFECTS.

A.2. THERMAL EFFECTS.

The thermal pulse of a nuclear weapon is characterized by an intense blinding flash and an immense thermal pulse. The great intensity of this flash can cause blindness to military personnel, either temporary or permanent. The thermal radiation is emitted in two distinct separate pulses. The first thermal pulse occurs as a result of x ray interaction with weapon materials and is insignificant since it contains very little energy. The second pulse contains tremendous energy and is considered militarily significant and generally the largest casualty producer of exposed personnel. The thermal pulses have two damage producing mechanisms which are: direct damage produced by generated heat and, secondary damage caused by fires and explosions from the ignition of surrounding materials. The blast wave may suppress some fires caused by these thermal pulses. The blast wave combined with the thermal pulse may create synergistic effects.

A.3. INITIAL NUCLEAR RADIATION EFFECTS.

The INR pulse of a nuclear weapon consists of gamma photons and neutrons, which are emitted within a few tens of nanoseconds after the event. These highly penetrative gamma photons and neutrons are extremely damaging to military personnel and electronics. The magnitude of this radiation at a given distance from GZ is dependent upon weapon yield, type and height of burst, terrain, and atmospheric conditions. For middle to high yield weapons, the damaging effects generated by the blast wave and thermal radiation greatly surpasses INR effects for unprotected equipment and personnel.

A.4. RESIDUAL RADIATION EFFECTS.

The residual radiation effects of a nuclear weapon consists of radioactive weapon debris, radioactive fallout, rainout, and Neutron Induced Activity (NIA), which sustains for longer than one minute after weapon detonation. The two militarily significant radiations which compose residual radiation are beta particles and gamma photons. Residual radiation has essentially no damaging effects on military systems, but presents major difficulties on military personnel in the area surrounding GZ, downwind of GZ, and troop movement through contaminated areas.

A.5. ELECTOMAGNETIC EFFECTS.

The electromagnetic environment of a nuclear weapon consists of the ionization of the surrounding atmosphere and EMP. The gamma photons, neutrons, beta particles, x rays, and positive ions emitted from the nuclear detonation causes electrons to be ejected from their perspective atoms, thus ionizing the atmosphere in the burst vicinity. This increase in electron density attenuates or refracts all electromagnetic signals from a few seconds to several hours depending on weapon yield and HOB. A nuclear detonation distributes approximately one millionth of its energy in the form of an intense EMP with a frequency content of a few hertz (Hz) to several hundred megahertz (MHz). The two EMP situations which are based upon weapon HOB are Endo-Atmospheric (SREMP) and Exo-Atmospheric (HEMP). SREMP occurs with an atmospheric event at an altitude of less than

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APPENDIX A. GENERAL NUCLEAR WEAPON EFFECTS.

40 km above sea level, possessing an extremely large electric and magnetic field over the burst vicinity. HEMP occurs from an event occurring at an altitude greater than 40 km above sea level and possesses a large electric and magnetic field over a diverse area. Of the two EMP situations, HEMP is considered the most militarily significant. In fact, HEMP is a line-of-sight phenomenon and can cause damage over hundreds of thousands, even millions, of square miles. HEMP has the greatest range of damage of all nuclear effects.

A.6. TIME HISTORY OF EFFECTS.

All effects produced by a nuclear weapon are dependent upon weapon yield, type of weapon, HOB, atmospheric conditions, and distance from GZ. See Figure A-2 for the sequence and time history of nuclear weapon effects from an example 27 kT weapon detonation, at a HOB of 180 m, at a distance of 1 km.

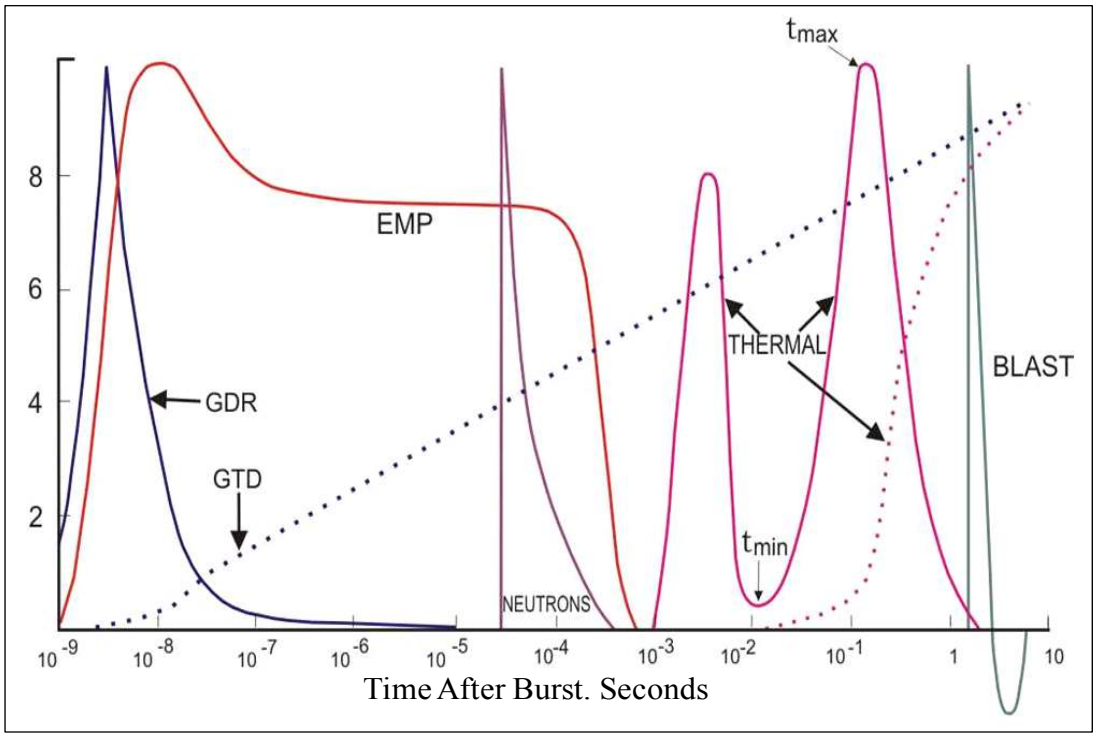


Figure A-2. Example time history from 27 kT nuclear weapon.

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APPENDIX B. NUCLEAR AIRBLAST ENVIRONMENT AND EFFECTS.

B.1. GENERAL.

a. Approximately fifty percent of the total energy generated by an Endo-Atmospheric detonation (HOB less than 40 km above sea level) of a normal type of nuclear weapon is in the formation of a tremendous blast or shock wave. Since the air density is relatively high up to 40 km above sea level, the nuclear weapon detonation generates extremely high velocity atoms which transfer tremendous energy to the closest layer of air, compresses a layer of surrounding air, and causes it to propagate outward from the center of the explosion. In the process of compressing a layer of air, a rarefaction occurs in the vacated space creating, in effect, a negative pulse, which propagates outward. Energy is then transferred to each successive air layer and upon emerging from the fireball, this energy release has assumed the characteristics of a blast or shock wave. Initially, exceeding the speed of sound, this transfer of energy and momentum forces the air layers to form the shock front of the blast wave.

b. The blast wave generates five significant damage parameters: static overpressure, dynamic pressure, impulse, duration, and negative overpressure. The first damage parameter is overpressure, which is defined as the transient pressure above the ambient pressure that acts on objects from all sides and tends to crush inwardly due to pressure differences. The variation of the overpressure with time depends on the energy yield of the explosion, the distance from the point of burst, and the medium in which the weapon is detonated. The second damage parameter is dynamic pressure, which is defined as the air pressure that results from the mass air flow or wind behind the shock front of the blast wave that tends to overturn, translate, tumble, or tear apart materiel. It is equal to the product of one half of the air density through which the blast wave passes and the square of the wind velocity behind the shock front as it impinges on the object or structure. The third significant damage parameter is impulse, which is defined as the product of the overpressure or dynamic pressure from the blast wave and the time during which it acts at a given point. More specifically, it is the integral, with respect to time of overpressure or dynamic pressure, between the time of arrival of the blast wave and where that parameter returns to zero at a given point. The fourth significant parameter is duration, which is the time in which a phase of the overpressure or dynamic pressure acts upon an object. The final damage parameter is negative pressure, which is defined as the transient pressure below the ambient pressure that oftentimes enhances damage by pulling back on objects that may be unstable and experiencing forces in the opposite direction. Blast damage will increase with an increase in any of the above parameters. A typical blast wave is illustrated in Figure B-1.

c. Peak static overpressure decreases with increasing range from GZ, but the duration of the positive and negative phases of the overpressure increases with range. Since target damage caused by overpressure is a function of pressure multiplied by the duration, the magnitude of potential damage does not decrease at the same rate as overpressure attenuation. Many military materials that have volume with thin walls, such as shelters, are damaged by peak static overpressure and are referred to as "diffraction sensitive" targets.

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APPENDIX B. NUCLEAR AIRBLAST ENVIRONMENT AND EFFECTS.

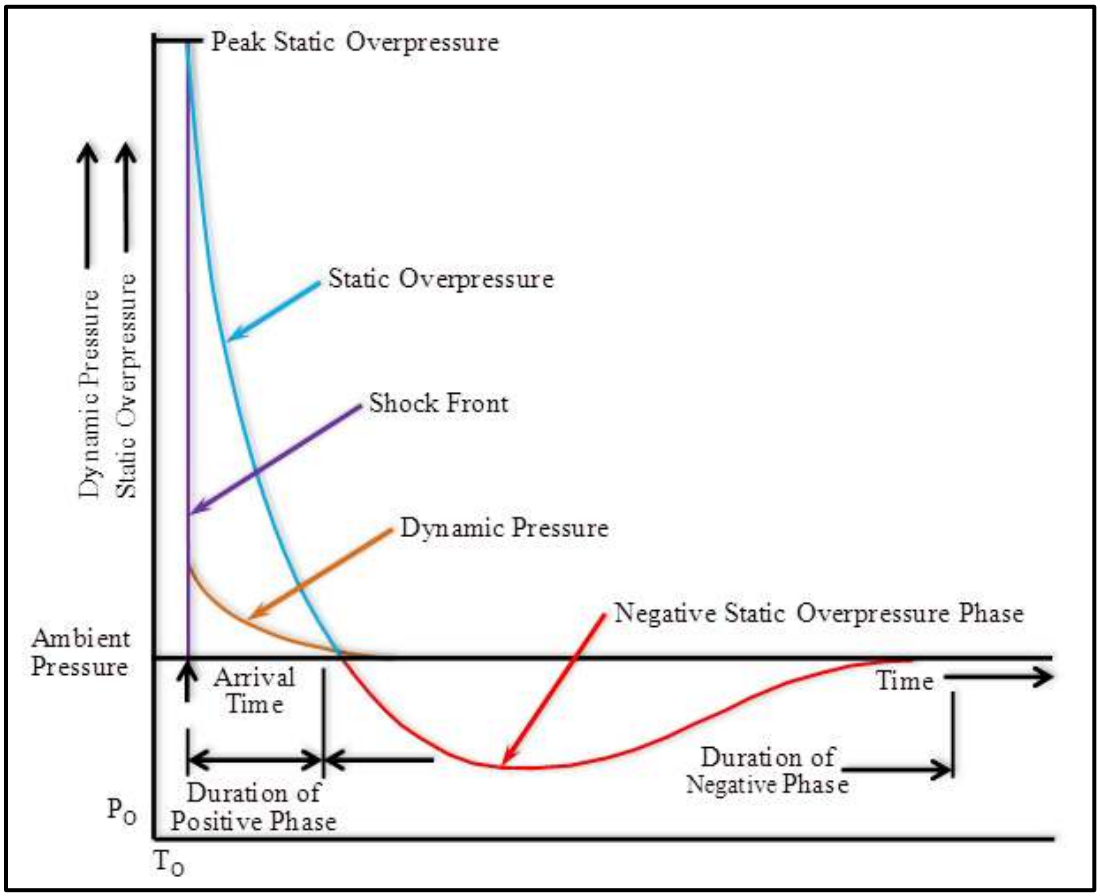


Figure B-1. Typical blast wave with respect to time at a fixed location.

d. The dynamic pressure is applied on a target for a longer duration than the overpressure because the moving air has mass and therefore momentum, which causes it to take longer to come to rest. Most military materiel targets are damaged primarily by dynamic pressure and are referred to as "drag sensitive" targets. The relationship between the peak dynamic pressure and the peak static overpressure is expressed by the Rankine - Huguenot equation, which reduces to the following equation (Equation B-1):

$$q = \frac{5}{2} \times \frac{P^2}{7Po + P} \qquad \text{(Equation B-1)}$$

where: q = dynamic pressure
P = overpressure
Po = ambient pressure

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APPENDIX B. NUCLEAR AIRBLAST ENVIRONMENT AND EFFECTS.

e. Table B-1 provides examples of the relationships between peak static overpressure, dynamic pressure, and wind velocity for an Ideal Shock Front.

TABLE B-1. OVERPRESSURE AND DYNAMIC PRESSURE RELATIONSHIPS

PEAK OVERPRESSURE (PSI / (kPa))	PEAK DYNAMIC PRESSURE (PSI / (kPa))	MAXIMUM WIND VELOCITY (MPH)
100 / (689.47)	123 / (848.05)	1,415
72 / (496.42)	74 / (510.21)	1,168
50 / (344.74)	41 / (282.69)	934
30 / (206.84)	17 / (117.21)	669
20 / (137.90)	8.1 / (55.85)	502
10 / (68.95)	2.2 / (15.17)	294
5 / (34.47)	0.6 / (4.14)	163
2 / (13.79)	0.1 / (0.69)	70

f. The HOB of weapon detonation is highly significant in maximizing blast damage. In order to inflict the greatest blast damage on a specific target, the optimum HOB is calculated, which will obtain the most effective Mach Stem. Near GZ, the incident and reflected ground shock fronts are separate. But, as the initial blast wave compresses and heats the surrounding air, it generates a swifter medium to which the reflected wave can propagate. Since the reflected wave travels faster because of this heated medium, it joins the initial blast wave and forms an almost vertical and reinforced shock front called the Mach Stem. The location where the incident and reflected waves converge is called the Triple Point. The formation of the Mach Stem produces an immediate and significant rise in exerted pressures which decreases with increasing range from GZ.

g. In addition, the shock front is also reflected by the face of the target. The pressure of this reflected target wave is added to the initial overpressure exerted on the target resulting in a greater overpressure than the original shock front. See Figure B-2 for Mach Stem formation and Figure B-3 for ground pressure against increasing range from GZ.

h. A nuclear detonation, particularly a burst that occurs under or near the surface, will transmit a shock wave through the surrounding Earth. Ground shock is important in damaging underground targets and shelters, but its effect on materiel targets located on the surface is insignificant in comparison to the effects caused by airblast. Thus, ground shock is not normally considered in the evaluation of survivability/ vulnerability of tactical military materiel.

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APPENDIX B. NUCLEAR AIRBLAST ENVIRONMENT AND EFFECTS.

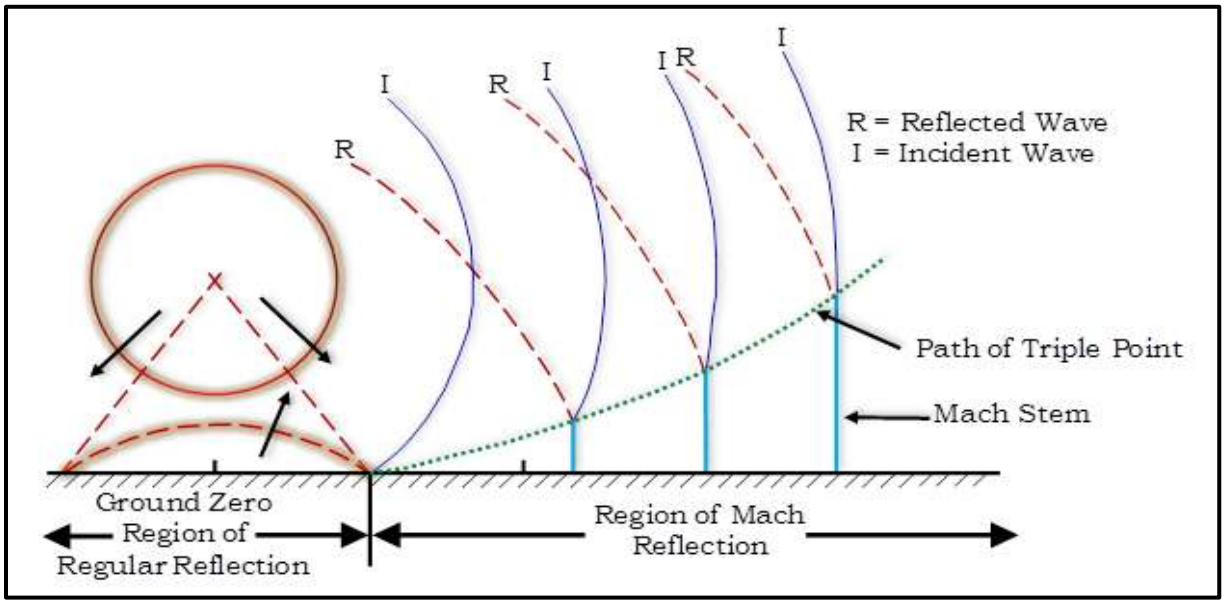


Figure B-2. Formation of the Mach Stem.

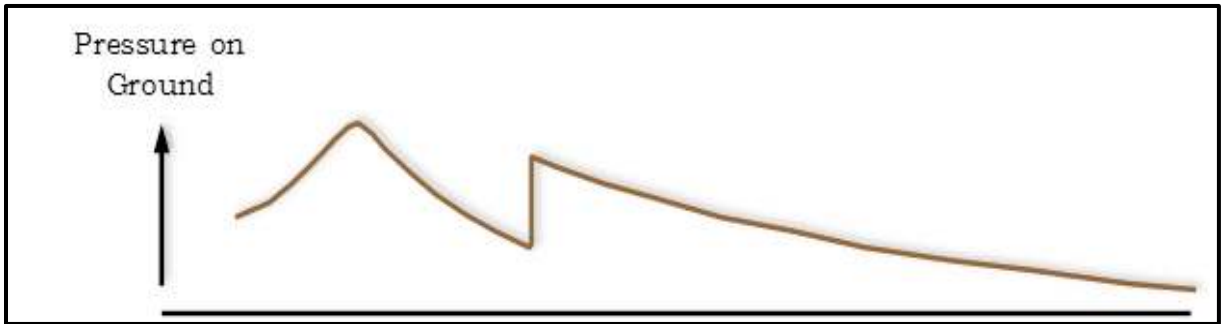


Figure B-3. Ground pressure versus increasing distance from GZ.

i. In conclusion, military targets which are exposed to the effects of a nuclear airblast will have the following two distinct sequential effects. First, the overpressure will attempt to crush the target. If the target contains openings, this pressure difference will quickly diminish until equilibrium is established and the crushing effect will be minimized; if the target is open, it will experience crushing forces for a short period of time. The combination of the blast wind and diffraction loading acting upon the object will exert a force on the object causing it to translate from GZ. Diffraction loading is the force of the static overpressure acting on the front face before the shock front envelops the target. Second, the dynamic pressure will apply drag loading on the object. Drag loading is the force on an object due to the transient winds accompanying the passage of the blast wave. Drag loading causes translation and overturning. Damage caused by drag loading depends upon the duration and strength of the positive phase of the blast wave.

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APPENDIX B. NUCLEAR AIRBLAST ENVIRONMENT AND EFFECTS.

j. Nuclear airblast testing requires the use of both experimental and analytical techniques to determine the response of systems and components to the blast wave. Adequate testing of a system requires the accurate simulation and analysis of the nuclear airblast environment in terms of overpressure, dynamic pressure, impulse, duration, and negative overpressure of the entire system under study. Detonation altitude, weapon yield and type, and system configuration, deployment, and mission essential functions shall be considered to adequately determine the damaging effects on military systems caused by the blast wave.

B.2. GENERAL FORCE APPLIED TO A SURFACE.

The numerical prediction method is; to project the vehicle surface onto a vertical plane, which is perpendicular to the incident blast wave, and then calculate the forces exerted on that surface. The defining equations used to predict the absolute worst-case forces on a surface are:

$$Force = P \times A$$

$$P = Pc + Pd + Pr$$

$$Pd \cong \frac{5}{2} \times \frac{Pc^2}{7Po + Pc}$$

$$Pr \cong 2Pc \frac{7Po + 4Pc}{7Po + Pc}$$

where:

P = pressure A=Area

Pc = criterion overpressure

Pd = peak dynamic pressure

Pr = peak reflected pressure

Po = atmospheric pressure [101.3 kPa (14.7 psi)]

kPa = kilo - Pascals

psi = pounds per square inch

B.2.1 Estimating the Structural Capability.

a. VEHICLE HULL EXAMPLE: The dimensions model was developed from Computer Aided Design (CAD) drawings and uses the side of the vehicle with the largest area as the air blast receiver. This area is calculated as the projection of the surface onto a vertical plane perpendicular to the blast force. If the surface area of the blast incident side is 14.5 m^2 then the calculated pressure is 313.7 kPa (.314 Mpa, 45.5 psi). This effort is performed only to determine if gross crushing or breaking could occur. The generic parameters of the structure materials are investigated using material databases and manufacturer information. Table B-2 shows generic material properties.

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APPENDIX B. NUCLEAR AIRBLAST ENVIRONMENT AND EFFECTS.

TABLE B-2. GENERIC MATERIAL PROPERTIES

ESTIMATED MATERIAL PROPERTIES				
Material	Yield Strength (Mpa)	Yield Strength Ultimate (Mpa)	Shear (Mpa)	Modulus of Elasticity (Mpa)
Ceramic	1400	303000	-----	448000
Steel	379	455	80000	200000
Kevlar	-----	3000	-----	112000

b. The predicted absolute maximum pressure on the incident surface of the system was 313.7 kPa. Data on the materials indicates that this pressure is below the elastic strength of the materials by 3 orders of magnitude. Therefore, there is expected to be no appreciable deformation or ruptures of the structure.

B.2.2 Gross System Acceleration.

a. Data from like systems and experience should be used to estimate system responses if possible. As a minimal prediction, the gross or overall acceleration of a system due to the air blast environment is primarily produced by the dynamic pressure (i.e., wind pressure). The gross low frequency acceleration of the system is then calculated using the equations below:

$$F = M \times A$$

$$\mathrm{A} = \frac{\mathrm{F}}{\mathrm{M}}$$

$$A(g's) = A / Ag$$

$$\mathrm{F} = \text{force}\,[\text{newtons}]$$

$$\mathrm{M} = \text{mass}\,[\text{kg}]$$

$$\mathrm{Ag} = \text{ acceleration due to gravity}\,[9.8\,\frac{\mathrm{m}}{\mathrm{s}^2}]$$

b. Utilizing the area of the vehicle as 14.7 m^2, the predicted force as 3.08e5 Newtons, and the mass of the vehicle expressed as 17,237 kg (38,000 pounds (lbs)); the resulting predicted low frequency base acceleration is 17.87 m/s^2 or 1.8 g. Data measured during testing include high frequency and impact accelerations; the base vehicle acceleration is obtained by filtering the data. Recorded data show, when using a 10Hz filter, the low frequency acceleration is between 1.8 g and 3 gs, which is in agreement with the calculation. However, the full frequency waveform had a maximum of 51 gs, which is a factor of 29 times greater than the base low frequency calculation. This implies that when making the gage predictions, a factor of

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50 (minimum value) should be used and in general a multiplication factor of 100 is preferred. When making prediction on internal rack mounted equipment, this factor should be increased by an additional multiplication factor of 2. Given this estimate, the major frame accelerations would be predicted to be 180 gs and the instrumentation would be set so that the gage output signal after amplification would be 50% of the data recorder maximum voltage, thereby providing for a quality recording of the signal from approximately 25 to 200% of the prediction. The same procedure would be used for the rack/vibration mounted equipment, except the original figure would be 200 instead of 100 or a prediction of 360 gs.

B.3 GENERAL GUIDELINES FOR FINITE ELEMENT ANALYSIS (FEA) MODELING FOR NUCLEAR BLAST.

a. Develop a three dimensional CAD computer model of the test item as displayed in Figure B-4, that can be used to perform FEA. The FEA can be performed utilizing any software that can perform static and dynamic overpressure calculations on the FEA model. The software used here was ALGOR.

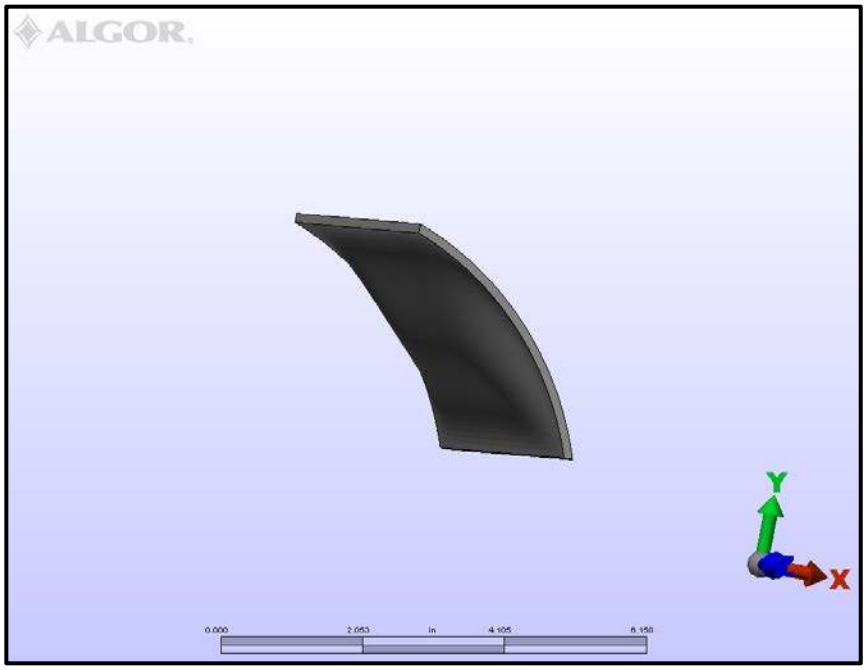


Figure B-4. Test item FEA analysis mode.

b. Equivalent Static overpressure and Dynamic pressure loads need to be calculated and applied to the outside surfaces on the blast incident source side of the test item three dimensional FEA model. Stress, strain, and deflection values can then be computed on the FEA model as displayed in Figure B-5 utilizing the software.

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APPENDIX B. NUCLEAR AIRBLAST ENVIRONMENT AND EFFECTS.

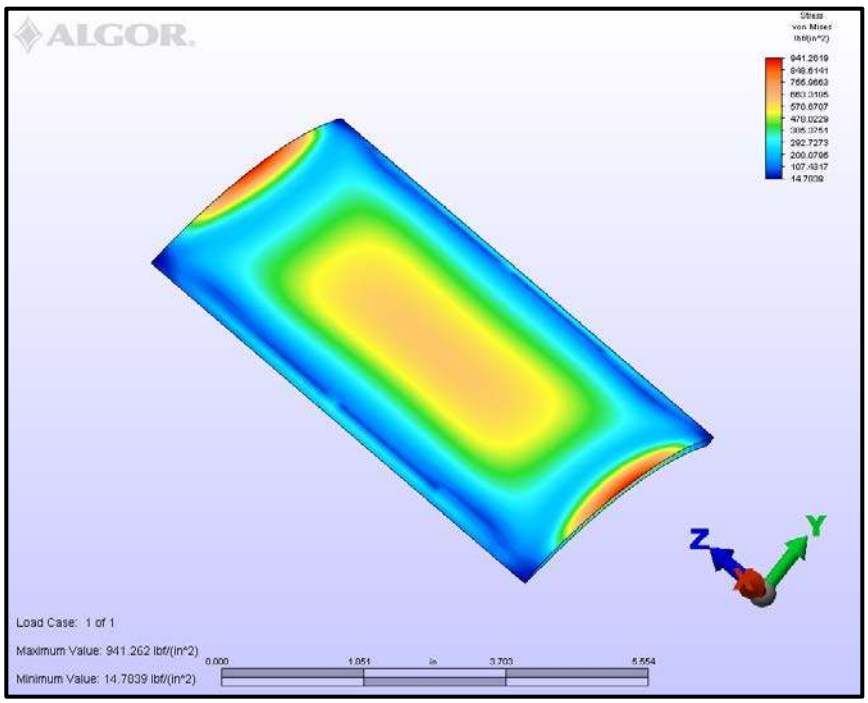


Figure B-5. Test item FEA stress analysis results.

c. The maximum stresses on the test item are determined from the FEA results and then compared to the Ultimate tensile strengths and the Yield strengths of the test item material properties. Design margins can then be calculated based on the FEA analysis results and material properties. Design margins of 1.5 or greater are desired so that the probability of the item failing under the load will be low.

d. Figure B-5 presents the stress for the test item calculated with the ALGOR FEA software. The test item will reach a maximum stress of 941 psi on the outer center edges of the window cover. The maximum stress values are represented by the red colored area in the FEA model. The majority of the test item experiences stress levels from approximately 14 psi to 570 psi, which is well below the Ultimate Tensile Strength of 5,510 psi for the window cover material. The safety margins of the test item compared to the Ultimate Tensile Strength of the test items material properties are presented in the Table B-3.

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APPENDIX B. NUCLEAR AIRBLAST ENVIRONMENT AND EFFECTS.

TABLE B-3. SAFETY MARGINS EXAMPLE COVER STRESSES

STRESS (psi)	ULTIMATE TENSILE STRESS (psi)	SAFETY MARGIN
14	5510	393.57
292	5510	18.86
663	5510	8.31
941	5510	5.85

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APPENDIX C. NUCLEAR THERMAL RADIATION ENVIRONMENT AND EFFECTS.

C.1. GENERAL.

a. Approximately thirty-five percent of the total energy generated by an endo-atmospheric detonation of a generic nuclear weapon is in the form of thermal radiation. After the detonation of a thermonuclear weapon, the core of the explosion attains an exceedingly high temperature in the tens of millions degrees. As a consequence of these exceedingly high temperatures, the core radiates electromagnetic energy which peaks to the soft x ray spectrum. The absorption of these x rays by the surrounding air produces tremendous heat which creates the fireball. This generated fireball radiates electromagnetic energy from 250 nanometers in the ultraviolet to 4000 nanometers in the infrared, and produces the damaging thermal effects upon military targets. These damaging effects have two governing mechanisms, the total heat delivered and the rate of delivery. As weapon yield is increased, the larger the fireball becomes, which significantly increases the total amount of energy that shall be dissipated and the duration of energy dissipation. Since this duration is significantly greater for higher yields, the delivery rate of thermal radiation is greatly reduced; therefore, reducing the damaging effects to a military target receiving similar thermal doses from different weapon yields.

b. In other words, for a specific thermal dose (total energy), a smaller yield weapon can inflict greater damage upon a specified target than a higher yield weapon because the smaller yield delivers the thermal radiation quicker. The three distinct factors which affect the damaging effects of thermal radiation are blast wave screening, inverse square law, and absorption and scattering due to the atmosphere.

c. The first factor is blast wave screening. As discussed earlier, the thermal radiation is emitted in two distinct separate pulses. The first pulse is generated by the interaction of the fireball radiating maximum energy and the departure of the shock front. As the fireball's radius increases, the shock front departs from the fireball heating the surrounding air to incandescence, which absorbs the thermal radiation from the fireball. This incandescence screens the fireball and causes the shock front to emit thermal radiation instead of the fireball. This process is called blast wave screening. The result is a thermal pulse of little energy and significance in regards to damage effects. This first pulse causes flash blindness. As the shock front expands, the cooling of the air allows the fireball to become visible and generates the second thermal pulse. Since this second pulse contains tremendous energy, it is militarily significant. See Figure C-1 for an example of the thermal radiation output of a nuclear weapon.

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APPENDIX C. NUCLEAR THERMAL RADIATION ENVIRONMENT AND EFFECTS.

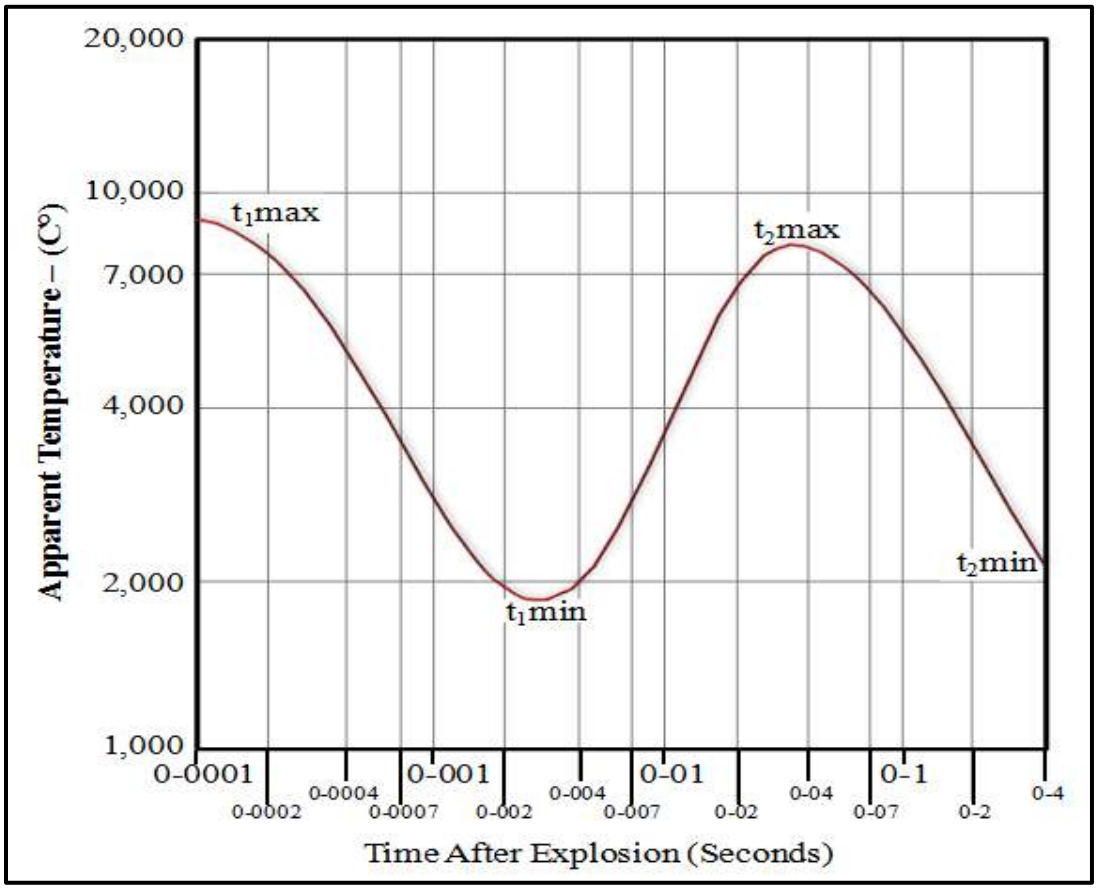


Figure C-1. Time after explosion.

d. The second factor affecting thermal radiation is the inverse square law effect. The inverse square law pertains to all forms of electromagnetic radiation and states that the thermal dose varies as the total energy emitted and inversely as the square of the range from the point of burst. See Figure C-2 for inverse square law equation and Figure C-3 for slant ranges for specified thermal exposures.

$$Q\ (\text{cals/cm}^2) = \frac{E\tau}{4\pi R^2}$$

Where: E = Total Energy Emitted
R = Range in centimeters
τ = Transmissivity of the Surrounding Atmosphere from the Burst to the Target

Figure C-2. Inverse Square Law.

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APPENDIX C. NUCLEAR THERMAL RADIATION ENVIRONMENT AND EFFECTS.

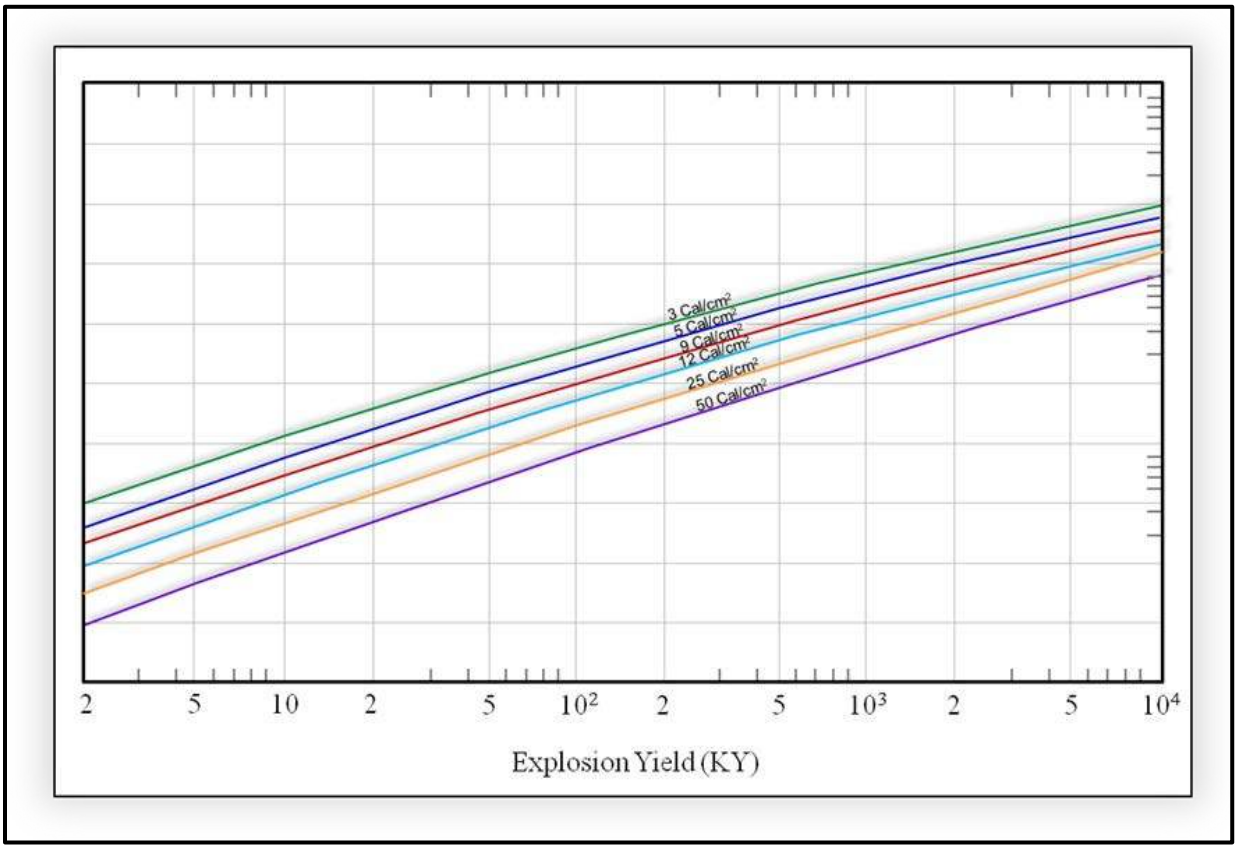


Figure C-3. Slant ranges for specified thermal doses.

e. The third factor affecting thermal radiation is absorption and scattering due to the atmosphere. As the thermal radiation passes through air, the air molecules absorb a portion of the energy in the x ray and ultra-violet frequency spectrum. Scattering of the thermal radiation occurs when air contains water droplets and/or dust particles. These droplets and particles reflect the thermal radiation from a single line of sight direction to a multitude of directions. This scattering effect only reduces the thermal radiation applied upon a target slightly, because the scattered energy from other directions reinforces this line of sight radiation. Visibility is an extremely important parameter in determining the effectiveness of thermal radiation upon its target. In poor visibility conditions such as fog or smoke, the effects of thermal radiation are greatly reduced if the weapon HOB is above these conditions. In this circumstance, the thermal radiation is not only affected by absorption, but also by reflection. In the case of ground bursts, the thermal radiation effects are significantly diminished by the absorption and scattering caused by greater amount of generated debris and the shielding effects of terrain and surface irregularities.

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APPENDIX C. NUCLEAR THERMAL RADIATION ENVIRONMENT AND EFFECTS.

f. The thermal effects of a nuclear weapon are an intense blinding flash and an immense thermal pulse. The great intensity of this flash can cause blindness to military personnel, either temporary or permanent. The severity of blindness is dependent on individual reflexes, flash intensity, pulse duration, and the protective posture of personnel. The thermal radiation is emitted in two distinct separate pulses. The first thermal pulse contains little energy, therefore it is insignificant. The second pulse contains tremendous energy, it is militarily significant and generally the largest casualty producer. The thermal pulse has two damage producing mechanisms, which are direct damage produced by generated heat, and secondary damage produced by fires and explosions caused by ignition of surrounding materials. The blast wave may suppress some fires caused by the thermal pulses. Also, smoking of the exposed object will attenuate the undelivered energy of the thermal pulse. See Figure C-4 for fireball power and thermal energy emitted versus normalized time.

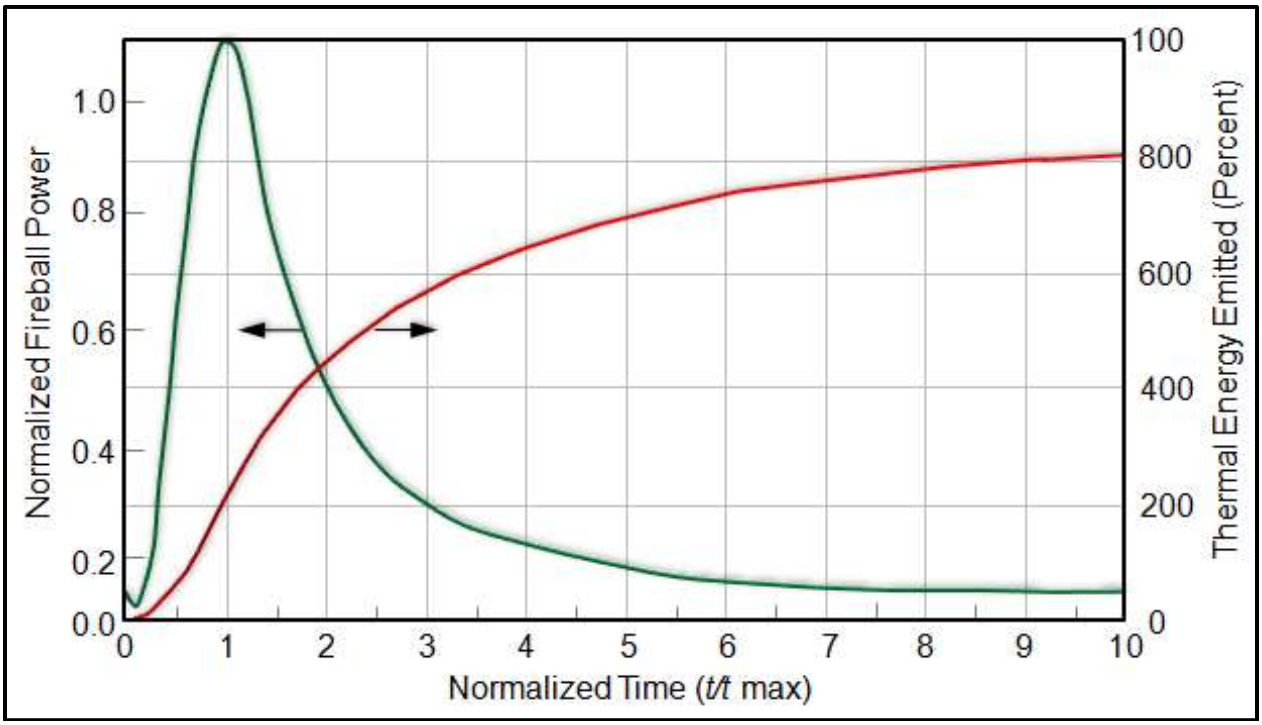


Figure C-4. Normalized fireball and thermal energy emitted versus normalized time.

g. To determine the resistance of an item of materiel to thermal radiation, the test system shall first be examined to determine potential problem areas. These are usually associated with combustible materials such as plastics, exposed ammunition, casing and containers, fabrics, rubber, and wood. Non-combustible materials include optics, sensors, and low melting point materials. Charring of paint is of no consequence, nor is the charring or discoloration of any

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material, provided that such damage does not interfere with the operation of the equipment. The fact that gasoline may spill out when motorized equipment is overturned should not be interpreted as suggesting that it will be ignited by thermal radiation, because the thermal pulse will have passed by the time the spilling occurs. Adequate resistance to thermal radiation can sometimes be confirmed by a visual examination that proves the item free of heat - sensitive areas. At other times, it may be necessary to expose components of the item to facilities which simulate the thermal pulse.

h. Typical problems that may be associated with thermal radiation are: destruction of insulation on wires, distortion of plastic moving parts, burning of rubber on tires and tracks, blackening and/or cracking of optical devices, weakening of materials, ignition of exposed propellants and ammunition, burning of tarpaulins and other fabrics, and ignition of combustibles which could lead to fire damage. Thermal damage is of little consequence when a system will suffer more severe damage from blast effects. See Table C-1 for examples of thermal radiation exposure required to ignite certain materials. Smaller yield weapons require less cal/cm^2 to ignite the same materials because of the faster rate or flux (cal/cm^2-sec) that the thermal energy is deposited on the material.

TABLE C-1. APPROXIMATE THERMAL RADIATION EXPOSURE REQUIRED FOR MATERIAL IGNITION

MATERIAL	RADIANT EXPOSURE (Cal/cm^2) FROM 1kT WEAPON	RADIANT EXPOSURE (Cal/cm^2) FROM 20 kT WEAPON	RADIANT EXPOSURE (Cal/cm2) FROM 1MT WEAPON
Tan Cotton Shirting	5	7	11
Newspaper	2	3	6
Battle Dress Uniform	15	20	30
NBC Suit	15	20	30
Truck Canvas Canopy	15	20	30
Dry Grass and Undergrowth	2 - 3	3 - 4	5 - 7
Cardboard	6	8	13
Plastics	4 - 6	8 - 10	9 - 13
Heavy Burlap	6	8	12

C.2. GENERAL GUIDELINES AND CALCULATIONS FOR NUCLEAR THERMAL.

a. The thermal energy response of a material can be modeled using electrical lumped nodal analysis techniques. Table C-2 provides the electrical to thermal analogies. Table C-3 provides the material properties.

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APPENDIX C. NUCLEAR THERMAL RADIATION ENVIRONMENT AND EFFECTS.

TABLE C-2. ELECTRIC TO THERMAL ANALOGY

GENERAL	ELECTRICAL	THERMAL
Flow Variable (Through Variable)	Current, $I = dq/dt$[Amps]	Heat flow, q[Joules/sec]
Potential Variable (Across Variable)	*Voltage, V[Volts]*	Temperature difference, *T[°C]*
Integrating Element (Delay Component)	Inductance, L [Henrys] Faraday's Law: $I = \phi V/dt/L$	Not Applicable
Proportional Element (Dissipative Component)	Resistance, $R = \rho L/A$[Ohms] Ohm's Law: $I = V/R$	Heat transfer resistance, R $q = T/R$
Differentiating Element (Accumulative Component)	Capacitance, $C = \varepsilon A/d$ *[Farads]* $I = C\ dV/dt$	Thermal heat capacity, $m\ c_p$ $q = m\ c_p\ dT/dt$
Other Variables	*Charge,* $q = \int I\ dt$ *[Coulombs]*	Heat, $Q = \int q\ dt$ [Joules]

TABLE C-3. MATERIAL PROPERTIES

Material	Density kg/m3	Specific Heat Cal/kg°C	Thermal Conductivity Cal/sm°C	Thermal Resistance sm°C/Cal	Width m	Simulation Area (vehicle side) m^2	Electrical Capacitance Model Value Cal/°C	Electrical Resistance Model Value s°C/Cal
HULL								
AIR	1.2	239.8	6.12E-03	1.63E+02	1.2	9.5	2.73E+03	20.63309
Kevlar	1440	81.2	9.55E-03	1.05E+02	0.0127	9.5	1.41E+04	0.1399542
Steel	7850	103.659	14.447	6.92E-02	0.00635	9.5	4.91E+04	4.627E-05
Ceralloy (Ceramic Armor)	3210	143.33	26.26	3.81E-02	0.0254	9.5	1.11E+05	0.0001018

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APPENDIX C. NUCLEAR THERMAL RADIATION ENVIRONMENT AND EFFECTS.

TABLE C-3. CONTINUED

Material	Density kg/m3	Specific Heat Cal/kg°C	Thermal Conductivity Cal/sm°C	Thermal Resistance sm°C/Cal	Width m	Simulation Area (vehicle side) m^2	Electrical Capacitance Model Value Cal/°C	Electrical Resistance Model Value s°C/Cal
FUEL TANK								
JP8 (FULL TANK)	1298	632.95	0.027467	3.64E+01	0.0254	0.285	5.95E+03	3.244723
JP8 (HALF TANK)	1298	632.95	0.027467	3.64E+01	0.0254	0.142	2.96E+03	6.5122962
AIR (HALF TANK)	1.2	239.8	6.12E-03	1.63E+02	0.0254	0.142	1.04E+00	29.218105

b. Calculation of the circuit modeling component parameters for a material are based on the equations provided below:

(1) Material properties:

(a) Thermal resistance of material:

$$R = \frac{A}{kL}$$

$A = \text{Surface Area}$

$k = \text{Thermal Conductivity}$

$L = \text{Thickness}$

(b) Thermal capacitance of material:

$$C = D \times L \times A \times Cp$$

$\text{D} = \text{Density}$

$\text{L} = \text{Thickness}$

$\text{A} = \text{Surface Area}$

$\text{Cp} = \text{Specific Heat}$

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(2) Dissipative elements:

(a) Radiation Resistance:

$$Rr = \frac{T}{A\sigma\varepsilon(T_1^4 - T_o^4)}$$

T = temperature $^{\circ}C$

T_o = ambient temperature $^{\circ}K$

T_1 = temperature $^{\circ}K$

A = surface area

σ = Stefan – Boltzmann constant $(1.35e^{-8}\frac{cal}{m^2 s^o C})$

ε = emissivity (assumed to be 0.2)

(b) Convection Resistance:

$$Rc = \frac{T}{A\beta(T_1 - T_o)^{\frac{5}{4}}}$$

T = temperature $^{\circ}C$

T_o = ambient temperature $^{\circ}K$

T_1 = temperature $^{\circ}K$

A = surface area

β = coefficient of heat transfer

$0.5\frac{\text{cal}}{\text{m}^2 s^o C}$ utilize (generic flat plate value)

c. Once the equivalent electrical parameters (component parameters) for a material are calculated, the system is modeled as a lumped node transmission line as in Figure C-5

d. The material layers are then combined to create an electrical circuit representation of their cross section, which can be driven by a thermal source. Modeling a section of the test item (using this method) and then performing the static calculation of temperature rise was performed as a check. This check was based on material, energy fluence, and included no losses. The modeling method resulted in a temperature rise of 2,085 °C, and the static calculation 2,045 °C, which is a difference of approximately 2 percent.

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APPENDIX C. NUCLEAR THERMAL RADIATION ENVIRONMENT AND EFFECTS.

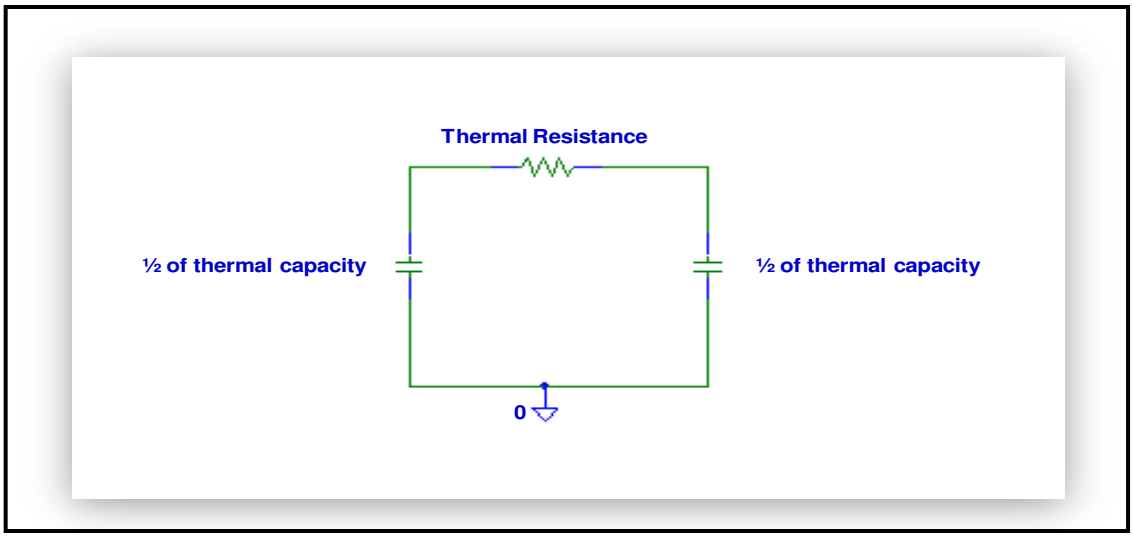


Figure C-5. Transmission line model of material.

e. Vehicle hub window and oil example.

(1) The thermal model of the HUB was developed by measuring the size of the check window and utilizing its area as the energy receiver. After the deposition of thermal energy the window size was then used to account for heat dissipation (no other materials or surfaces were included). The thermal parameter values of the materials are provided in Table C-3 , no values for gear oil could be found and therefore the parameters for JP8 were used (this should be conservative and the resulting electrical schematic is presented as Figure C-6. R5 is used to simulate the HUB radiation of energy back to the environment. The thermal source used for this stimulation is based on the piece-wise-linear model provided in the nuclear survivability criterion document. Please note that the distance from the window into the gear oil was varied to estimate temperature versus distance and is provided as Figure C-7.

(2) HUB: As can be seen from the simulation provided in Figure C-7 the thermal time constant for the HUB is large and the temperature lags the thermal input significantly. The gear oil located within 11 mm of the sight glass will reach the boiling temperature of JP8 (86 °C). This indicates that the gear oil, which is over heated, will be changed and become a contaminant. The predicted volume, which will be overheated, is 7.39 cc. The HUB contains 900 cc; therefore, the contaminant will be approximately 0.8 percent of the volume. The model does not include the following; the reduction in received energy due to ablation (the creation of smoke from the heated materials (i.e., plastic)), the material surrounding the sight glass, or the transport of energy due to boiling. Therefore, because of the small amount of gear oil being changed/contaminated, it is not expected that the HUB performance will be affected.

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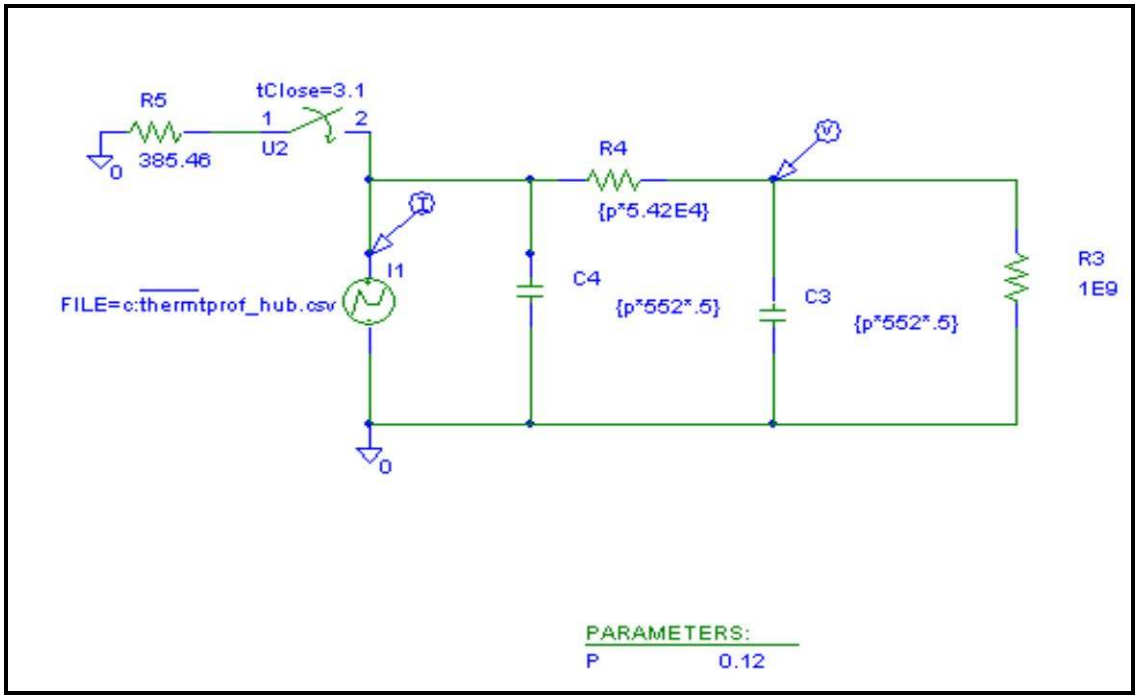


Figure C-6. HUB Thermal Model

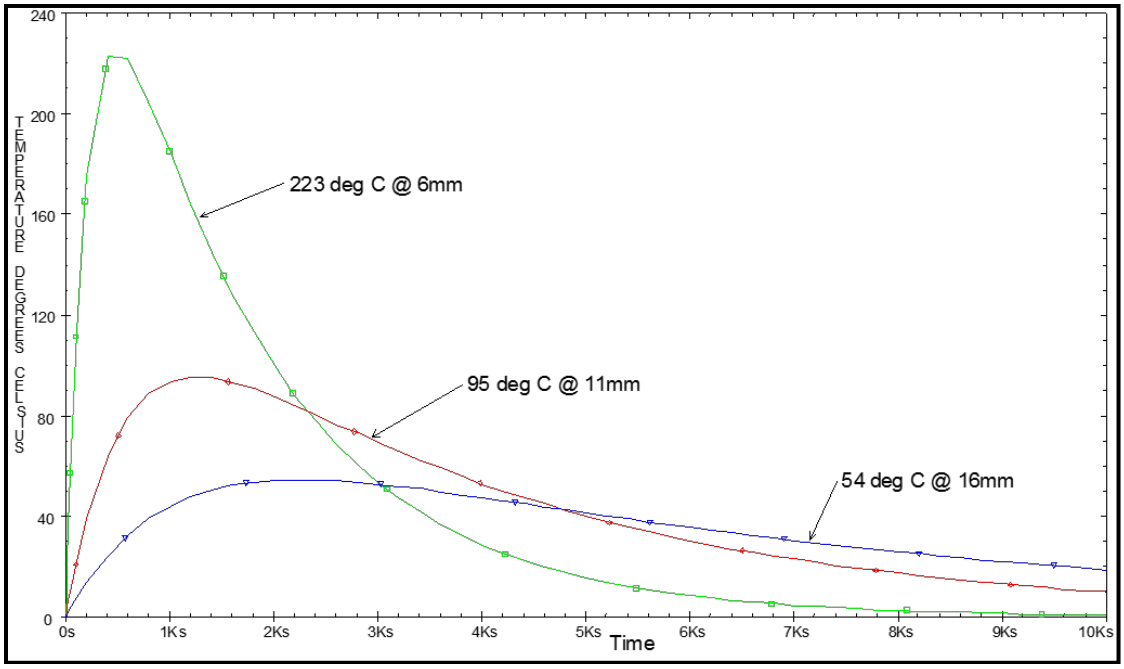


Figure C-7. Hub Oil Temperature.

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APPENDIX C. NUCLEAR THERMAL RADIATION ENVIRONMENT AND EFFECTS.

C.3. GENERAL GUIDELINES FOR NUCLEAR THERMAL FEA MODELING.

a. Develop a three dimensional CAD computer model of the test item as displayed in Figure C-8 that can be used to perform the Thermal FEA analysis. The Thermal FEA analysis can then be performed utilizing any software that can perform transient thermal calculations on the thermal FEA model.

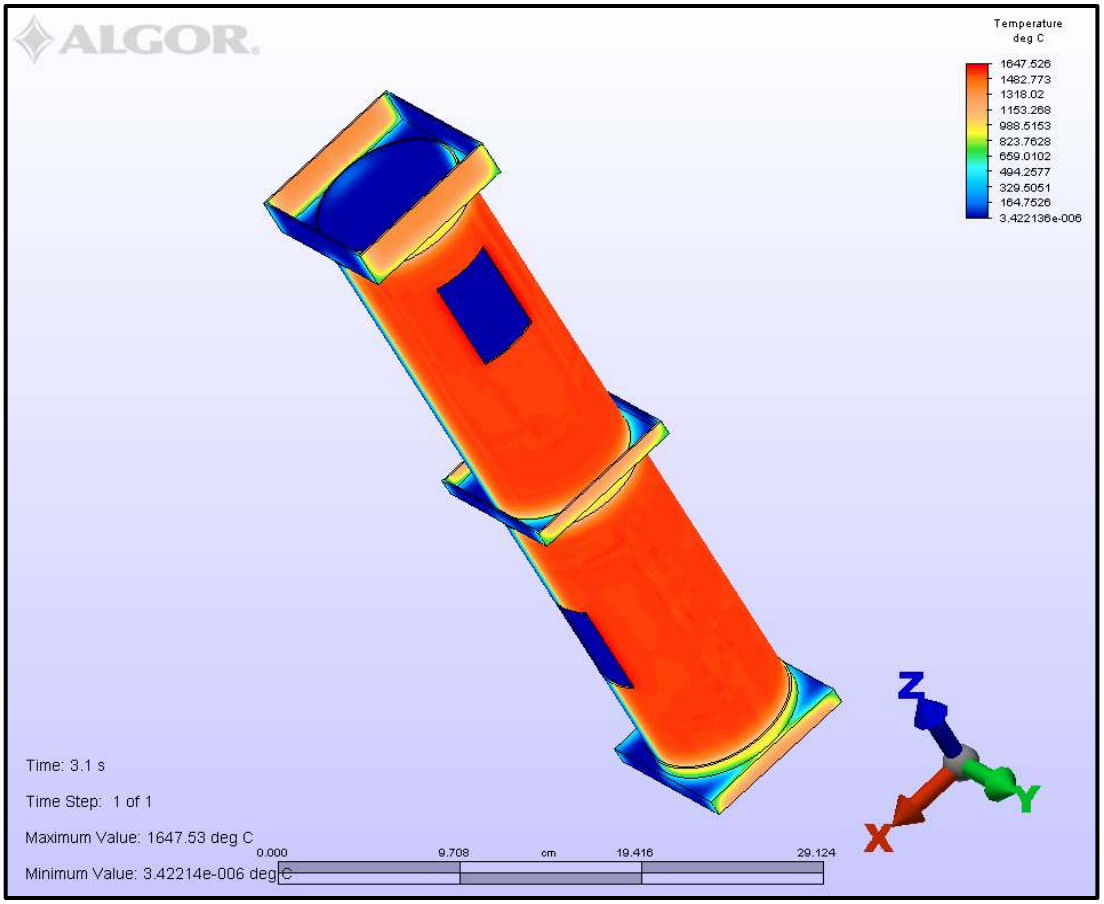


Figure C-8. Test item ALGOR thermal FEA temperature results.

b. The required thermal loads can then be applied to the outside surfaces, on the thermal incident source side of the test item three dimensional FEA model. Temperature and Heat Flux values can then be computed for the FEA model as displayed in Figure C-8, which are the temperature results utilizing ALGOR FEA software. Temperature values should also be calculated for the three different materials of the test item at various applied thermal energies and compared to thermal test results.

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c. The Thermal Finite Element Analysis performed using ALGOR shows the steel canister reaching temperatures of 1,647 °C. Comparing the value calculated as in Section 1.2 (which is 1,520 °C), to the results from FEA, the values are within 8%. At these temperatures, any polymer or material that has a lower melting temperature in the test item will ignite. The test item will be close to its melting temperature of 1,440 - 1,520 °C.

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APPENDIX D. NUCLEAR ELECTROMAGNETIC ENVIRONMENT AND EFFECTS.

D.1. GENERAL.

a. The electromagnetic environment produced by a nuclear weapon consists of the ionization of the atmosphere and generation of an EMP. The gamma rays, neutrons, beta particles, x rays, and positive ions emitted from the nuclear detonation causes electrons to be ejected from their perspective atoms, thus ionizing the atmosphere in the burst vicinity. This increase in electron density attenuates or refracts all electromagnetic signals from a few seconds to several hours depending on weapon yield and HOB. Radio communications depend on propagation of transmitted waves through the atmosphere. Depending on the specific frequency, this propagation occurs in one of two paths, ground or sky waves. Low frequencies utilize the ground wave path, while the high frequency band utilizes the sky wave path which is reflected back to Earth by the ionosphere. Very High Frequency (VHF) and Ultra High Frequency (UHF) penetrate the ionosphere; therefore, any disturbance in the ionosphere does not affect communications in these frequency bands. See Table D-1 for frequency band effects caused by atmosphere ionization.

TABLE D-1. FREQUENCY BANS EFFECTS CAUSED BY ATMOSPHERE IONIZATION

BAND	FREQUENCY RANGE	EFFECTS ON COMMUNICATIONS
Very Low Frequency (VLF)	3 – 30 kHz	Limited Effects
Low Frequency (LF)	30 – 300 kHz	Drastic Reduction of Sky Wave Path, but No Effects on Ground Wave Path
Middle Frequency (MF)	300 – 3000 kHz	Same as LF
High Frequency (HF)	3 – 30 MHz	Considerable Effects
VHF	30 – 300 MHz	Limited Effects, but Propagation Enhancement Possible
UHF	300 – 3000 MHz	Limited Effects
RADAR	3000 – 10000 MHz	Attenuated and Refracted

b. A nuclear detonation distributes approximately one millionth of its energy in the form of an intense EMP with a frequency content of a few Hz to several hundred MHz. The area affected by EMP and the characteristics of the pulse, is a function of burst altitude, weapon design, and yield. Typical EMP intensity is in the order of tens of thousands of volts/meter. This compares with the order of 200 volts/meter for nearby radars, 10 volts/meter for communication equipment, and 0.01 volts/meter for typical metropolitan area ambient. Two characteristics of EMP which result in a threat to electrical equipment are field amplitude and broad frequency spectrum. There are three basic mechanisms for EM coupling to a conducting structure: electrical induction, the basic mechanism for linear conductors; magnetic induction, the principal mechanism when the conducting structure forms a closed loop; and Earth transfer

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impedance for buried conductors. Devices which may be susceptible to functional damage due to electrical transients include active electronic devices, passive electronic components, semiconductor devices, squibs and pyrotechnic devices, meters, and power cables. Operational upset can be expected in digital processing systems, memory units, guidance systems, and power distribution systems. Damage mechanisms include dielectric breakdown, thermal effects, and interconnection failures. The two EMP situations which are based upon burst altitude are (Endo-Atmospheric) SREMP and (Exo-Atmospheric) HEMP.

c. The first EMP situation, SREMP, occurs within the atmosphere at an altitude of less than 40 km above sea level, and possesses an extremely large electric and magnetic field over the burst vicinity. Of particular concern are events at or within 1 km of the surface. Only within these limits are tactical surface systems close enough to the event to have the potential to be adversely affected by SREMP. SREMP is generated by collisions between photons from gamma radiation and molecules of the atmosphere. These highly energetic photons eject electrons from the surrounding air molecules, producing ionized air molecules. This immense separation of charge creates an intense E-Field of several 100,000 volts/meter and a large associated H-Field of 500 ampere-turns / meter. Ninety percent of its energy is contained in the 100 Hz to 10 kHz range. See Figure D-1 for an example of the SREMP waveform and Figure D-2 for relative energy versus frequency for an Endo-Atmospheric Burst.

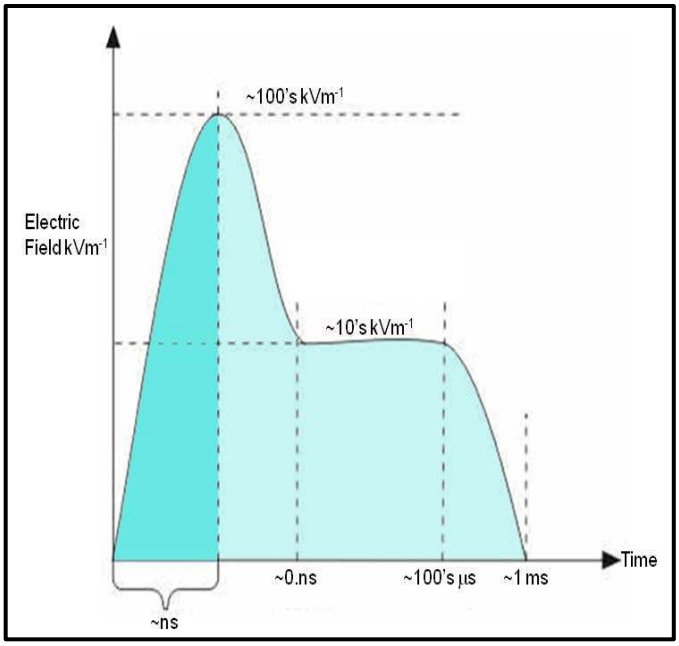


Figure D-1. Endo-Atmospheric EMP waveform.

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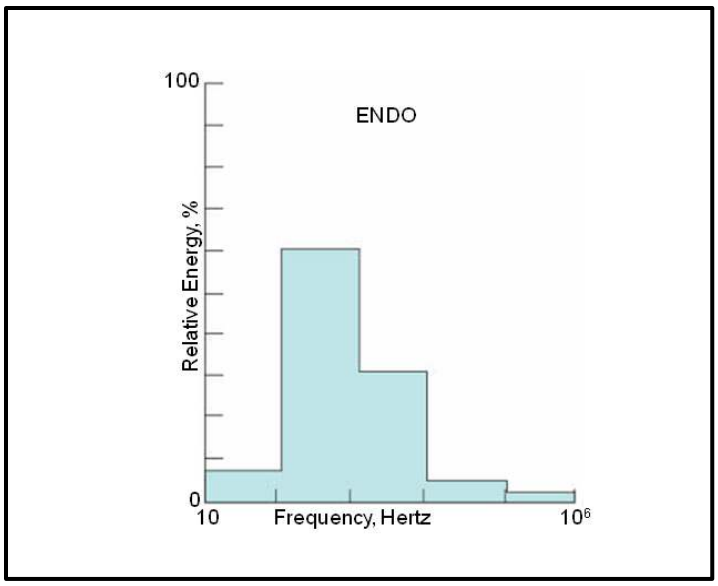


Figure D-2. Endo-Atmospheric relative energy versus frequency.

d. The second EMP situation, HEMP, occurs at an altitude greater than 40 km above sea level, and possesses a large electric and magnetic field over a diverse area. This tremendous area of effects is the reason HEMP is considered militarily significant and the most damaging of the two EMP situations. The HEMP is generated by gamma photons being absorbed by the atmospheric molecules at altitudes from 20 to 40 kilometers. This absorption causes electrons to be deflected by the earth's magnetic field into a spiral path about the field lines, causing them to radiate electromagnetic energy. See Figure D-3 for formation of HEMP and Figure D-4 for the detailed geometry of this phenomenon.

e. The waveform and frequency content of HEMP is drastically different from its SREMP counterpart. This electron radiated energy creates a large, diverse E-Field in the range of tens of kilovolts/meter and an associated H-Field in the range of 10 to 100 ampere-turns/meter. Ninety percent of its energy is contained in the 100 kHz to 10 MHz range. See Figure D-5 for an example of the HEMP waveform and Figure D-6 for relative energy versus frequency for an Exo-Atmospheric Burst. See Figure D-7 for an example of the diverse coverage in area and corresponding generate E-Field contours by an Exo-Atmospheric burst.

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APPENDIX D. NUCLEAR ELECTROMAGNETIC ENVIRONMENT AND EFFECTS.

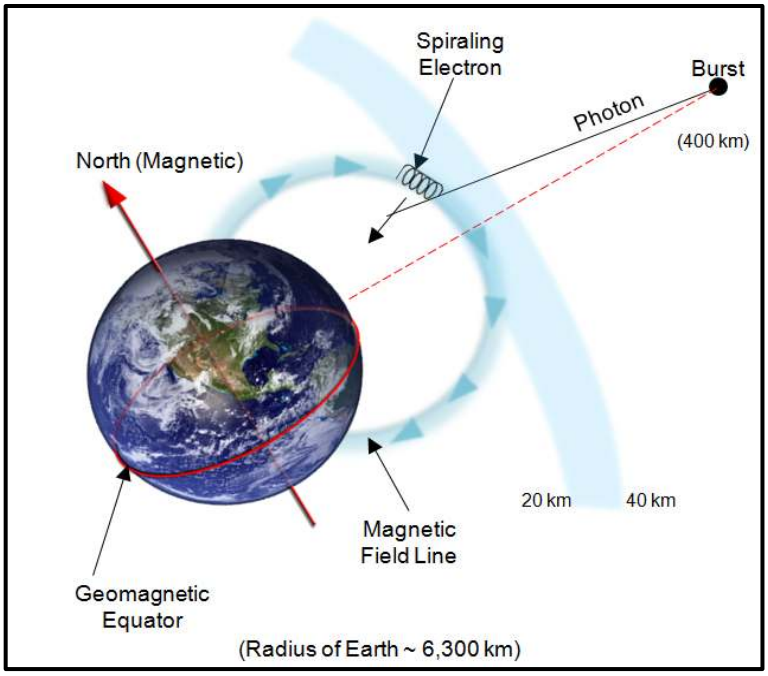


Figure D-3. Formation of Exo-Atmospheric EMP.

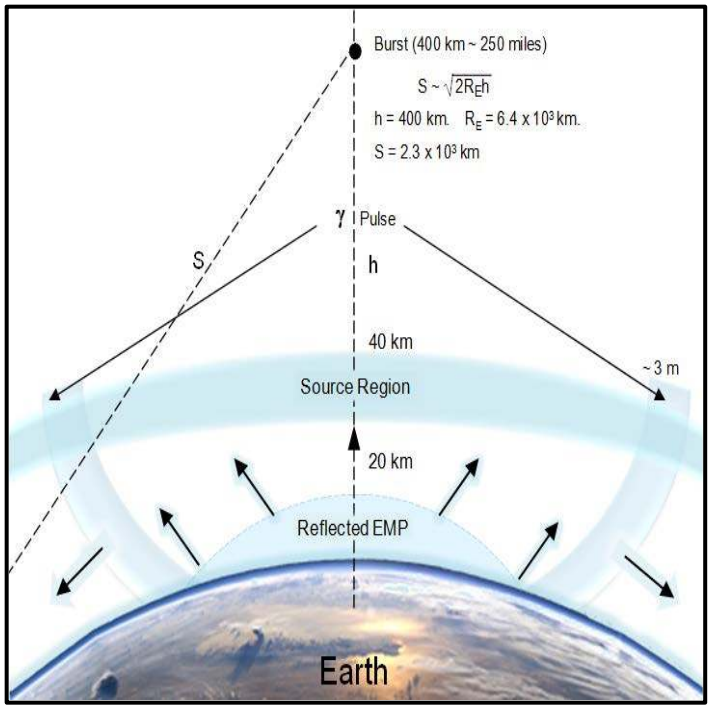


Figure D-4. Detailed geometry for Exo-Atmospheric burst.

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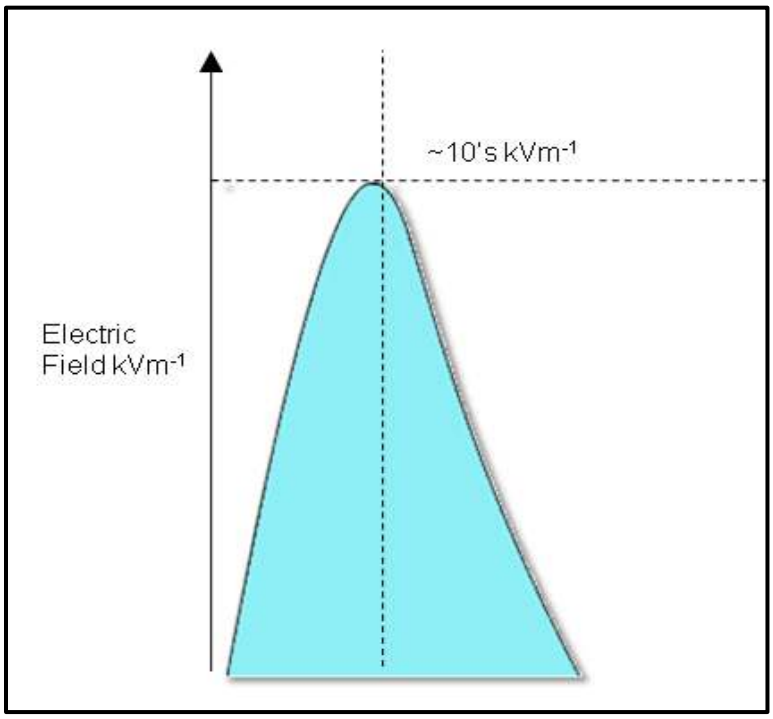


Figure D-5. Exo-Atmospheric EMP waveform.

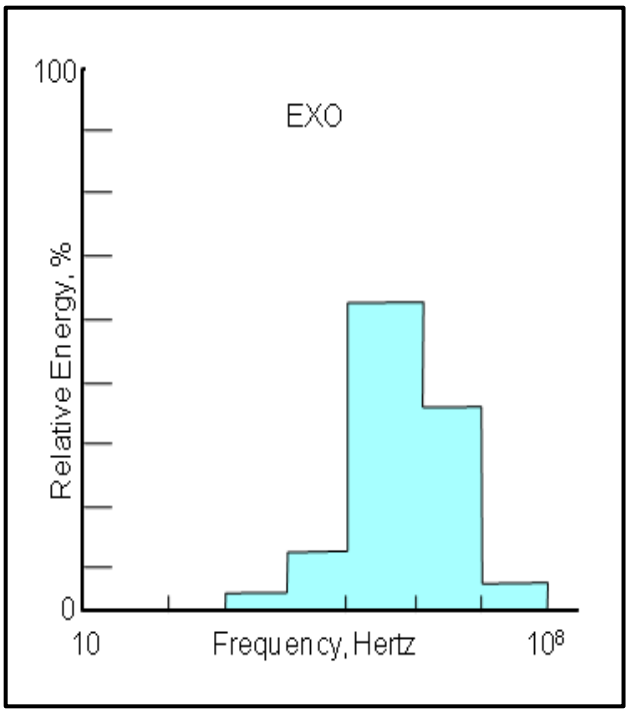


Figure D-6. Exo-Atmospheric relative energy versus frequency.

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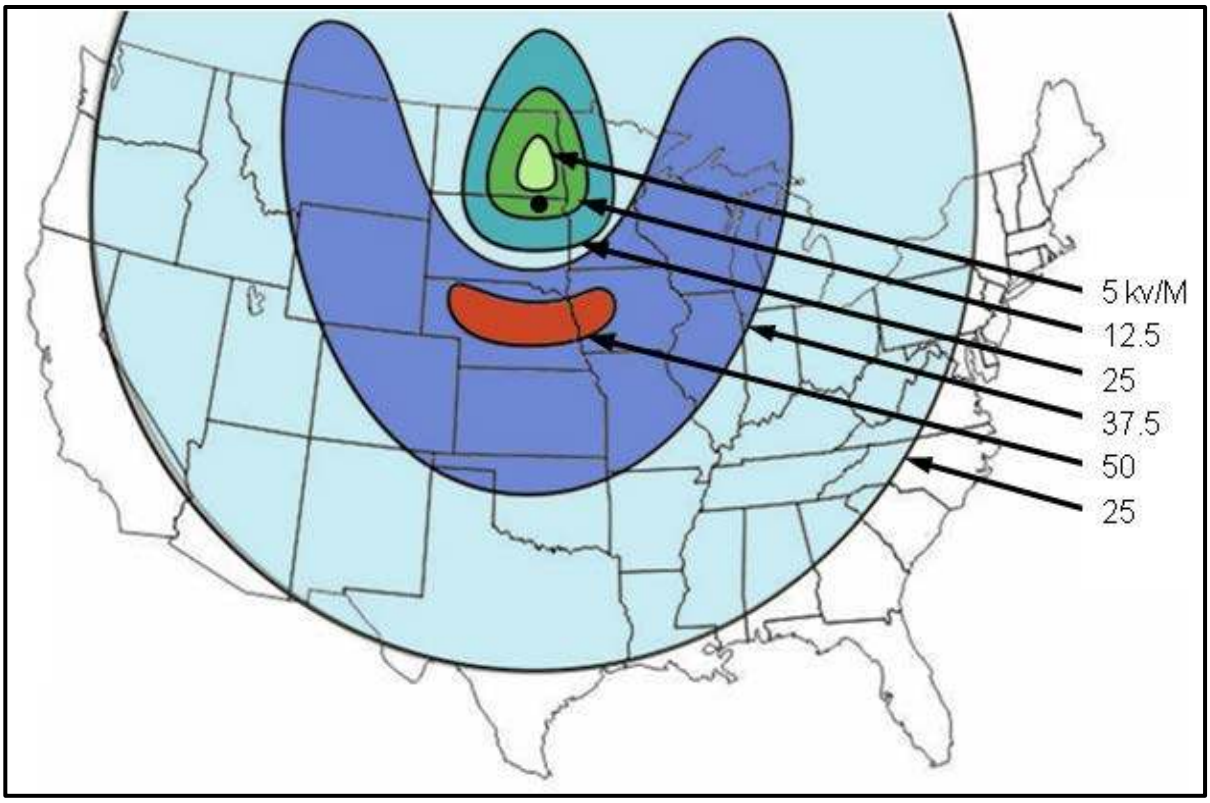


Figure D-7. Generated E-Field contours at the Earth's surface from a HEMP.

f. EMP testing requires the use of both experimental and analytical techniques to determine the response of systems and components to the EMP. Adequate testing of a system requires simulation of the EMP environment in terms of amplitude, time and geometrical effects of the entire system under study. Detonation altitude, angles of arrival and polarization of the field shall be considered. Frequency domain calculations may be applied to determine critical resonant frequencies inherent to the test system. Current injection techniques shall be utilized for distributed systems as an integral part of the EMP test. Current injection is greatly beneficial in the context of determining safety margins and, enhancing and verifying HEMP simulator results. But, current injection should not be the primary means of obtaining accurate HEMP data.

g. Also, deliberate hardening devices like terminal protection devices shall be analyzed and tested if necessary, to determine safety margins. Likewise, the attenuation afforded by enclosures shall be analyzed so that the effects on the survivability of the enclosed electronics can be quantified.

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APPENDIX D. NUCLEAR ELECTROMAGNETIC ENVIRONMENT AND EFFECTS.

D.2 OVERVIEW OF EMP TEST INSTRUMENTATION AND MATHEMATICS.

Provided is an over view of the data processing that occurs in EMP testing to form a basic knowledge foundation. It should be noted that this is not an in-depth presentation and that the procedures used and data accuracy provided by the instrumentation are in general the responsibility of the test facility.

D.2.1 Data Measurements.

a. For testing at pulse/high frequency facilities most sensors and data links are self calibrated by the instrumentation specialist. The traceability back to a laboratory calibration is through the use of a network analyzer which has calibration performed at the factory or a standard calibration laboratory. The instrumentation is basically divided into three sections which will be discussed separately below:

b. Sensors. In general the facility calibration of sensors is limited to the ones which measure voltage or current, sensors for measuring fields (electric or magnetic) or for measuring surface currents are factory calibrated. The current probes are available with different current measuring capabilities and different through hole sizes. In most cases (except for very small very specific probes) clamp on devices are used. A special adapter is required for each size probe for the facility calibration to be performed. The probe adapter (Figure D-8) allows for the placement of probe around the center conductor and inside the external conductor. The probe is then connected to a network analyzer, the network analyzer output feeds both the forcing function and the reference measurement and measures the voltage output of the current probe. This forcing function is driven across the frequency spectrum for which the current probe is being calibrated. The current probe voltage output is divided by the current through the fixture; the current through the fixture is based on the measured reference voltage and the fixed load impedance. The result is the transfer function of the current probe versus frequency which is called the probe transfer impedance since the units are volts/ampere. The voltage probes are calibrated in the same manner with the exception that the transfer function units are volts/volt. Experience has indicated that 400 measurements across the frequency spectrum tend to be adequate for calibration. This frequency calibration is used as follows: the data produced by the current or voltage probe is recorded on fast oscilloscopes as voltages in the time domain. The recorded data then requires correction for the probe transfer function (frequency response) and conversion to the correct units of measure. This is accomplished using convolution in the time domain, the process however takes place in the frequency domain since in the frequency domain convolution reduces to simple point by point division. The process for a current probe is that the recorded time domain waveform is converted to the frequency domain using FFT. The resulting frequency data is then divided on a per frequency basis by the probe transfer function at that frequency, interpolation between probe calibration frequency points is used to match the exact frequency resulting from the FFT. Once the convolution has taken place the FFT is now corrected for probe response and the inverse FFT results in the corrected time response with the appropriate units.

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Figure D-8. Probe adapter 1.

D.2.2 Fiber Optic Data Links.

In general, the facility calibration of data links is performed as matching pairs (i.e., the optical transmitter and the optical receiver are matched into a data link). These matched pairs are always used together during data acquisition. The DATA links are available from several manufactures one of the most common is NANO-FAST. The link is connected to a network analyzer, the network analyzer output feeds both the forcing function and the reference measurement and measures the voltage output of the link. This forcing function is driven across the frequency spectrum for which the link is being calibrated. The link voltage output is divided by the measured reference voltage across the frequency range being calibrated. The result is the transfer function of the link versus frequency for which the units are volts/volt. Experience has indicated that 400 measurements across the frequency spectrum tend to be adequate for calibration. This frequency calibration is used as follows: the data produced by the sensor are input to the fiber transmitter and received by the fiber receiver and the receiver output voltage is recorded on fast oscilloscopes as voltages in the time domain. The recorded data then requires correction for the link transfer function (frequency response). This is accomplished using convolution in the time domain, the process however takes place in the frequency domain since in the frequency domain convolution reduces to simple point by point division. The process for a link is that the recorded time domain waveform is converted to the frequency domain using FFT. The resulting frequency data are then divided on a per frequency basis by the link transfer function at that frequency. Interpolation between probe calibration frequency points are used to match the exact frequency resulting from the FFT. Once the convolution has taken place the FFT is now corrected for the link response and the inverse FFT results in the corrected time response.

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APPENDIX D. NUCLEAR ELECTROMAGNETIC ENVIRONMENT AND EFFECTS.

D.2.3 Fiber.

The fiber is not calibrated versus frequency since it possesses no elements which would result in a change in frequency spectrum transmission. Most fiber links provide for a standard signal being transmitter over the fiber to verify its acceptability. This measurement being unacceptable indicates two potential faults; the first is that the ends of the fiber are dirty, and the second is that the fiber is broken.

D.2.4 Demonstration of Measurement Conversion.

The data starts as a raw voltage measurement recorded on a digitizing oscilloscope. These raw data are processed through frequency domain convolution using the instrumentations factors indicated in Section D.2.1 into the final corrected measurement. Figure D-9 shows graphically the dramatic changes that occur in the data as the units are converted and the frequency response of the instrumentation is accounted for.

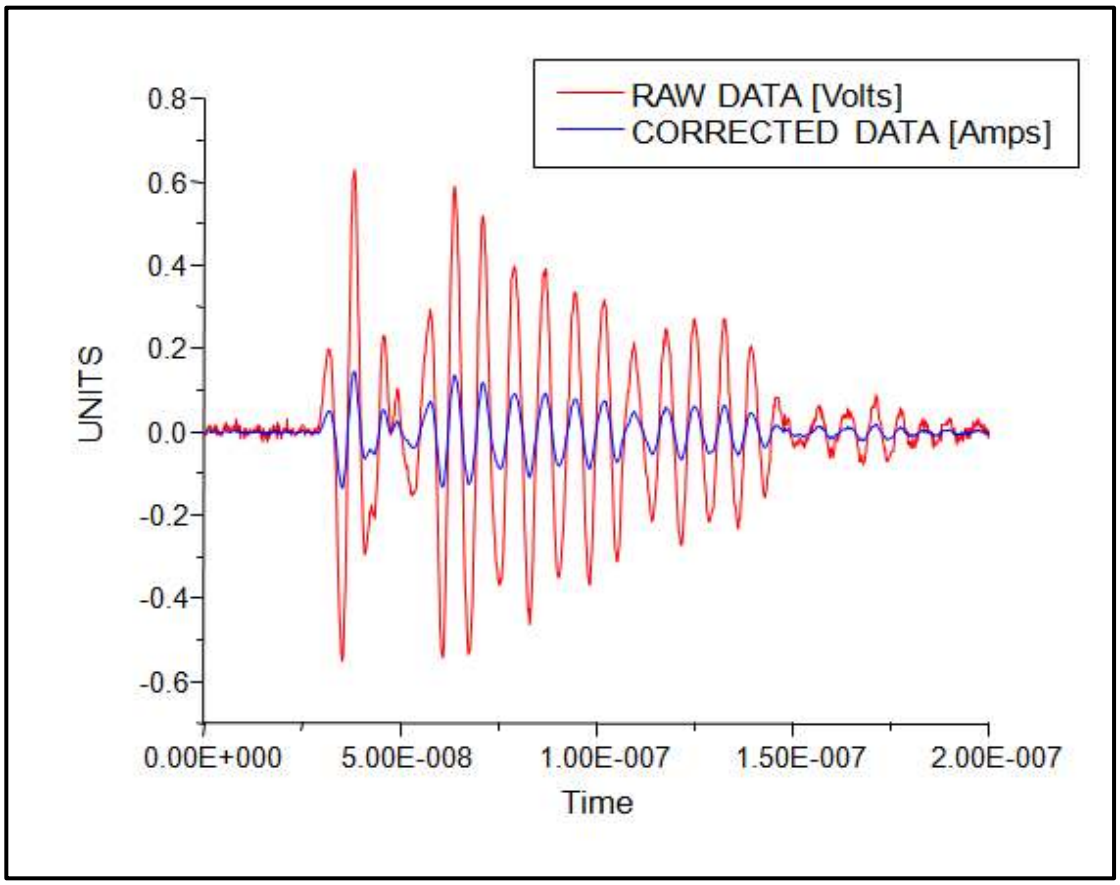


Figure D-9. Raw data correction.

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APPENDIX D. NUCLEAR ELECTROMAGNETIC ENVIRONMENT AND EFFECTS.

D.2.5 Data Calculations.

a. The data manipulation and calculations at pulsed/high frequency facilities are usually performed by the facility data specialist. Provided is a basic over view of the methods and results. The purpose of the calculations is to develop a set of scalars or NORMS, which can be used to describe the characteristics of the induced signals and provide engineering data for use in the correction of deficiencies if required. Provided in Table D-2 are examples of the SCALAR quantities for damped sinusoids and Figure D-10 is the FOURIER transform from which they were generated.

TABLE D-2. EXAMPLE OF CALCULATED DAMPED SINUSOID SCALAR VALUES

TEST POINT ID	ORIENT	SHOT NO.	PEAK I (AMP)	RESONANT FREQUENCY	LOWER FREQUENCY	HIGHER FREQUENCY	BANDWIDTH	ALPHA	Q
1	2-front	5170	0.1478	1.29E+08	1.26E+08	1.33E+08	8.18E+06	2.38E+07	14.15
2	2-front	5170	0.1030	1.25E+08	1.20E+08	1.30E+08	9.80E+06	3.26E+07	15.67
3	2-front	5170	0.1551	1.63E+08	1.50E+08	1.70E+08	2.00E+07	2.85E+07	15.84

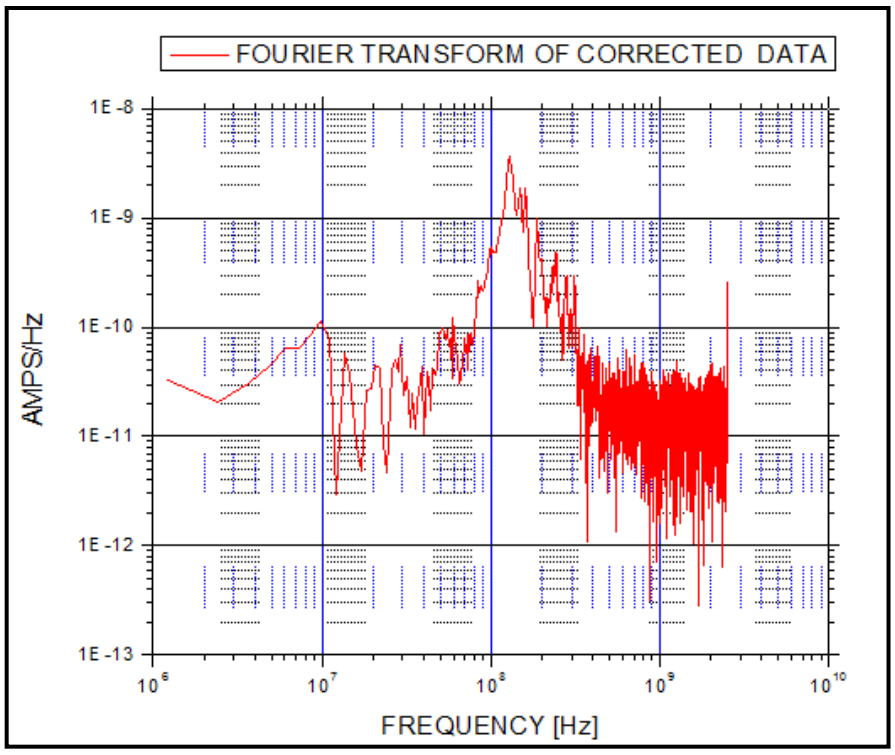


Figure D-10. Induced Current Fourier Transform.

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b. The values are calculated in the time domain representation of the waveform as follows:

(1) The resonant frequency (Fo) is the first pole in the frequency domain after the DC level and is displayed on the graph at about 1.3E8.

(2) The low (Fl) and high (Fh) frequencies are measured at the 3 dB down points from the peak, with the low frequency being to the left and the high frequency being to the right. If necessary the slope is used to calculate the 3 dB point on signals containing more noise.

(3) The bandwidth is the difference between the high and low frequency.

(4) Alpha (the exponential attenuation rate) is calculated by: $\alpha = \pi(Fh - Fl)$

(5) Q (quality factor) is calculated by: $Q = \frac{Fr\ (Resonant\ Frequency)}{\Delta f\ Half{-}Power\ Bandwidth}$

(6) Df (Damping factor) is calculated by: $Df = \frac{1}{2Q}$

D.2.6 Example Technical Analysis.

From the scalars presented in Table D-2, which were generated from the measured EMP induced current, the following is known; a) the induced current is at a frequency of 129 MHz, b) the bandwidth is approximately 10 MHz, c) the Q is large and the damping factor is small indicating the signal is over damped and will decay quickly, and d) the amplitude of the induced current was 0.15 Amps. Therefore, if the component connected to this cable was experiencing EMP induced problems, it could most likely be eliminated by designing a simple low pass filter (or a more complicated filter based on the circuit signal requirements) with an upper cutoff frequency of 119 MHz and which is capable of carrying the normal signal currents plus approximately 0.2 Amperes.

D.3 EMP ORDNANCE TEST INFORMATION.

Provided are some example procedures for EMP testing of materials and equipment, which contain ordnance.

D.3.1 Most facilities will not test all the presence of full up ordnance during any type of electromagnetic testing and this includes EMP. The types of systems which contain ordnance and require EMP testing are those which are electrically controlled and actuated and not mechanically actuated. These systems utilize Electrically Initiated Devices (EIDs) to perform explosive train functions, such as initiating the battery, arming the device etc. In order to

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perform a EMP assessment on a system that uses these EID, the main charge requires replacement with inert material and the testing needs to be performed on two system configurations;

a. The first and foremost configuration (i.e., TYPE 1) is when the EIDs are replaced with an equivalent resistor and a current probe. The equivalent resistor provides the same impedance to the system under test and the current probe is routed such that the measurement of the induced current can be recorded by an external DAS system. In all ordnance systems the safety margin of the explosives with respect to the environment are of upmost importance. The EMP EID induced current measurement is used in performing safety margin calculations based on the EID specifications.

b. The second configuration (i.e., TYPE 2) is with the main charge removed and the EIDs installed. Safety procedures shall be developed and strictly followed to insure that projectiles are not created should the EID function, which could injure personnel or damage the facility or equipment. Potential for damage with only the EIDs installed usually is provided in the System Safety Assessment Report (SSAR). In some case the statement will be directly made so that the functioning of the EID cannot deform the surface or create projectiles. In other cases, the indication could be that the item requires enclosure in a plastic or plywood (i.e., non-conductive container for EMP) of a specific thickness to insure that possible projectiles are contained. There shall also be a method of determining if the EID has functioned. The test configuration is the basic GO-NOGO configuration and provides an indication as to whether the calculations and measurements from the first configuration were correct.

D.3.2 EID specifications contain valuable pieces of data. The first is the device maximum no-fire current (i.e., the EID will not function with the continuous application of 20mA). The second is the device all fire current (i.e., the EID will function with the application of 1A). The third is the timing for the EID to function (i.e., some will indicate a square pulse with an amplitude and a time and others may indicate all EIDs will fire within 10 milli-seconds when a 103.4 μ-Farad capacitor charged to 1.6 Volts is discharged through the EED.) From these specifications two important EMP parameters are determined, the first is that for the best case the EMP induced current (regardless of time) is less than the no-fire current; and the second is that the EMP induced energy should provide for a minimum safety factor of 16.5 dB.

D.3.3 Provided are some possible measurements and calculations. Table D-3 provides examples of reduced EMP test data for a system with three EIDs. Figure D-11 provides a frequency and time corrected induced current. Figure D-12 provides the energy delivered to the EID by the EMP pulse. The energy is acquired by rectifying the induced current, then calculating instantaneous power at each data point utilizing the EID impedance and finally integrating over time to convert to energy.

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APPENDIX D. NUCLEAR ELECTROMAGNETIC ENVIRONMENT AND EFFECTS.

TABLE D-3. EID INDUCED CURRENTS

TEST POINT ID	ORIENT	SHOT NO.	PEAK I (Amp)	PEAK DERIVATIVE	PEAK IMPULSE	RECTIFIED IMPULSE	ROOT ACTION INTEGRAL	TOTAL ENERGY 1 Ohm	TOTAL ENERGY 3 Ohms
1	1-front	5166	0.0858	8.56E+07	1.66E-10	4.04E-09	1.18E-05	1.38E-10	4.15E-10
2	1-front	5167	0.1030	1.29E+08	4.80E-10	7.44E-09	1.80E-05	3.26E-10	9.77E-10
3	1-front	5169	0.1551	1.63E+08	2.37E-10	5.21E-09	1.69E-05	2.85E-10	8.55E-10
1	2-front	5170	0.1473	1.74E+08	1.79E-10	6.73E-09	2.01E-05	4.03E-10	1.21E-09
2	2-front	5171	-0.1471	1.72E+08	1.03E-09	8.22E-09	2.12E-05	4.51E-10	1.35E-09
3	2-front	5173	0.1012	9.85E+07	1.19E-10	3.64E-09	1.07E-05	1.14E-10	3.42E-10
1	2-back	5175	0.0125	1.98E+07	2.04E-11	6.42E-10	1.72E-06	2.94E-12	8.83E-12
2	2-back	5176	0.0142	2.12E+07	1.92E-11	7.52E-10	1.94E-06	3.78E-12	1.13E-11
3	2-back	5177	-0.0168	2.25E+07	4.15E-11	6.63E-10	2.00E-06	4.00E-12	1.20E-11
1	2-side	5180	-0.0343	5.30E+07	8.81E-11	1.13E-09	3.43E-06	1.18E-11	3.54E-11
2	2-side	5181	0.0459	7.25E+07	1.44E-10	2.47E-09	6.59E-06	4.34E-11	1.30E-10
1	2-side	5182	-0.0324	5.06E+07	9.51E-11	1.12E-09	3.40E-06	1.16E-11	3.47E-11
3	2-side	5183	-0.0506	6.09E+07	7.10E-11	1.75E-09	5.46E-06	2.99E-11	8.96E-11
1	2-front	5184	-0.0553	8.63E+07	1.81E-10	1.67E-09	5.20E-06	2.71E-11	8.12E-11

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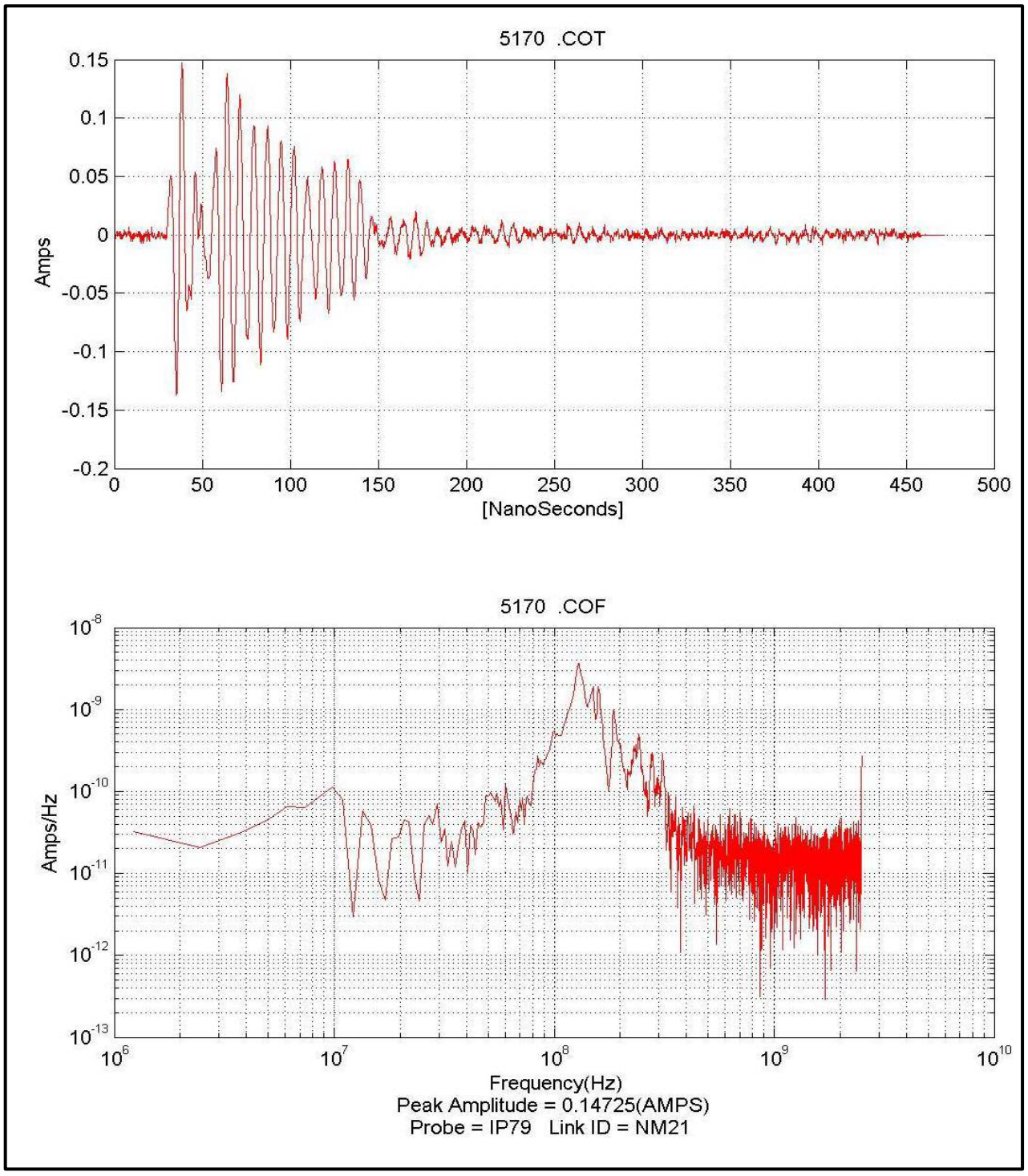


Figure D-11. EID current example.

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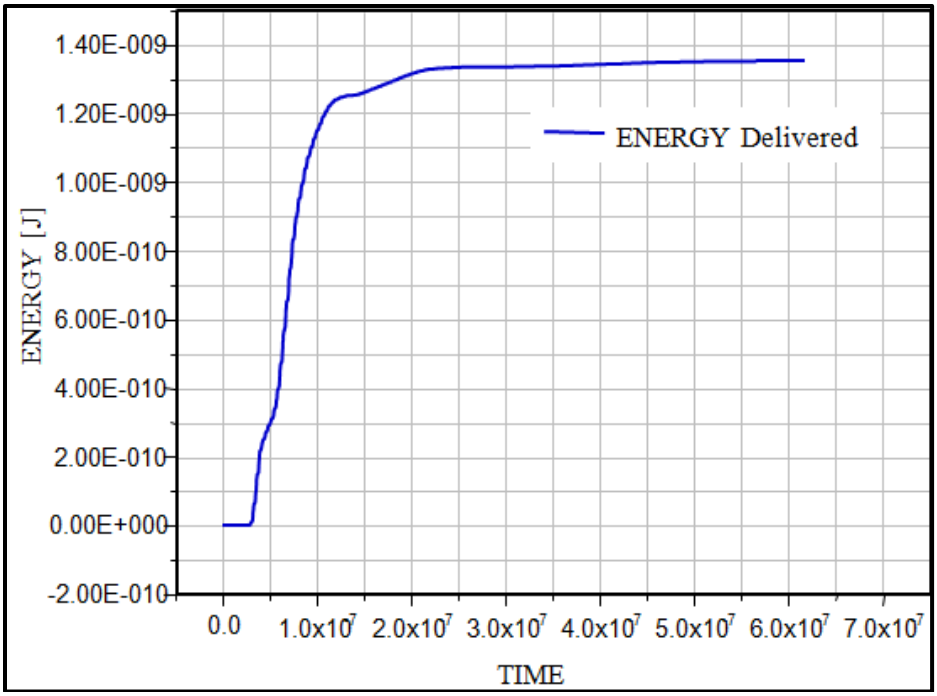


Figure D-12. EID EMP Induced Energy.

D.3.4 Example EID Technical Analysis.

The following is based on an EID with the following specifications provided:

a. The EID will not function with the application of 20 mA for 1 minute.

b. The EID will fire within 10 milli-seconds when a 103.4 µ-Farad capacitor charged to 1.6 Volts is discharged through the EID.

c. The EID impedance is 3 Ohms.

D.3.4.1 At no point during testing did any of the ten **TYPE 2** units enable or fire as a result of the EMP exposures. The test results initial indication is that insufficient energy was delivered to the EID for it to function.

D.3.4.2 The **TYPE 1** EID EMP induced current measurements indicate that a maximum of 1.35E-9 Joules are available to the EID at the specified impedance (three Ohms). Using the total length of recorded current waveform (600 nano-seconds) as time, the EID EMP induced energy deposition rate is calculated to be 2.25E-3 J/Sec. The specification indicates that all EIDs will fire within 10 milli-seconds when a 103.4 µ-Farad capacitor charged to 1.6 Volts is discharged

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through the EID. This correlates to an energy level of 1.32E-4 Joules as calculated by 1/2 CV^2 and a deposition rate of 1.32E-2 J/sec (based on the 10 milli-seconds time period).

The specifications also indicates that no EID shall fire when subjected to a current of 20 milli-amperes for a period of one minute, which corresponds to an energy of 7.2E-2 Joules at a deposition rate of 1.2E-3 J/sec. Therefore, based on energy deposition rates during firing, the EID has a conservative safety factor of five (5) as calculated by dividing the all fire energy deposition rate (1.32E-2 J/sec) by the EMP induced deposition rate (2.25E-3 J/sec). Based on total energy delivered for no-fire the EID has a large safety factor as calculated by dividing the No-fire energy delivered (7.2E-2 J) by the EMP energy delivered (1.35E-9 J). This agrees with the **TYPE 2** results in that no EIDs were discharged as a result of the HEMP environments.

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

E.1. GENERAL RADIATION CHARACTERISTICS.

Approximately fifteen percent of the energy delivered by the detonation of a nuclear weapon is produced as (non-thermal) radiation. There are four types of radiation which are worthy of attention.

a. The first type of radiation is called alpha radiation and is comprised of a stream of alpha particles which are essentially the nuclei of helium atoms. Most are absorbed or interact with the air molecules before arriving at a survivable system. This type of radiation has exceedingly slight penetrating power causing essentially no effects on material or on military personnel because it can be stopped by a single sheet of paper or the skin of the personnel. Military casualties are only produced by alpha radiation when it is internally introduced into the body. Therefore, alpha radiation is not considered militarily significant to equipment or personnel.

b. The second type of radiation is called beta radiation and is comprised of a stream of beta particles, which are electrons. This type of radiation has slight penetrating power and can just barely penetrate the skin. The only effects caused by initial beta radiation on military personnel are skin burns called "Beta Burns." Since there are essentially no notable effects on material or on military personnel, initial beta radiation has little military significance.

c. The third type of radiation is called gamma radiation, which is comprised of photons or electromagnetic waves very similar to x rays; but have shorter wavelengths, are usually more energetic and, therefore, have substantially greater penetrating power. Since gamma radiation can travel great distances through air and pass through thick low density materials, gamma radiation is extremely significant to military personnel and systems containing electronics.

d. The last type of radiation is called neutron radiation and is comprised of a particle stream of neutrons. Neutrons are one of the elementary particles which comprise the nucleus of all atoms and possess no charge. Since neutrons usually have much greater penetrating power than gamma radiation, neutrons are extremely significant to military personnel and systems containing electronics.

E.2. RESIDUAL RADIATION ENVIRONMENT AND EFFECTS.

The residual radiation effects of a nuclear weapon consists of radioactive weapon debris and fallout, and NIA, which sustains for longer than one minute after weapon detonation. The effects of residual activity can cover an extensive area and can persist for long durations which are dependent upon weapon yield and design, HOB, soil composition, and atmospheric conditions. The radiation composition of residual radiation consists of alpha particles, beta particles, and gamma radiation. A more detailed examination of the two primary residual radiation effects, NIA and fallout follow:

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a. The first effect, NIA occurs in the vicinity of GZ and is caused by neutron capture in the system and various materials which exist on the earth's surface near GZ. This process of neutron capture produces radioactive isotopes that emit beta and/or gamma radiation. The duration of significant radiation emission is dependent upon the isotopes created. Typically, at distances where main battle tanks (MBTs), armored personnel carriers (APCs), and shelters are survivable, NIA is not a major concern. Although the primary construction material (iron, aluminum, carbon fiber composites) do capture neutrons, and produce gamma photons, majority of the resulting isotopes have short half-lives.

b. The normally predominant effect, fallout, is highly dependent upon the type of weapon burst. In air bursts, the vaporized radioactive products condense into small particles in the range of 20 μm in diameter. These extremely small particles are conveyed high in the atmosphere and descend to earth over an exceedingly prolonged time. Rain or snow accelerates fallout. By this process, the fallout has significantly decayed and has been greatly dispersed by winds resulting in a very low radioactivity level which is militarily insignificant.

c. On the other hand, during ground or near ground bursts, vast quantities of soil and debris are drawn into the fireball forming condensation centers for vaporized radioactive products. This results in a massive cloud of radioactive particles with diameters up to 500 μm. The rate at which these particles descend to Earth is based upon increasing mass and atmospheric conditions. In other words, the denser the particle, the quicker it will fall to Earth unless adverse atmospheric conditions prevail. The outcome of this type of burst is steadily increasing radioactivity closer to GZ the measurement point because of the decrease in decay time and the increase in radioactive particles can cause hot spots that can be harmful.

d. In particular; dust, snow, or mud collection re-suspension can be a problem. Residual radiation has essentially no damaging effects on nuclear survivable military systems, but presents major difficulties to military personnel in the area surrounding GZ and downwind of GZ. Gamma radiation is the primary concern when providing protection for military personnel against residual radiation. The effects of both alpha and beta radiations may be made insignificant for military concerns by thin layers of materials. Combat vehicles; particularly tanks and armored personnel carriers, because of their armor and massive metal construction, inherently provide a considerable amount of shielding against residual radiation.

e. The amount of shielding provided is almost a direct function of the mass of material between the individual and the radiation source. The effectiveness of material in attenuating radiation may be represented by its "half-value thickness", the thickness of the particular material which absorbs half of the gamma radiation incident upon it. The amount of shielding provided for each crew member is expressed in terms of "protection factor" defined as follows:

$$\textit{Protection Factor} = \frac{\textit{Free Field Radiation Level At 3 Feet Above The Ground}}{\textit{Radiation Level At Personnel Location}}$$

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

f. The protection factor is the ratio of the radiation dose a person would receive if he were standing in the open in a fallout field to the dose he would receive in the vehicle at the same location. Some tanks may provide protection factors against residual radiation as high as 20, whereas a ¼-ton truck may provide a protection factor of 1.25. However, hot spots can drastically reduce the gamma protection factors. The term "transmission factor" is also used. It is essentially the inverse of the protection factor and is defined as:

$$Transmission\ Factor = \frac{Dose\ Inside\ Shield}{Dose\ Outside\ Shield}$$

g. Typical transmission factors for shelters and vehicles against initial and residual radiation are contained in Table E-1.

TABLE E-1. TYPICAL TRANSMISSION FACTORS FOR NUCLEAR RADIATION EFFECTS

SHIELDING ITEM	RESIDUAL RADIATION TRANSMISSION FACTOR	NEUTRONS INITIAL RADIATION TRANSMISSION FACTOR	GAMMA INITIAL RADIATION TRANSMISSION FACTOR
Armored Personnel Carrier	0.6	0.7	0.7
Light Tank	0.2	0.3	0.2
Medium Tank	0.1	0.3	0.1
High Mobility Multipurpose Wheeled Vehicle (HMMWV), 1/4-Ton Truck	0.8	1.0	1.0
3/4-Ton Truck	0.7	1.0	1.0
2 1/2-Ton Truck	0.6	1.0	1.0
4- To 7-Ton Truck	0.5	1.0	1.0
Foxhole	0.1	0.3	0.2
Open Trench	0.1	0.3	0.2
Shelter with 3 feet of Earth Cover	0.005	0.05	0.02

E.3 INITIAL NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

a. Initial nuclear radiation is defined as that nuclear radiation which is emitted by a nuclear explosion within the first minute after the nuclear weapon burst. Initial nuclear radiation is composed of alpha particles, beta particles, gamma radiation and neutrons. Only the highly penetrative gamma rays and neutrons are extremely damaging to military personnel and electronics in military systems. The magnitude of this radiation is dependent upon weapon yield, type, HOB, and distance from GZ. For middle to high yield weapons, the damaging effect on systems generated by the blast wave and thermal radiation surpasses INR effects. However, for

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small weapon yields, the damaging effects to the electronics, electro-optic systems, and personnel of tanks, APCs, and similar systems is INR.

b. The gamma radiation produced by a nuclear detonation comes from various sources, such as the initial fission reaction, fission product decay, and NIA in the warhead debris and in the surrounding air molecules. Prompt gamma photons are those produced during fission and as a result of neutron interactions with weapon materials in the first µsec. Capture gamma photons are those emitted as the result of the capture of a neutron by a nucleus. Other examples of gamma radiation sources are delayed and inelastic gamma photons.

c. The neutron radiation produced by a nuclear detonation comes from two main sources, prompt neutrons and delayed neutrons. Prompt neutrons constitute over 99 percent of the total neutron production and are released in the initial µsec during the warhead fission or fission-fusion processes. Delayed neutrons are emitted within the first minute and are produced by the interaction of the prompt neutrons with atoms in their path. Small quantities of neurons are also produce by the action of high energy gamma photons on the weapon materials.

d. As the distance from GZ increases, the effects of INR are significantly reduced by two factors. These factors are that the intensity of INR decreases according to the inverse square law, and air scattering and absorption. The inverse square law is the principle factor for decreases in INR levels. The factor of air scattering and absorption has very little effect on neutron dose, but affects gamma radiation dose significantly. Figure E-1 is an example of initial radiation effects from a 1 kT airburst and Figure E-2 provides INR scaling factors versus weapon yield.

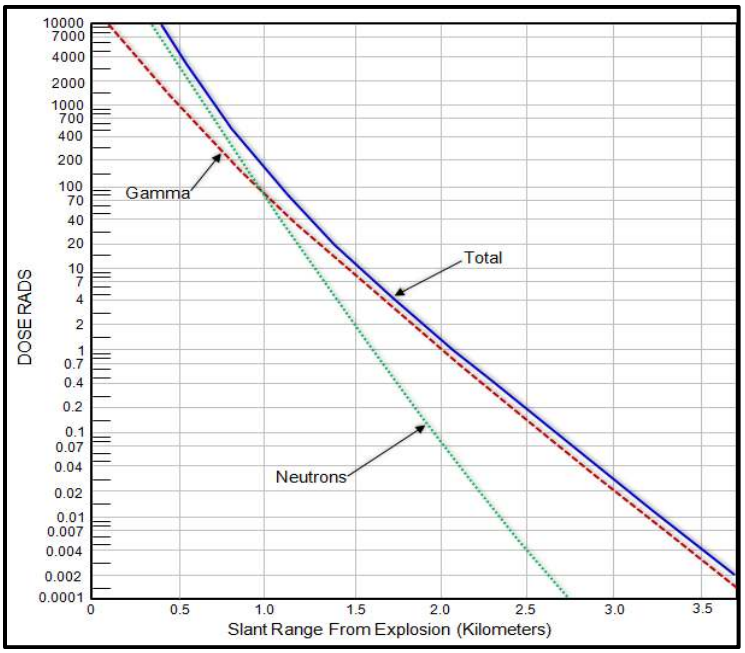


Figure E-1. Initial radiation effects from a 1 kT air burst.

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

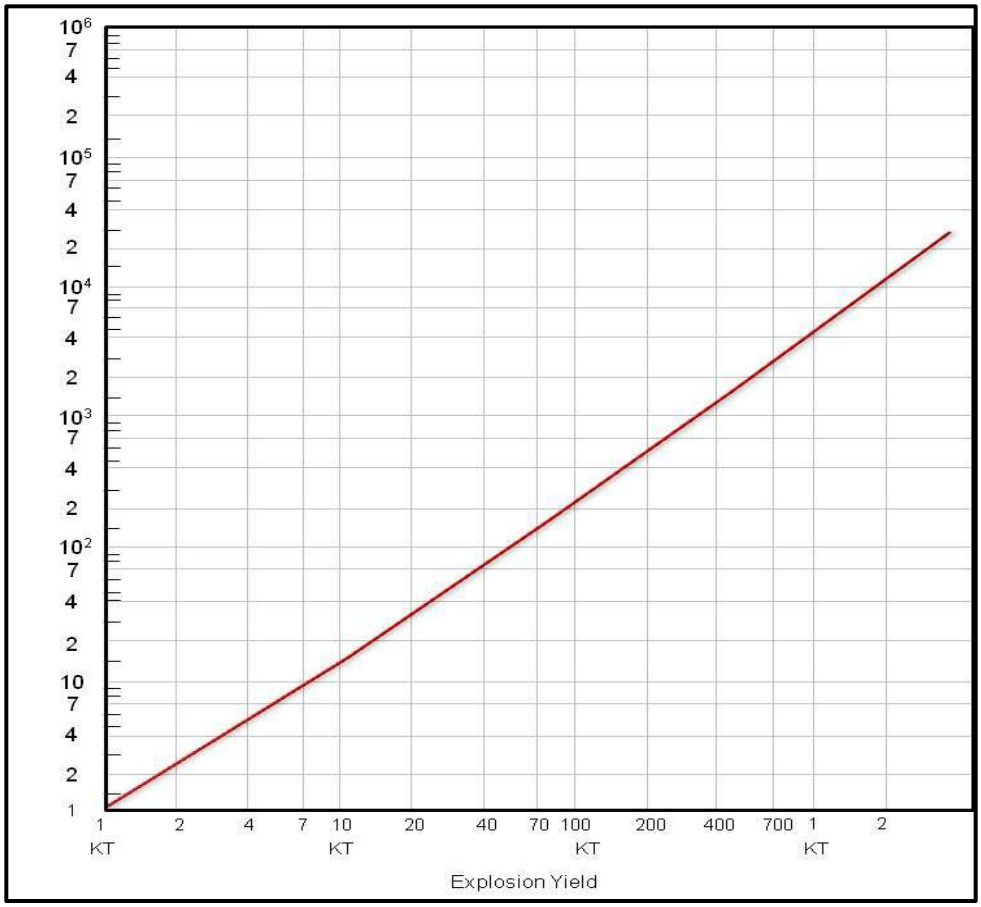


Figure E-2. INR scaling factors versus weapon yield.

e. The INR shielding tests involve the exposure of materiel to neutron and gamma radiation sources to measure the protection afforded to the crew and the most vulnerable internal electronic components by the walls of the vehicle. These radiation sources shall have spectra that collectively approximate the spectra from nuclear weapons. See Table E-1 for typical transmission factors of INR effects on military equipments and shelters.

f. The INR tests are only concerned with the effects on military personnel and equipment from gamma radiation and neutrons. The damaging effects produced by INR are only militarily significant in the consideration of the drastic effects it induces on electrical properties of semiconductor devices. To adequately test and analyze these drastic effects, three test environments based upon the time history of nuclear weapon radiation effects have been developed. These tests are: gamma dose rate, neutron fluence, and gamma total dose.

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g. The gamma dose rate test determines the effects of the initial gamma pulse created by a nuclear detonation on powered semiconductor devices primarily of the Metal-Oxide-Semiconductor (MOS) technology. The gamma dose rate pulse generates damaging photocurrents, which produces transient upsets, latchup, and burnout in semiconductor devices.

h. The neutron fluence test determines the effects of lattice displacement damage generated by neutron fluence on semiconductor devices primarily of the bipolar technology. The semiconductor piece-parts of a system generally receive the most damage by neutrons. Neutron damage results in a decrease of the minority carrier lifetime and an increase in bulk resistivity of the semiconductor material. These effects severely alters the electrical characteristics of the piece-parts, and in some cases, the induced damage is severe enough to cause complete device failure or failure in its circuit application.

i. The gamma total dose test determines the effect of total ionizing dose deposited on semiconductor devices. The production of hole-electron pairs through ionization creates trapped charge in the semiconductor material. The total dose effects in semiconductors are exhibited either as a change in electrical parameters or as a catastrophic failure. Of particular concern is N-channel MOS technology, which is sensitive to gamma dose.

E.4. ENVIRONMENTAL TEST CONSIDERATIONS.

E.4.1 Gamma Dose Rate.

a. Detailed description of the method of producing the gamma dose rate test environment: including; photographs of the test facility setup showing test system location relative to the gamma radiation source. The Test Method should be either electron beam or Bremsstrahlung and source specifications. With current technology, digital photographs are considered the best method of archiving a test setup / procedure. In addition to photographs, the archive and report generation should utilize one of the many computer graphics programs available that can generate clear and concise drawings of the physical test setup.

b. Complete set of pre-test mapping data in radiation absorbed dose in silicon (cGy(Si)) and absorbed dose in tissue (cGy (tissue)) ±10% for each test location. This requirement is met by using existing facility data to estimate the STT distances and then refining these STTs by taking pre-test GDR irradiations. For each of these characterization irradiations, a series of CaF_2(Mn) TLDs are placed on the centerline axis of the simulator and exposed. These pre-test mapping dosimeter readings are then utilized to calculate each required STT distance for a required GDR environment test level. This calculation process shall be performed for each electronic item to be exposed. The basic equation is presented as Equation E-1 (gamma dose rate inverse square equation) and should be verified against the pre-test mapping data. An area equal to the area of the LRU or test item's active electronics shall be mapped at each STT. This is required in order to define the exposure level and gradients associated with the test, for each test item, no matter of size or requirements.

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$$\frac{\dot{\gamma}_{req}}{D^2_{traget}} = \frac{\dot{\gamma}_{map}}{D^2_{map}} \qquad \text{Equation E-1}$$

where: D_{map} = *the distance from the source to the dosimetry point*
D_{target} = *the distance from the source to the target*
$\dot{\gamma}_{map}$ = *gamma dose rate at mapping point*
$\dot{\gamma}_{req}$ = *gamma dose rate required*

Re-arranging terms, Equation E-2 (target distance determination) provides the distance required to achieve a given gamma dose rate environment.

$$D_{target} = D_{map}\sqrt{\frac{\dot{\gamma}_{req}}{\dot{\gamma}_{map}}} \qquad \text{Equation E-2}$$

c. Information required prior to test execution. Required pre-test information includes test item or system familiarity and target information, such as sizes and locations of mission critical electronic elements and knowledge gained in the testing of other similar military hardware. Of primary importance is the ability to utilize these data in developing a coherent test procedure. This procedure should include but are not limited to using the electrical schematics to determine possible susceptible areas, insure that all Mission Critical (MC) electronics are exposed, and insure that the gamma dose rate protection devices (if present) are included and evaluated. In addition, it is necessary to ensure that:

(1) Each target area, MC electronic assemblage of the test item will be properly exposed to the GDR environment while in a powered operational state.

(2) The function of the target area will be baselined before testing, and that this baseline will be repeated after each GDR exposure.

(3) Each target area will have mounted to the incident surface dosimetry, such as CaF_2(Mn) TLDs, for recording the incident gamma dose environment; and dosimetry, such as PIN or Compton Diode, for recording the free-field dose rate received during the test. The dose recording dosimetry shall be changed after every exposure. This dosimetry, at a minimum, should be placed on all vertical surfaces of test item under test, with the largest dimension of the TLD being perpendicular to the incident environment. These dosimeters are used to determine the incident environment on the surface of the electronics-under-test and can be used to estimate the internal gamma dose rate environment.

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(4) On large test items gamma dose dosimetry should also be located inside the test item on or near the target area(s).

E.4.2 Neutron Fluence.

a. Detailed description of the method of producing the NF test environment including photographs of the test facility setup showing test system location relative to the neutron radiation source. This requirement is best attained by being familiar with available facilities. If not, then reviewing the facilities listed in the DTRA "A Complete Guide to Nuclear Weapons Effects Simulator Facilities and Applications" (2013 Edition) and requesting a facilities brochure, which should contain a description of the radiation producing method (a FBR is the preferred source for NF irradiation) and specifications. With current technology, digital photographs are considered the best method of archiving a test setup / procedure. In addition to photographs, the archive and report generation should utilize one of the many computer graphics programs available that can generate clear and concise drawings of the physical test setup.

b. Complete set of pre-test mapping data expressed in 1 MeV equivalent (damage in silicon) neutron/cm^2. Because of the extreme control required in the operation of most reactors, the facility will maintain a set of control parameters and dosimetry that can be used to accurately predict experimental exposures. Once a familiarity with the facility has been established there is usually little need for a neutron mapping exposure. Neutron mapping is required (for each of these characterization exposures a series of Sulfur activation dosimetry and TLDs are setup at the centerline axis of the proposed experiment position). These dosimeter readings are then utilized to calculate the required source-to-target distance for a given neutron environment (for a pulse operation), or the exposure time required to achieve a given neutron fluence (for a power operation). The method (pulse or power) of receiving NF is not of importance as long as the received fluence is converted to 1 MeV equivalent damage in silicon. The 1 MeV equivalent conversion is used as a normalization factor between the different types of reactors and operations. Reactors in general are of the water moderated style, or the free air mass moderated style, with operations being either pulsed or continuous power depending on experiments requirements. The best source found for determining experiment location for a given fluence is the reactor operations crew.

c. Information required prior to test execution. Required pre-test information includes test item familiarity and target information such as sizes and locations of mission critical electronic elements and knowledge gained in the testing of other similar military hardware. Of primary importance is the ability to utilize these data in developing a coherent test procedure. This procedure should include, but are not limited to, using the electrical schematics to determine possible susceptible areas (insure that all MC electronics are exposed). In addition, it is necessary to insure that: (1) the function of all electronic test items/assemblies will be baselined before testing, and that this baseline will be repeated after each exposure to the neutron environment, and (2) each target area will have mounted to each surface dosimetry for recording the incident neutron and

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gamma dose environment. This dosimetry should be placed on the target area assemblages at a point on the side which corresponds to the mid-plane of the active electronic microcircuits (this assumes disassembly to the circuit card assembly level).

d. Dosimetry from the NF environment. The sulfur pellets and TLDs (CaF_2(Mn) recommended) should be placed as indicated in the section above and their location used to determine the overall neutron exposure and gamma dose; respectively for each target area. The primary method of recording the neutron environment is through the use of sulfur pellets, which measure the total neutron energy greater than 3 MeV received. The dosimetry shall be changed after every exposure. The sulfur measurement (neutrons/cm^2 greater than 3 MeV) is converted to 1 MeV equivalent damage in silicon by a conversion factor (approximately 6.7 for an experiment at 26 inches for the WSMR FBR and changes as a function of distance). This conversion factor is dependent on the neutron spectrum produced by the reactor and the distance at which the test is performed. Therefore, the conversion to 1 MeV equivalent is not a constant and shall be obtained from the facility dosimetry specialist.

e. Electronics under test orientation and data. The test items shall be oriented such that all of the active MC electronics are exposed to the NF environment. This should be accomplished at the lowest assembly level possible, in order to reduce neutron dose gradients and unnecessary material activation. The STT distance and neutron level shall be known before each exposure to insure that the required test levels are received. Once all test items have received the initial level, the test items should be positioned to receive the 200% level. Again, it is emphasized that the primary goal is to expose all the MC electronics to test levels which will account for: all error in dosimetry and data recording, response differences in microcircuits due to manufacturing processes, and to account for the small sample size evaluated (usually one for assemblies and ten for devices). The operational data taken on the items under test should be performed using system functional check sheets, which include a description of the operation to be performed and the nominal results. A functional data sheet should be completed after each neutron exposure. Dosimetry from all test items which have been exposed to the NF environment should be removed and evaluated for the NF achieved.

E.4.3 Gamma Total Dose.

a. Detailed description of the method of producing the GTD test environment including photographs of the test facility setup showing test system location relative to the gamma radiation source. This requirement is best attained by being familiar with available facilities. If not, then reviewing the facilities listed in the DTRA "A Complete Guide to Nuclear Weapons Effects Simulator Facilities and Applications" (2013 Edition), and requesting a facilities brochure, which should contain a description of the radiation producing method (Co_{60} is the preferred source for low dose rate gamma radiation) and specifications. With current technology, digital photographs are considered the best method of archiving a test setup / procedure. In addition to photographs, the archive and report generation should utilize one of the many computer graphics programs available that can generate clear and concise drawings of the physical test setup.

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b. Complete set of pre-test mapping data in Radiation Absorbed Dose (RAD) in silicon (RAD(Si)) = (cGy(Si)) and absorbed dose in tissue (cGy(tissue)) ±10% for each test location. The pre-test mapping requirement is met by using existing facility data to estimate STT distances and then refining these STTs by taking pre-test GTD environment exposures. For each of these characterization exposures a series of TLDs are placed on the centerline axis of the simulator and exposed. These dosimeter readings are utilized to calculate the required STT distance for a given dose rate and the exposure time required to achieve the given GTD. This calculation process shall be performed for each test item to be exposed. The basic dose rate equation is presented as Equation E-1 and should be verified against the pre-test mapping data. Equation E-2 can be used to select the STT distance for a given dose rate. The GDR for the GTD test should be selected between 30 and 300 cGy(Si)/sec and result in an exposure time of less than one minute. The physical dimensions of the items under test will determine what the highest dose rate used will be. The position of the items-under-test is selected to minimize the gamma dose gradients, which are calculated using Equation E-2. Once the STT distance (which maximizes GDR and minimizes dose gradient) is determined, the time of exposure is calculated using Equation E-3 (time of exposure).

$$t_{exposure} = \frac{\gamma_{req}}{\dot{\gamma}_{map}} \qquad \text{Equation E-3}$$

where: γ_{req} = GTD requirement
$\dot{\gamma}_{map}$ = GDR at mapping point

c. Information required prior to test execution.

(1) Required pre-test information includes test item familiarity and target area information such as sizes and locations of mission critical electronic elements and knowledge gained in the testing of other similar military hardware. Of primary importance, is the ability to utilize these data in developing a coherent test procedure. These procedures should include, but are not limited to, using the electrical schematics to determine possible susceptible areas, insure that all MC electronics are exposed, and insure that the hardened critical process(es) for GDR protection devices (if present) are included and evaluated. In addition, it is necessary to insure that: (1) every test item will be properly exposed to the GTD environment while in a powered operational state, (2) the function of all items under test will be baselined before testing and that this baseline will be repeated after each exposure to the GTD environment, and (3) that each test item will have mounted to each of its surfaces dosimetry for recording the incident GTD environment.

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(2) If GDR power removal has been employed as a circuit protection method, then the GTD test should be designed so that the percentage of gamma dose received after the power down removal time, is received with the test item in a powered OFF non-operational state. The dosimetry, at a minimum, should be placed on all vertical surfaces (all surfaces if physically possible) of the items under-test, with the largest dimension of the TLD being perpendicular to the incident environment. These dosimeters are used to determine the incident environment on the surface of the items under test and can be used to estimate the internal gamma dose environment.

E.5. ELECTRONIC COMPONENT (PIECE PART) CONSIDERATIONS.

a. See Table E-2 for testing requirements on generic part families.

TABLE E-2. INR TESTING REQUIREMENTS FOR GENERIC PART FAMILIES

GENERIC PART FAMILY	GAMMA DOSE RATE TESTING	GAMMA TOTAL DOSE TESTING	NEUTRON FLUENCE TESTING	COMMENTS
1. Diodes	No	No	No	
2. PIN Diodes	Yes	No	No	
3. Temperature Compensated Diodes	No	No	Yes	
4. Zener Diodes	No	No	No	Special Circuit Situations
5. High Fτ (> 50 MHz) Transistors	No	No	No	Special Circuit Situations
6. Low Fτ (< 50 MHz) Transistors	No	No	Yes	
7. Power Transistors	No	No	Yes	
8. Crystals	No	Yes	Yes	
9. Crystals Oscillators	Yes	Yes	Yes	Technology Dependent
10. Operational Amplifiers	No	No	Yes	
11. Comparators	Yes	No	Yes	
12. Complementary Metal Oxide Semiconductor (CMOS) Analog Switches	Yes	Yes	Yes	Certain Manufacturers
13. Fixed Regulators	No	No	Yes	
14. DC to DC Converters	Yes	Yes	Yes	Technology Dependent
15. ADC	Yes	Yes	Yes	Technology Dependent
16. DAC	Yes	Yes	Yes	Technology Dependent
17. JFETs	Yes	No	Yes	
18. MOSFETs	Yes	Yes	Yes	
19. Discrete Timers	No	No	No	

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-2. CONTINUED

GENERIC PART FAMILY	GAMMA DOSE RATE TESTING	GAMMA TOTAL DOSE TESTING	NEUTRON FLUENCE TESTING	COMMENTS
20. Linear Timers	No	No	Yes	
21. SCRs	Yes	No	Yes	
22. Unijunction Transistors	No	No	Yes	
23. Discrete Opto-Electronics	No	No	Yes	
24. Opto-Couplers	No	No	Yes	
25. EE PAL	Yes	Yes	No	Technology Dependent
26. Transistor-Transistor Logic (TTL) PAL	No	No	No	
27. Ultraviolet (UV) PAL	No	Yes	No	
28. EE PROM	Yes	Yes	No	Technology Dependent
29. UV PROM	Yes	No	No	Technology Dependent
30. TTL PROM	No	No	No	
31. N-Channel Metal-Oxide-Semiconductor (NMOS) PROM	No	Yes	No	
32. Static RAMs	Yes	Yes	No	
33. IDT RAMs	Yes	Yes	No	

b. Generic Damage Levels:

(1) Gamma Dose Rate.

(a) Upset < 1E9 cGy(Si)/sec.

(b) Damage < 1E9 cGy(Si)/sec.

(c) Based Upon 20 nsec Pulsewidth.

(2) Gamma Total Dose: < 1600 cGy(Si).

(3) Neutron Fluence: < 1E12 n/cm^2.

E.5.1 Gamma Dose Rate.

a. Environment Induced Effects. The primary effect of the GDR environment is ionization and transient damage of the material that the environment is applied upon. In the case of electronics, this material is some form of semiconductor material, predominantly silicon. The ionization produces the generation of gamma photon induced transient currents called photocurrents. These photocurrents produce secondary effects that include:

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(1) Error generation in logic and analog circuits.

(2) Secondary photocurrents.

(3) Photocurrent induced burnout.

(4) Latch-up.

(5) Memory Device Corruption.

(6) Electronic Microcircuit Damage.

b. As can be seen from this variety of effects, the GDR induced response can be as small as an unnoticed error, or as catastrophic as a completely inoperable system. For electronic device comparison GDR threshold values, see Table E-3.

TABLE E-3. GAMMA DOSE RATE (GDR) CONCERN THRESHOLDS

PIECE-PART TYPE	TECHNOLOGY	GDR UPSET THRESHOLD (cGy(Si)/sec)	GDR LATCHUP / BURNOUT THRESHOLD (cGy(Si)/sec)	COMMENTS
Analog to Digital and Digital to Analog Converters	BICMOS	1E7 – 1E8	1E8 – Above	Extremely sensitive to latchup and burnout / Current limiting and power removal critical to survivability / circuit acceptability / Shall limit power to device safe region
Analog Switches	CMOS	1E7 – 1E8	1E8 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability / Shall limit power to device safe region
Analog Switches	MOS/JFET	1E7 – 5E8	5E8 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Charge Coupled Devices (CCDs)	CCD	5E6 – 5E8	5E8 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Comparators	Bipolar / Linear	1E7 – 5E8	5E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability. Change of state

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-3. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GDR UPSET THRESHOLD (cGy(Si)/sec)	GDR LATCHUP / BURNOUT THRESHOLD (cGy(Si)/sec)	COMMENTS
Comparators	CMOS	1E7 – 5E8	5E8 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability/ Change of state
Crystal Oscillators	Hybrid	1E7 – 1E9	1E9 – Above	Slight effects on frequency / Low Risk @ tactical GDR levels – Concern with control circuitry
Crystals	Linear	1E7 – 1E9	1E9 – Above	Slight effects on frequency / Low Risk @ tactical GDR levels
Darlingtons	Bipolar / Linear	1E7 – 1E9	1E9 – Above	Potential concern @ tactical GDR levels / photocurrent generation will occur. Configuration dependent / Shall limit power to device safe region
DC/DC Converter with Magnetic Feedback	Hybrid	1E7 – 1E9	1E9 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
DC/DC Converters with Optical Feedback	Hybrid	5E6 – 1E9	1E9 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Digital Devices	Current Mode Logic (CML)	1E8 – 1E10	> 1E10	Not concern @ tactical GDR levels / photocurrent generation will occur
Digital Devices	Emitter Coupled Logic (ECL)	5E8 – 1E10	> 1E10	Not concern @ tactical GDR levels / photocurrent generation will occur. Change of state
Digital Devices	Gallium Arsenide	1E8 – 1E10	> 1E10	Not concern @ tactical GDR levels / photocurrent generation will occur. Change of state
Digital Devices	Current Injection Logic (I^2L)	1E8 – 5E9	5E9 – Above	Not concern @ tactical GDR levels / photocurrent generation will occur. Change of state
Digital Devices	TTL	1E7 – 1E9	1E9 – Above	Low Risk concern @ tactical GDR levels / photocurrent generation will occur / Change of state
Digital Potentiometers	CMOS	5E7 – 1E9	1E9 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability / Data loss

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-3. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GDR UPSET THRESHOLD (cGy(Si)/sec)	GDR LATCHUP / BURNOUT THRESHOLD (cGy(Si)/sec)	COMMENTS
Diodes – High Frequency	Bipolar / Linear	1E6 - Above	N/A	Not concern @ tactical GDR levels / photocurrent generation will occur
Diodes – Light Emitting (LEDs)	Bipolar / Linear	1E6 - Above	N/A	Not concern @ tactical GDR levels / photocurrent generation will occur
Diodes – Light Emitting (LEDs)	Gallium Arsenide Variants	1E6 - Above	N/A	Not concern @ tactical GDR levels / photocurrent generation will occur
Diodes – Photo	Bipolar / Linear	1E6 - Above	N/A	Not concern @ tactical GDR levels / photocurrent generation will occur
Diodes – Rectifier	Bipolar / Linear	1E6 - Above	N/A	Not concern @ tactical GDR levels / photocurrent generation will occur
Diodes – Reference / Zener / Suppression	Bipolar / Linear	1E6 - Above	N/A	Not concern @ tactical GDR levels / photocurrent generation will occur
Diodes – Signal	Bipolar / Linear	1E6 - Above	N/A	Not concern @ tactical GDR levels / photocurrent generation will occur
Fully Programmable Gate Array (FPGA)	CMOS	1E7 – 5E8	1E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability. Loss of data
Insulated Gate Bipolar Transistors (IGBTs)	IGBT	1E7 – 1E9	1E9 – Above	Significant photocurrent generation will occur / Circuit Application is critical to acceptability. Configuration dependent
Logic Devices	TTL	1E7 – 1E9	1E9 – Above	Low Risk concern @ tactical GDR levels / photocurrent generation will occur / Change of state
Memory - DRAM	CMOS	5E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability Change of state
Memory – EEPROM & Flash	CMOS	5E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability Change of state

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TABLE E-3. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GDR UPSET THRESHOLD (cGy(Si)/sec)	GDR LATCHUP / BURNOUT THRESHOLD (cGy(Si)/sec)	COMMENTS
Memory – EEPROM & Flash	CMOS	5E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability Change of state
Memory - FIFO	CMOS	5E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability / Change of state
Memory - PROM	TTL	1E7 – 1E9	1E9 – Above	Low Risk concern @ tactical GDR levels / photocurrent generation will occur / Change of state
Memory - ROM	TTL	1E7 – 1E9	1E9 – Above	Low Risk concern @ tactical GDR levels / photocurrent generation will occur / Change of state
Memory – SDRAM	CMOS	5E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability / Change of state
Memory – SRAM	CMOS	5E7 – 5E8	5E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability / Data loss
Micro-Controllers	CMOS	5E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability / Change of state
Micro-Processors	Bipolar	1E7 – 1E9	1E9 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability / Change of state
Micro-Processors	CMOS	5E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability / Change of state
MOSFETs – N-Channel	NMOS	1E7 – 1E9	1E9 – Above	Significant photocurrent generation will occur / Greatest concern – Push / Pull Configuration Burnout

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-3. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GDR UPSET THRESHOLD (cGy(Si)/sec)	GDR LATCHUP / BURNOUT THRESHOLD (cGy(Si)/sec)	COMMENTS
MOSFETs – P-Channel	P-Channel Metal-Oxide-Semiconductor (PMOS)	1E7 – 1E9	1E9 – Above	Significant photocurrent generation will occur / Greatest concern – Push / Pull Configuration Burnout
Multiplexers	CMOS	1E7 – 1E8	1E8 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability / Shall limit power to device safe region
Multiplexers	MOS/JFET	1E7 – 5E8	5E8 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Operational Amplifiers	BICMOS	1E7 – 5E8	5E8 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Operational Amplifiers	BIMOS	1E8 – 1E9	1E9 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Operational Amplifiers	Bipolar / Linear	1E7 – 5E8	5E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Operational Amplifiers	CMOS	1E7 – 5E8	5E8 – Above	Extremely sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Opto-Couplers – Photo-Diodes	Bipolar / Linear	1E6 – 1E10	> 1E10	Not concern @ tactical GDR levels / photocurrent generation will occur
Opto-Couplers – Photo-Transistor / Diode	Bipolar / Linear	1E6 – 1E10	> 1E10	Not concern @ tactical GDR levels / photocurrent generation will occur
Opto-Electronics Solid State Relays	Hybrid	1E6 – 1E9	1E9 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-3. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GDR UPSET THRESHOLD (cGy(Si)/sec)	GDR LATCHUP / BURNOUT THRESHOLD (cGy(Si)/sec)	COMMENTS
Passive Devices	Many	N/A	> 1E13	Not concern @ tactical GDR levels / Applicable to secondary effects, corrective actions and circuit acceptability
Programmable Logic Device – EEPROM	CMOS	1E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability / Change of state
Programmable Logic Device – SRAM	CMOS	1E7 – 5E8	5E8 – Above	Sensitive to latchup, memory corruption and burnout / current limiting and power removal critical to survivability / circuit acceptability / Change of state
Silicon Controlled Rectifiers SCRs/ Thyristor	Bipolar / Linear	N/A	1E6 – 1E9	Significant photocurrent generation will occur due to Four Layer Path / Will turn On due to photocurrent generation / primary Nuclear Event Detection Device
Timers	Bipolar / Linear	1E7 – 5E8	5E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Timers	CMOS	1E7 – 5E8	5E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Transistors GBWP < 50 MHz	Bipolar / Linear	1E7 – 1E9	1E9 – Above	Not concern @ tactical GDR levels / photocurrent generation will occur / Configuration dependent
Transistors GBWP > 50 MHz	Bipolar / Linear	1E7 – 1E9	1E9 – Above	Not concern @ tactical GDR levels / photocurrent generation will occur / Configuration dependent
Transistors - Junction Field Effect (JFET)	FET	1E7 – 1E9	1E9 – Above	Not concern @ tactical GDR levels / photocurrent generation will occur / Configuration dependent
Transistors – Photo	Bipolar / Linear	1E7 – 1E9	1E9 – Above	Not concern @ tactical GDR levels / photocurrent generation will occur / Configuration dependent
Uni-Junction Transistors (UJTs)	Bipolar / Linear	1E7 – 1E9	1E9 – Above	Not concern @ tactical GDR levels / photocurrent generation will occur / Configuration dependent

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-3. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GDR UPSET THRESHOLD (cGy(Si)/sec)	GDR LATCHUP / BURNOUT THRESHOLD (cGy(Si)/sec)	COMMENTS
Voltage References	Bipolar / Linear	1E7 – 1E9	E9 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Voltage References	CMOS	1E7 – 5E8	5E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Voltage Regulators – Linear – Fixed and Adjustable	Bipolar / Linear	1E7 – 1E9	1E9 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Voltage Regulators – Linear – Fixed and Adjustable	CMOS	1E7 – 5E8	5E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Voltage Regulators – Switching	Bipolar / Linear	1E7 – 1E9	1E9 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability
Voltage Regulators – Switching	CMOS	1E7 – 5E8	5E8 – Above	Sensitive to latchup and burnout / current limiting and power removal critical to survivability / circuit acceptability

c. General Protection Methodology. The primary response to the GDR environment is the production of photocurrents. In silicon, the hole/electron pair produced by 1 cGy(Si) is approximately $4.2X10^{13}$ pairs/cm^3. It is apparent from this pair production rate that the GDR environment is capable of producing large photocurrents, particularly when hundreds or thousands of cGy(Si) are involved. The photocurrents generated are limited by the size of the microcircuit active area, the magnitude, pulsewidth and energy of the GDR environment, and the ability of the electronic circuit design to provide paths for these induced currents. Intuitively, the solutions for the reduction in photocurrents are the reduction in:

(1) The Microcircuit's Active Area.

(2) The GDR environment.

(3) The Applicable Circuit's Current Capacity.

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d. Reduction in the microcircuit's active area. This is accomplished by the use of specially made integrated circuits, which use insulator materials as substrates, such as Silicon-On-Sapphire (SOS) or Silicon-On-Insulator (SOI), and some present interest towards Silicon-On-Diamond. The second reduction technique in the GDR environment can be accomplished by the addition of high density material such as lead and is called shielding. However, utilizing shielding as mitigation to the GDR environment is not effective or practical, due to the impact on the system's weight, space claim, and performance degradations. The last solution, reduction in Applicable Circuit's Current Capacity, is accomplished by utilizing one or more of the following four methods:

(1) Resistive Current Limiting.

(2) Inductive current Limiting.

(3) NF Irradiation to reduce/eliminate parasitic paths.

(4) Power Removal (crowbarring (manual) and circumvention (automatic)). (Recommend < 250 μsec to first time constant)

e. Each of these four methods, require different circuit implementation that shall be adequately considered prior to the selection of the mitigation / protection technique. Normally, the primary method utilized in systems that do not have an "Operate Through" requirement is the GDR induced power removal technique (circumvention). The proper integration / implementation of the power removal methodology throughout a system are perhaps the most critical and cost effective of all INR survivability tasks. This integration shall be implemented ASAP in a system design.

E.5.2 Neutron Fluence.

a. NF Environmental Induced Effects. The primary effect of the NF environment is crystal lattice damage is the semiconductor material which the environment is imposed upon. Neutrons can also produce ionization as a secondary effect and produce transient responses. These transient responses are however, short lived, annealing occurs within minutes and is not considered critical to most Army tactical systems. It shall be noted that these transient and displacement effects could become more critical as device layout technology becomes smaller. This effect has been observed in highly integrated CMOS devices such as Synchronous Dynamic Random Access Memories (SDRAMs). The crystal lattice displacement is evidenced in the reduction of microcircuit parameters. These microcircuit parameters are primary influenced by the reduction in minority carrier life time which directly influences the gain of bipolar devices and increased leakage that affects storage time for SDRAMS. This gain reduction in bipolar devices affects other circuit parameters such as output current, input current, and operation. For electric device comparison NF threshold values see Table E-4.

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-4. NEUTRON FLUENCE (NF) CONCERN THRESHOLDS

PIECE-PART TYPE	TECHNOLOGY	NF THRESHOLD RANGE (n/cm^2)	COMMENTS
Analog to Digital and Digital to Analog Converters	BICMOS / CMOS	> 1E12	
Analog Switches	CMOS	> 3E12	
Analog Switches	MOS/JFET	> 3E12	
Charge Coupled Devices (CCDs)	CCD	2E12 - Above	Increase in dark current and Decrease in Charge Transfer Efficiency (CTE)
Comparators	Bipolar / Linear	1E12 - Above	Increase in Bias Currents, Vos and Decrease in Open Loop Gain and Sink Current
Comparators	CMOS	> 5E12	
Crystal Oscillators	Hybrid	> 5E12	Slight effects on frequency / Low Risk @ tactical NF levels – Concern with control circuitry
Crystals	Linear	> 5E12	Slight effects on frequency / Low Risk @ tactical NF levels
Darlingtons	Bipolar / Linear	1E12 - Above	Messenger-Spratt cannot account for actual Darlington NF degradation
DC/DC Converter with Magnetic Feedback	Hybrid	2E12 - Above	Slight Increases / Decreases in Output Voltage
DC/DC Converters with Optical Feedback	Hybrid	5E11 - Above	The optical feedback design is the most sensitive to the NF / Output Voltage goes to Zero or Input Voltage
Digital Devices	CML	> 1E13	Not concern @ tactical NF levels
Digital Devices	ECL	> 1E13	Not concern @ tactical NF levels
Digital Devices	Gallium Arsenide	> 1E13	Not concern @ tactical NF levels
Digital Devices	I2L	> 1E13	Not concern @ tactical NF levels
Digital Devices	TTL	> 1E13	Not concern @ tactical NF levels
Digital Potentiometers	CMOS	1E12 - Above	Device will not re-program due to charge pump failure
Diodes – High Frequency	Bipolar / Linear	> 1E13	Not concern @ tactical NF levels
Diodes – Light Emitting (LEDs)	Bipolar / Linear	5E11 - Above	Substantial Decrease in Light Output / Generally not an issue due to diagnostic circuit application
Diodes – Light Emitting (LEDs)	Gallium Arsenide Variants	8E11 - Above	Substantial Decrease in Light Output
Diodes – Photo	Bipolar / Linear	3E11 - Above	Substantial Decrease in Light Detection/ Increases in Dark Current / Circuit application is critical
Diodes – Rectifier	Bipolar / Linear	> 1E13	Not concern @ tactical NF levels
Diodes – Reference / Zener / Suppression	Bipolar / Linear	5E12 - Above	Decrease in Output Voltage
Diodes – Signal	Bipolar / Linear	> 1E13	Not concern @ tactical NF levels

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-4. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	NF THRESHOLD RANGE (n/cm^2)	COMMENTS
Fully Programmable Gate Array (FPGA)	CMOS	> 1E13	Not concern @ tactical NF levels
Insulated Gate Bipolar Transistors (IGBTs)	IGBT	8E11 - Above	Increase in VCE(Sat)
Logic Devices	TTL	> 1E13	Not concern @ tactical NF levels
Memory – EEPROM & Flash	CMOS	5E11 - Above	Device will not re-program due to charge pump failure , some multi-bit cell technology can lose data
Memory – EPROM	CMOS	> 5E12	Not concern @ tactical NF levels
Memory - FIFO	CMOS	> 1E13	Not concern @ tactical NF levels
Memory - PROM	TTL	> 1E13	Not concern @ tactical NF levels
Memory - ROM	TTL	> 1E13	Not concern @ tactical NF levels
Memory – DRAM, SDRAM, DDR	CMOS	2E11 – Above	Data Corruption through insufficient refresh rate / Missing Bits
Memory – SRAM	CMOS	> 1E13	Not concern @ tactical NF levels
Micro-Controllers	CMOS/BICMOS	> 3E12	
Micro-Processors	Bipolar	> 1E13	Not concern @ tactical NF levels
Micro-Processors	CMOS	> 1E13	Not concern @ tactical NF levels
MOSFETs – N-Channel	NMOS	> 1E13	Not concern @ tactical NF levels
MOSFETs – P-Channel	PMOS	> 1E13	Not concern @ tactical NF levels
Multiplexers	CMOS	> 3E12	
Multiplexers	MOS/JFET	> 3E12	
Operational Amplifiers	BICMOS	1E12 - Above	Increase in Bias Currents, Vos and Decrease in Open Loop Gain
Operational Amplifiers	BIMOS	1E12 - Above	Increase in Bias Currents, Vos and Decrease in Open Loop Gain
Operational Amplifiers	Bipolar / Linear	1E12 - Above	Increase in Bias Currents, Vos and Decrease in Open Loop Gain
Operational Amplifiers	CMOS	> 5E12	
Opto-Couplers – Photo-Diodes	Bipolar / Linear	1E12 - Above	Substantial Decrease in Current Transfer Ratio
Opto-Couplers – Photo-Transistor / Diode	Bipolar / Linear	5E11 – Above	Substantial Decrease in Current Transfer Ratio
Opto-Electronics --Solid State Relays	Hybrid	5E11 – Above	Device will not Turn ON, due to insufficient drive
Passive Devices	Many	> 1E14	Not concern @ tactical NF levels / Applicable to secondary effects, corrective actions and circuit acceptability
Programmable Logic Device – EEPROM	CMOS	1E12 – Above	Device will not re-program due to charge pump failure
Programmable Logic Device – SRAM	CMOS	> 1E13	Not concern @ tactical NF levels

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-4. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	NF THRESHOLD RANGE (n/cm^2)	COMMENTS
Silicon Controlled Rectifiers (SCRs) / Thyristors	Bipolar / Linear	5E11 – Above	Substantial Increase in Holding Current requirement
Timers	Bipolar / Linear	1E12 – Above	Increase in Reset Time / Circuit application is critical
Timers	CMOS	> 1E13	Not concern @ tactical NF levels
Transistors GBWP < 50 MHz	Bipolar / Linear	5E11 - Above	Substantial Decrease in Gain
Transistors GBWP > 50 MHz	Bipolar / Linear	3E12 - Above	Substantial Decrease in Gain
Transistors - Junction Field Effect (JFET)	FET	> 1E13	Not concern @ tactical NF levels
Transistors – Photo	Bipolar / Linear	3E11 - Above	Extremely sensitive to NF degradation
Uni-Junction Transistors (UJTs)	Bipolar / Linear	5E11 - Above	Substantially Increases RB1 and RB2
Voltage References	Bipolar / Linear	7E11 - Above	Significant shift in Output Voltage (Increase or Decrease) / Substantial Increase in Line & Load Regulation
Voltage References	CMOS	> 5E12	
Voltage Regulators – Linear – Fixed and Adjustable	Bipolar / Linear	7E11 - Above	Significant shift in Output Voltage (Increase or Decrease) / Substantial Increase in Line & Load Regulation
Voltage Regulators – Linear – Fixed and Adjustable	CMOS	> 5E12	
Voltage Regulators – Switching	Bipolar / Linear	7E11 - Above	Significant shift in Output Voltage (Increase or Decrease) / Substantial Increase in Line & Load Regulation
Voltage Regulators – Switching	BiMOS / BiCMOS	7E11 - Above	Significant shift in Output Voltage (Increase or Decrease) / Substantial Increase in Line & Load Regulation
Voltage Regulators – Switching	CMOS	> 3E12	

b. General Protection Methodology.

(1) There are no general overall hardening procedures to follow (as there were for the GDR environment), which will reduce the effects of the NF environment. Since the neutron induced degradation is primarily the result of crystal lattice damage, there is little or no difference in the degradation effects if the microcircuit is powered or unpowered. The prime method of improving the NF survivability of microcircuits is through the use of technologies, which are less susceptible and by utilizing devices with higher operating speeds and less active silicon lattice

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volume to be disrupted by the neutrons. A device specification parameter which can be used in the selection of less susceptible bipolar transistors is the Gain Bandwidth Product (GBP or f_t). A generic value of GBP to be used as a cutoff for device selection can be determined by utilizing the Messenger-Spratt transistor degradation equation. The calculation involves determining a maximum generic degradation and then determining the GBP at the required NF. The Messenger-Spratt equation is presented as Equation E-4, including temperature as Equation E-5 and the GBP equation being presented as Equation E-6.

$$\frac{1}{\beta} = \frac{1}{\beta_o} + \frac{K_d \, \Phi_n}{2\pi \, GBP} \qquad \text{Equation E-4}$$

$$\beta_\Phi = \frac{\beta_o}{\left[\frac{T}{298}\right]^{-2.6} + \frac{\beta_o K_d \Phi}{2\pi G_{bwp}}\left[1.1 - 0.1\left(\frac{T}{298}\right)^4\right]} \qquad \text{Equation E-5}$$

$$GBPe = \frac{K_d \phi_n (\beta \times \beta_0)}{2\pi(\beta_0 - \beta)} \qquad \text{Equation E-6}$$

Where:

K_d = $1.35X10^{-6}$ sec/cm^2 for general silicon
GBPe = use β_o=100 and β = 90 for a 10% reduction
β_o = pre-exposure HFE
β = post-exposure HFE
GBP = Gain Bandwidth Product
Φ_n = Neutron fluence
π = Pi = 3.14159
T = temperature degrees Kelvin

(2) The primary method of accounting for NF degradations in microcircuits is to design the electronic circuits so that the overall circuit performance is not degraded. As an example, design the circuits to ensure that the required gain is less than the post-neutron value with an acceptable DM.

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

E.5.3 Gamma Total Dose.

The GTD environment includes the effects of both the gamma and x ray spectrums, with x rays being produced when an excited electron falls to its normal valance shell and emits a photon of energy. This photon is similar to electromagnetic waves which are produced at atomic dimension frequencies. The x ys are produced as an electron passes near an atom and is changed in direction. This change in direction causes the electron to radiate significant energy. The radiated energy is called Bremsstrahlung or braking radiation and consists of high energy x ray photons. For ground based or near surface systems, x rays are not of direct concern because they are absorbed within a few meters of the detonation. For burst at high altitudes where the air density is low, the soft x-rays travel long distances before they are degraded and absorbed. Therefore, these x-rays are a critical factor in survivability of high altitude or space based systems.

a. GTD Environment Induced Effects. The GTD environment is significant in electronic microcircuits which contain highly resistive isolation techniques such as the Metal Oxide Semiconductor (MOS) devices. The GTD environment produces hole-electron pairs that have sufficient energy (especially in the presence of an electric field (i.e., the device powered)) to be separated in and from the highly resistive isolation and not allowed to recombine (hole becomes trapped), resulting in a net positive microcircuit charge. This residual charge results in a shift in the activation and deactivation threshold levels for the microcircuit. If the shift is significant, the microcircuit will no longer respond as required WRT its datasheet specifications. Additionally, this residual charge can result in the change in logic of a memory storage device resulting in false instructions being generated and is normally manifested as once a “1” is written, the location remains a “1” even after a “0” instruction. For electronic device comparison GDR threshold values see Table E-5.

TABLE E-5. GAMMA TOTAL DOSE (GTD) CONCERN THRESHOLDS

PIECE-PART TYPE	TECHNOLOGY	GTD THRESHOLD RANGE (cGy(Si)	COMMENTS
Analog to Digital and Digital to Analog converters	BICMOS / CMOS	5K - above	Shifts
Analog Switches	CMOS	1K – Above	Increased Rds(ON) and leakage currents
Analog Switches	MOS/JFET	2K – Above	Increased Rds(ON) and leakage currents
Charge Coupled Devices (CCDs)	CCD	10K - Above	Not concern @ tactical GTD levels
Comparators	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-5. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GTD THRESHOLD RANGE (cGy(Si)	COMMENTS
Comparators	CMOS	3K - Above	Increase in Vos
Crystal Oscillators	Hybrid	5K - Above	Slight effects on frequency / Low Risk @ tactical GTD levels – Concern with control circuitry
Crystals	Linear	10K - Above	Slight effects on frequency / Low Risk @ tactical GTD levels
Darlingtons	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
DC/DC Converter with Magnetic Feedback	Hybrid	10K - Above	Not concern @ tactical GTD levels
DC/DC Converters with Optical Feedback	Hybrid	3K - Above	The optical feedback design is the most sensitive to the GTD
Digital Devices	CML	10K - Above	Not concern @ tactical GTD levels
Digital Devices	ECL	10K - Above	Not concern @ tactical GTD levels
Digital Devices	Gallium Arsenide	> 20K	Not concern @ tactical GTD levels
Digital Devices	I2L	10K - Above	Not concern @ tactical GTD levels
Digital Devices	TTL	10K - Above	Not concern @ tactical GTD levels
Digital Potentiometers	CMOS	1.5K - Above	The primary concern is the EEPROM input becoming all “1s” / Data corruption and shift in Logic Thresholds
Diodes – High Frequency	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Diodes – Light Emitting (LEDs)	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Diodes – Light Emitting (LEDs)	Gallium Arsenide Variants	> 20K	Not concern @ tactical GTD levels
Diodes – Photo	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Diodes – Rectifier	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Diodes – Reference / Zener / Suppression	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Diodes – Signal	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Fully Programmable Gate Array (FPGA)	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
Insulated Gate Bipolar Transistors (IGBTs)	IGBT	2K - Above	Increase in VCE(Sat)
Logic Devices	TTL	10K - Above	Not concern @ tactical GTD levels
Memory - DRAM	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
Memory – EEPROM & Flash	CMOS	1.5K - Above	The primary concern is the EEPROM input becoming all “1s” / Data corruption and shift in Logic Thresholds / Data Loss can occur @ 8K Rads(Si)

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TABLE E-5. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GTD THRESHOLD RANGE (cGy(Si)	COMMENTS
Memory - EPROM	CMOS	1.5K - Above	The primary concern is the EEPROM input becoming all "1s" / Data corruption and shift in Logic Thresholds / Data Loss can occur @ 8K Rads(Si)
Memory - FIFO	CMOS	5K – Above	Data corruption and shift in Logic Thresholds
Memory - PROM	TTL	10K - Above	Not concern @ tactical GTD levels
Memory - ROM	TTL	10K - Above	Not concern @ tactical GTD levels
Memory – SDRAM	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
Memory – SRAM	CMOS	2K – Above	Data corruption and shift in Logic Thresholds
Micro-Controllers	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
Micro-Processors	Bipolar	10K - Above	Not concern @ tactical GTD levels
Micro-Processors	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
MOSFETs – N-Channel	NMOS	1.5K - Above	Decrease in Vgs(th) and increase in Rds(ON)
MOSFETs – P-Channel	PMOS	1.5K - Above	Increase in Vgs(th) and increase in Rds(ON)
Multiplexers	CMOS	1K – Above	Increased Rds(ON) and leakage currents
Multiplexers	MOS/JFET	2K – Above	Increased Rds(ON) and leakage currents
Diodes – Light Emitting (LEDs)	Gallium Arsenide Variants	> 20K	Not concern @ tactical GTD levels
Diodes – Photo	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Diodes – Rectifier	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Diodes – Reference / Zener / Suppression	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Diodes – Signal	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Fully Programmable Gate Array (FPGA)	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
Insulated Gate Bipolar Transistors (IGBTs)	IGBT	2K - Above	Increase in VCE(Sat)
Logic Devices	TTL	10K - Above	Not concern @ tactical GTD levels
Memory - DRAM	CMOS	3K – Above	Data corruption and shift in Logic Thresholds

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TABLE E-5. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GTD THRESHOLD RANGE (cGy(Si)	COMMENTS
Memory – EEPROM & Flash	CMOS	1.5K - Above	The primary concern is the EEPROM input becoming all "1s" / Data corruption and shift in Logic Thresholds / Data Loss can occur @ 8K Rads(Si)
Memory - EPROM	CMOS	1.5K - Above	The primary concern is the EEPROM input becoming all "1s" / Data corruption and shift in Logic Thresholds / Data Loss can occur @ 8K Rads(Si)
Memory - FIFO	CMOS	5K – Above	Data corruption and shift in Logic Thresholds
Memory - PROM	TTL	10K - Above	Not concern @ tactical GTD levels
Memory - ROM	TTL	10K - Above	Not concern @ tactical GTD levels
Memory – SDRAM	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
Memory – SRAM	CMOS	2K – Above	Data corruption and shift in Logic Thresholds
Micro-Controllers	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
Micro-Processors	Bipolar	10K - Above	Not concern @ tactical GTD levels
Micro-Processors	CMOS	3K – Above	Data corruption and shift in Logic Thresholds
MOSFETs – N-Channel	NMOS	1.5K - Above	Decrease in Vgs(th) and increase in Rds(ON)
MOSFETs – P-Channel	PMOS	1.5K - Above	Increase in Vgs(th) and increase in Rds(ON)
Multiplexers	CMOS	1K – Above	Increased Rds(ON) and leakage currents
Multiplexers	MOS/JFET	2K – Above	Increased Rds(ON) and leakage currents
Operational Amplifiers	BICMOS	3K – Above	Increase in Vos and decrease in Large Signal Voltage Gain
Operational Amplifiers	BIMOS	3K – Above	Increase in Vos and decrease in Large Signal Voltage Gain
Operational Amplifiers	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Operational Amplifiers	CMOS	3K – Above	Increase in Vos and decrease in Large Signal Voltage Gain
Opto-Couplers – Photo-Diodes	Bipolar / Linear	5K – Above	Decrease in transmissivity of material between diodes
Opto-Couplers – Photo-Transistor / Diode	Bipolar / Linear	5K – Above	Decrease in transmissivity of material between diodes / transistor pair
Opto-Electronics --Solid State Relays	Hybrid	2K – Above	Device will not Turn ON, due to threshold shifts

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-5. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GTD THRESHOLD RANGE (cGy(Si)	COMMENTS
Passive Devices	Many	> 20K	Not concern @ tactical GTD levels / Applicable to secondary effects, corrective actions and circuit acceptability
Programmable Logic Device – EEPROM	CMOS	1.5K - Above	The primary concern is the EEPROM input becoming all "1s" / Data corruption and shift in Logic Thresholds
Programmable Logic Device – SRAM	CMOS	2K – Above	Data corruption and shift in Logic Thresholds
Silicon Controlled Rectifiers (SCRs) / Thyristors	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Timers	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Timers	CMOS	2K – Above	Changes in Logic Thresholds
Transistors GBWP < 50 MHz	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Transistors GBWP > 50 MHz	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Transistors - Junction Field Effect (JFET)	FET	> 20K	Not concern @ tactical GTD levels
Transistors – Photo	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Uni-Junction Transistors (UJTs)	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Voltage References	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Voltage References	CMOS	2K - Above	The primary concern is charge pumps internal to the reference
Voltage Regulators – Linear – Fixed and Adjustable	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Operational Amplifiers	BICMOS	3K – Above	Increase in Vos and decrease in Large Signal Voltage Gain
Operational Amplifiers	BIMOS	3K – Above	Increase in Vos and decrease in Large Signal Voltage Gain
Operational Amplifiers	Bipolar / Linear	> 20K	Not concern @ tactical GTD levels
Operational Amplifiers	CMOS	3K – Above	Increase in Vos and decrease in Large Signal Voltage Gain
Opto-Couplers – Photo-Diodes	Bipolar / Linear	5K – Above	Decrease in transmissivity of material between diodes
Opto-Couplers – Photo-Transistor / Diode	Bipolar / Linear	5K – Above	Decrease in transmissivity of material between diodes / transistor pair
Opto-Electronics --Solid State Relays	Hybrid	2K – Above	Device will not Turn ON, due to threshold shifts

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APPENDIX E. NUCLEAR RADIATION ENVIRONMENT AND EFFECTS.

TABLE E-5. CONTINUED

PIECE-PART TYPE	TECHNOLOGY	GTD THRESHOLD RANGE (cGy(Si)	COMMENTS
Passive Devices	Many	> 20K	Not concern @ tactical GTD levels / Applicable to secondary effects, corrective actions and circuit acceptability
Programmable Logic Device – EEPROM	CMOS	1.5K - Above	The primary concern is the EEPROM input becoming all "1s" / Data corruption and shift in Logic Thresholds
Programmable Logic Device – SRAM	CMOS	2K – Above	Data corruption and shift in Logic Thresholds
Silicon Controlled Rectifiers (SCRs) / Thyristors	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Timers	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Timers	CMOS	2K – Above	Changes in Logic Thresholds
Transistors GBWP < 50 MHz	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Transistors GBWP > 50 MHz	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Transistors - Junction Field Effect (JFET)	FET	> 20K	Not concern @ tactical GTD levels
Transistors – Photo	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Uni-Junction Transistors (UJTs)	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Voltage References	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Voltage References	CMOS	2K - Above	The primary concern is charge pumps internal to the reference
Voltage Regulators – Linear – Fixed and Adjustable	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Voltage Regulators – Linear – Fixed and Adjustable	CMOS	3K - Above	
Voltage Regulators – Switching	Bipolar / Linear	10K - Above	Not concern @ tactical GTD levels
Voltage Regulators – Switching	BiMOS / BiCMOS/ CMOS	2K - Above	

b. General Protection Methodology. There are no general overall hardening GTD procedures to follow (as there were for the GDR environment) that will reduce the effects of the GTD environment. Based upon experience and historical test data, it has been observed in many microcircuits that the exposure of the electronics while in an unpowered / unbiased state will considerably improve their GTD survivability threshold since the freed electrons are not all

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removed from the insulator and can recombine. This implies that very fast responding GDR induce power removal is one method for improving microcircuit GTD survivability. However, the primary method of improving GTD susceptibility threshold is through the use of technology and device selection processes. The avoidance of NMOS microcircuits is highly recommended since their response has proven to be the worst of all technologies. This is normally not difficult, since pure NMOS devices are rare in today's technology, which is predominantly CMOS technology. As with all testing, the GTD survivability of an electronic assembly or circuit is determined by the weakest building block, which is normally at the device and circuit design level. It is critical to the qualification testing that power removal is taken into account when determining survivability. As shown in Figure E-3, electronic assemblies and devices received dose shall account for unbiased exposure based upon a reasonable power removal time. As an example, if an assembly can remove power within 250 µsecs, the electronic item is exposed in a biased condition to 50% and in an unbiased condition for the remaining 50% of the GTD exposure.

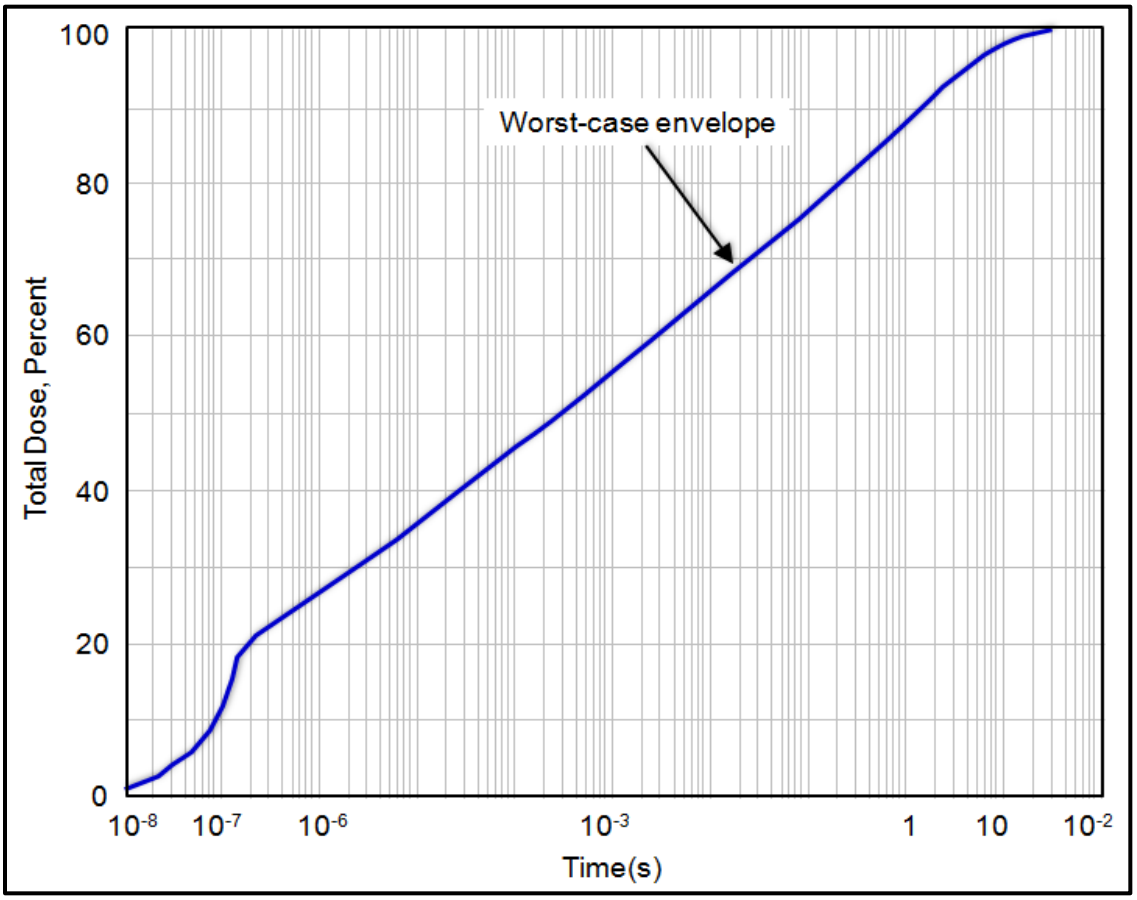


Figure E-3. GTD percentage applied based upon power removal.

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APPENDIX F. NUCLEAR RADIATION PROTECTION FACTOR.

Provided are general tasks for accomplishing radiation protection factor measurements.

F.1. TASK 1: DETERMINE THE POWER LEVEL CALIBRATION.

F.1.1 Subtask 1.1: Determine the Neutron Spectrum at a Well Defined Location.

F.1.1.1 Measurement.

a. Activation foils at 24 inches from core centerline at core mid-plane - will require at least one full day of experimental time. Reactor operation should be at 5 kW for a minimum of 40 minutes. Data collection from the activation experiment requires a minimum of one month post-exposure decay for accuracy. Data analysis may require one to two additional months.

b. Select a set of reference power level monitors (recommend both S-32 activation monitors and Ni-58) to be placed at the location of the power channel instrumentation. A similar set of monitors will be placed at the 24 inch reference location.

F.1.1.2 Calculation.

Simulate the experiment with a relatively detailed model of the close-in geometry. Geometry will need to be extended to 3 - 5 mean free paths to adequately evaluate the near-field environment. The calculation provides a neutron source description normalized to the absolute number of fission neutrons inside the core. The neutrons are transported by Monte Carlo methods to the experimental positions. The distribution will be calculated at the reference 24 inch position for the activation foils as well as the power level monitor positions.

F.1.2 Subtask 1.2: Determine the Power Level Normalization.

F.1.2.1 Measurement.

Determine the number of neutrons for a defined portion of the neutron energy spectrum at the reference position as a function of indicated reactor power. Establish the appropriate reference monitor relationship between the reference position and the power channel location.

F.1.2.2 Calculation.

From the high fidelity Monte Carlo calculation, Section F.1.1.2, estimate the number of neutrons arriving at the power channel location per fission occurring in the reactor core. This provides the power channel response per kW.

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APPENDIX F. NUCLEAR RADIATION PROTECTION FACTOR.

F.2 TASK 2: DETERMINE THE RELAXATION LENGTH FOR THE FAST NEUTRON COMPONENT (< 3 Mev), AND BUILDUP OF THERMAL NEUTRON AND GAMMA COMPONENT (DEPARTURE FROM $1/r^2$).

F.2.1 Measurement.

Requires one 20 kW-hr (presumes the power level calibration from Task 1 is complete or can be estimated) operation. The data are for direct comparison with the previous test. Sulfur activation monitors at 10 meter intervals to 170 meters, Au activation monitors at ~20 meter intervals to 170 meters, and TLDs at 10 meter intervals to 170 meters. If conditions permit, a 50 kW-hr operation would be more desirable and yield improved data.

F.2.2 Calculation.

Simulate the experiment with Monte Carlo methods. The preferred method is to use a fixed source term (which could be generated during step F.1.1.2) and detailed modeling to approximately 300 meters (within the full extent of at least 3000 meter geometry). This will allow close comparison between calculated and measured data at 170 meters. This approach is expensive in terms of computer time.

F.3 TASK 3: DETERMINE THE FREE FIELD ENVIRONMENT AT PRE-SELECTED POINTS OF INTEREST.

F.3.1 Subtask 3.1: Neutron Spectrum.

a. Instrumentation needed include: rotating spectrometer, 4 proton recoil chambers, and data logger/PC.

b. The proposed locations are 170 meters (for comparison with previous measured data), 400 meters (the range of the previously designated NATO Standard Position at APRF) and 635 meters (current Safety exclusion zone fence at WSMR's FBR). At each position, 120 V AC is required, weather protection, and accurately surveyed positions. Measurements can be made at one meter height.

c. For each position:

(1) Need minimum of three good independent measurements at each position for a total of twelve operations, plus false starts and aborts.

(2) Proton Recoil System – Determines the neutron spectrum from 50 keV to approximately 4 MeV.

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APPENDIX F. NUCLEAR RADIATION PROTECTION FACTOR.

(3) Need a best estimate of the entire neutron spectrum to account for the portion of the spectrum the proton recoil system does not detect. The 1980's vintage calculation will be used until the calculation is updated.

(4) Project order-of-magnitude doses, power levels and run times needed, based on previous experience.

F.3.2 Subtask 3.2: Measure Neutron Dose at Each Position.

a. Instrumentation needed include: rotating spectrometer, 4 proton recoil chambers, data logger/PC, and Eberline NRD Remmeters.

b. The proton recoil system will provide absorbed dose and ambient dose equivalent for each spectrum it measures. However, the system is only responsive to the portion of the neutron spectrum between 50 keV and 4 MeV.

c. Additionally, the neutron absorbed dose at each position can be estimated by folding the neutron spectrum determined by the proton recoil system, supplemented by the old DOT-III transport calculations, with the KERR neutron response function. This would be a first order adjustment to improve the absorbed dose estimate.

d. An independent estimate of the neutron dose can be obtained with NRD Remmeters. Accuracy will be somewhat dependent on the distance from the source and the deviation of the neutron spectrum from the ^{252}Cf spectrum use for calibration. Same cautions as described in the preceding section.

e. A second independent estimate can be obtained from evaluating Subtasks 3.3 and 3.4.

F.3.3 Subtask 3.3: Measure Total Dose at Each Position.

a. Need minimum of three good independent measurements at each position for a total of twelve operations, plus false starts and aborts. Instrumentation needed includes:

(1) Far West Tissue Equivalent Chambers – 3 instruments available (1 cc, 80 cc, and1000 cc) but operate one at-a-time with the calibrated electrometer.

(2) Keithly 6514 electrometer.

(3) Data recorder/logger or PC

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APPENDIX F. NUCLEAR RADIATION PROTECTION FACTOR.

b. Using Far West Tissue Equivalent chambers, measure the total neutron plus gamma dose at each selected position. Run times and power levels are probably somewhere intermediate between the more sensitive NRD Remmeters and the proton recoil system. With some trial and error during the proton recoil measurements, by the tissue equivalent system can be fielded. However, this could extend the time required for experimental setup by trying to dial-in two separate instruments at the same time. Also, an increase the number of false starts and repeats because of out-of range selections is likely.

F.3.4 Subtask 3.4: Measure Gamma Dose at Each Position.

a. Instrumentation needed includes:

(1) Far West Ionization Chambers - 2 instruments available (2 cc graphite and GM-1 counter) operating singly with the Keithly 6514 Electrometer.

(2) Keithly 6514 Electrometer.

(3) Possibly could use a Fluke/Cardinal Health set of ion chambers also as an independent measurement system.

(4) Can also use various hand held HP instruments.

b. Using Far West ionization chambers, the gamma only dose can be measured with only small interferences from the neutron contribution. Run time should be similar to Subtask 3.3; but, cannot do both at the same time because both uses the same power supply, electrometer and data logger.

F.4. TASK 4: SHIELDING OR REDUCTION OR PROTECTION FACTOR ESTIMATES.

a. Definitions.

(1) Protection Factor (PF):

$PF = D_o/D_i$

D_o = Free-field Radiation Dose

D_i = Dose due to radiation penetrating the shield

(2) Neutron Protection Factor (NPF):

$NPF = D_{no}/(D_{ni} + D_{gn})$

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APPENDIX F. NUCLEAR RADIATION PROTECTION FACTOR.

D_{no} = Free Field (1m height) Neutron Dose

D_{ni} = Shielded Neutron Dose

*D_{gn} = Secondary Gamma Dose

(3) Neutron Reduction Factor (NRF):

NRF = D_{no}/D_{ni}

D_{no} = Free Field (1m height) Neutron Dose

D_{ni} = Shielded Neutron Dose

*D_{gn} = Secondary Gamma Dose

(4) Gamma Protection Factor (GPF):

GPF = D_{go}/D_{gg}

D_{go} = Free Field (1m height) Gamma Dose

D_{gg} = Dose due to primary gammas penetrating shield

b. Thus, to completely characterize each test position as described in AEP-14, we need:

(1) Neutron plus Gamma Source (Reactor):

(a) D_o proton recoil dose plus graphite or GM1 ion chamber dose outside and/or total dose from tissue equivalent chambers.

(b) D_i proton recoil dose + graphite or GM1 ion chamber dose inside and/or from tissue equivalent chambers.

(c) D_{no} proton recoil dose outside and/or Eberline NRD dose.

(d) D_{ni} proton recoil dose inside and/or Eberline NRD dose.

(e) D_{go} ion chamber dose free field (1 m height).

**D_{gn} = Secondary Gamma Dose is a derived quantity from total gamma dose measurements made inside the shield and subtracting the dose component due to direct penetration of the dose from the outside. It is not a direct measurement.*

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APPENDIX F. NUCLEAR RADIATION PROTECTION FACTOR.

(f) (D_{gg} + *D_{gn}) shielded ion chamber dose.

(2) Pure Gamma Source (Gamma Range):

(a) D_{go} Ion Chamber Dose Free Field (1m height).

(b) D_{gg} Ion Chamber Dose due to primary gammas penetrating shield.

F.4.1 Task 4.1: Measure the Shielding/Reduction Factors for the NATO 2 meter Box at 400 Meters.

This follows the same methods outlined above and provides confidence in measured values inside a shielded structure. The test will require free field measurements of all quantities outside the box and then again in the same environmental conditions inside the box. The evaluation is compared with a direct calculation of the shielding geometry. Pure gamma protection factors can be obtained at the gamma range facility. The value of this test is that this is likely to be the only configuration in which measurement and calculation can be easily compared. The simplified geometry of the box eliminates the complexity of the test item where geometry is complex and/or material composition is not available. If this path is taken then there is also a separate subtask to evaluate the environment inside the box with Monte Carlo methods, discrete ordinates methods, or combinations of both.

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

G.1. HIGH ALTITUDE ELECTROMAGNETIC PULSE (HEMP).

G.1.1 Objectives.

a. To assess the survivability of the M1A2 V2 when exposed to the HEMP environment specified in MIL-STD-2169B, and using the MIL-STD-464C E-Field parameter.

b. To update the M1 Abrams LCN & ES database and identify the baseline configuration of the M1A2 V2 for the Life-Cycle/management and control as specified in AR 70-75, and MIL-STDs 2169B and 464C. This will be accomplished by entering into the Life-Cycle database pertinent data, results, and information from this HEMP test.

G.1.2 Criteria.

G.1.2.1 HEMP Levels.

The M1A2 V2 shall perform all its mission essential operational performance functions following exposure to the HEMP environment specified in SA-SA00001D. The tank shall remain combat effective following HEMP illumination without component replacement. The tank shall be subjected to Early time (E1) HEMP waveform defined in MIL-STD-2169B. The M1 A2V2 Abrams does not have an operate-through requirement; instead, it is allowed to experience upsets that can be re-set by the crew to achieve full operational capability within the allowable downtime of seven minutes after the HEMP event. The HEMP criteria levels for the M1A2 V2 are:

a. E-field = Omitted [volts/meter]

b. H-field = Omitted [amp-turns/meter]

c. Rise Time = Omitted [seconds]

G.1.2.2 Omission.

The two E1 HEMP timing levels are extracted from the MIL-STD-2169B and SA-SA00001D, which are classified SECRET. The E1 HEMP timing criterion levels are, therefore, omitted from this document in order to maintain its UNCLASSIFIED status. The E1 HEMP peak E-Field of 50 kV/m is extracted from MIL-STD-464C. The E1 HEMP criteria of MIL-STD-2169B are available by contacting TEDT-WS-V, WSMR, or by obtaining a copy from the U.S. Army Nuclear and Combating Weapons of Mass Destruction Agency (USANCA). The E1 HEMP criterion levels will be provided in the classified Detailed Test Report (DPT).

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

G.1.2.3 HEMP Description.

a. The M1A2 V2 will only be subjected to the Early-Time Waveform of the HEMP environment produced by an exo-atmospheric nuclear detonation. The HEMP environment has two additional waveform components; E2 for Intermediate-Time, and E3 for Late-Time. These two waveforms will not be considered for the M1A2 SEP V2 because they are applicable for systems connected by very long cables (E2) or to systems connected to the power grid or communications lines (E3).

b. The E2 and E3 HEMP waveforms will contribute some energy coupling to the M1A2 V2; but, the amount of energy coupled by these two waveforms to this discrete system will be insignificant relative to E1. Also, energy coupled by these two waveforms will not be additive since they will be out of time phase with each other as well as with E1.

G.1.2.4 LCN&ES.

IAW AR 70-75, MIL-STDs 2169B and 464C, a life-cycle program shall be established and implemented for mission critical systems such as the M1A2 V2 Tank. In addition, the M1A2 V2 Tank's operational performance requirements shall be met throughout its rated life cycle. The production, operation, maturity, maintenance, storage, upgrades, enhancements, ambient environment, and DMS solutions and technology insertions, shall not introduce any HEMP susceptibilities or unacceptable levels of degradation into the M1A2 V2 Abrams tank.

G.2. TEST PROCEDURES.

G.2.1.1 General Procedures.

The HEMP survivability program for the M1A2 V2 will include testing at Survivability, Vulnerability and Assessment Directorate (SVAD) HEMP simulator, the Advanced Fast EMP. The survivability of the M1A2 V2 to its HEMP criteria level will be assessed by:

a. Performing a pre-test energy coupling analysis.

b. Establishing its complete performance baseline prior to HEMP testing, using baseline self-test checks and diagnostic tests.

c. Performing detailed bulk current measurements on cables identified in the pre-test analysis (external cables, POE, and internal cables).

d. Testing the system-under-test (SUT) in different configurations. The M1A2 V2 will be tested in two hull orientations with respect to the E field vector (i.e., longitudinal axis parallel to E field vector, and perpendicular to the E field vector). For each hull orientation, the turret will be

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tested in two positions: 1) gun over the drivers hatch, and 2) gun over the curbside. In each of these four hull-turret orientation-positions (configurations), the SUT will be illuminated by a series of HEMP pulses. At each E-Field level for each test configurations, if no failures occur, the SUT will be illuminated, at a minimum, twice more or until all data acquisition has been completed.

e. Illuminating the SUT in four configurations to 75%, 100% and 120% of its E-Field criterion level.

f. Illuminating the SUT in a fully operational mode (i.e., the tank will be fully operational with engine in TAC idle), fire control in the NORMAL mode, and communications, etc., powered ON. The final drive will be disconnected as deemed necessary. All hatches except the drivers will be opened during exposures.

g. Repeating the necessary pre-test baseline checks on the V2 SUT after each illumination. Illuminating the SUT multiple times. The number of test pulses performed will depend on how many tank harness shields are monitored for HEMP induced currents. It is planned to measure all accessible cables; no physical changes will be made to access data points. Unacceptable effects will be investigated to quantify, determine the cause, and identify fixes.

h. Diagnosing all effects. Most HEMP responses are manifested as system upsets. In the event of an upset, the system power will be cycled to determine if normal operation can be restored. If normal operation is restored, the illumination will be repeated several times to verify the effect. If system operation is not restored, further investigation will be performed to determine the affected LRU.

i. Documenting upsets, failures, downtime, and corrective actions; most problems induced will be transient upsets and will be correctable by cycling power OFF/ON.

j. Identifying and classifying all failures to the electronic devices/component level.

G.2.1.2 Test Equipment.

The following test equipment is scheduled for this test.

a. Simulator: AFEMP Free field HEMP simulator and antenna.

b. Data transmitter Links: Nanofast® OP300 fiber-optic.

c. Inductive current probes: EATON® 91550-2.

d. Environment reference probe: MGL-2 D-dot free field probe.

e. Environment monitoring probe: EG&G ADC-4 Free-Field probe.

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f. Environment data recording device: Tektronix® TDS 7154B Digital Phosphor Oscilloscopes.

G.2.1.3 Test Facility.

The HEMP testing performed on the M1A2 V2 SUT will utilize the SVAD AFEMP facility, which generates the E1-type HEMP waveform defined in MIL-STD-2169B. The AFEMP provides a horizontally polarized EM Environment (EME), which is ideal for a horizontally coupling system like the M1A2 V2 Tank.

G.2.1.4 Pre-Test Analysis.

A pre-test analysis will be conducted to:

a. Evaluate and incorporate pertinent test data and results from previous tests.

b. Analyze drawings and circuits to identify potentially harmful energy paths.

c. Identify test system's internal configuration.

d. Identify and determine all energy coupling POEs.

e. Analyze grounding schemes and identify EM barrier features such as shielded cables to include connectors / backshells.

f. Evaluate deliberate hardening devices and techniques.

g. Define DAS requirements.

h. Identify cables for measurements.

i. Identify test levels, orientations, configurations, and operational modes based on the results of the hardening determination.

G.2.1.5 System Setup.

a. Prior to testing, the SUT will be functionally checked to ensure proper operation. Problems will be documented, reported, and corrected if detrimental to the HEMP survivability assessment program on the M1A2 V2.

b. The SUT in a pre-test analysis selected configuration will be positioned in the HEMP facility near the center of the test volume for the desired test level (defined by the peak E-Field). The SUT will then be powered and a functional check performed.

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c. A bulk current probe will be placed near (but not attached to) the longest cable length in the SUT, and an EM noise measurement taken to establish the data collection base. This type of base-line measurement will be made for all of the current probes.

d. Bulk current measurements will be obtained on cables identified in the pre-test analysis as being potential paths for harmful levels of HEMP induced energy. Pin current measurements using a BOB will only be collected if a failure is identified. The pin current data will be utilized in a failure analysis and/or to perform corrective actions.

G.2.1.6 System Test.

The SUT will be illuminated by a transverse electromagnetic wave whose E-Field magnitude is approximately 75% of the E1 HEMP criterion value. After illumination, the SUT will again be checked to establish the functional status of the system. At the completion of each successful test series (all cables measured), the SUT will be changed to account for energy coupling into different cable layouts and functions in the system. Once the series of four hull-turret orientation-positions and modes have been completed, then the E-Field magnitude will be increased to the next E-Field level and the test procedures repeated. These procedures will be repeated for the third E-Field test level.

G.2.1.7 Effects Procedure.

a. If an effect / anomaly occur, it will be documented and diagnosed. Testing will not be continued until the problem is understood, and its effect on the SUT has been assessed as well as potential impacts on the SUT results if testing is continued. If an upset occurs, the SUT's power will be cycled OFF/ON. If the SUT fully recovers, testing will be repeated at the same level and test orientation to determine whether the problem was EME induced or an anomaly. Borderline cases may require an additional test exposure or CI testing to explicitly establish whether the effect was environmentally induced.

b. If the SUT does not recover, then follow-up checks, measurement review, and review of the pre-test analysis will be used to identify the energy path and the affected electronic devices/component. If the effect is a failure, diagnostic checks will be performed to determine energy path and victim(s). If the operational status of the SUT can be restored, an engineering judgment will be made of potential risk to the SUT if testing is continued. Every effort will be made to complete testing.

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

G.2.1.8 Environment Measurements.

Measurements of each illumination will be made using an Electric Flux Density per unit time (D-dot) probes, so that the magnitude of the E-Field and pulse shape can be determined. This information will be digitized, reviewed, and stored for later environment compliance analysis. Injected current signals will be measured using a calibrated bulk current probe, reviewed and then stored for later stress level compliance analysis and upset / problem evaluations.

G.2.2 Data Required.

a. Detailed description of each HEMP environment to include photographs of the test facility setup showing SUT position relative to the HEMP antenna array.

b. Complete set of pre-test mapping data of each HEMP illumination level with the E-Field expressed in Volts/meter (V/m) (±5%), rise time and pulse width expressed in nanoseconds (nsec) (±5%), frequency expressed in Hertz (Hz)(±5%), and Magnetic Field (H-Field) amplitude expressed in Amp-turns/meter (±5%).

c. Detailed description of the M1A2 V2 functional checks used to baseline the SUT to determine its post-illumination capabilities.

d. Visual inspection, logs, test conductor notes, and photographs.

e. Detailed description and serial numbers of the M1A2 V2 subsystems.

f. Detailed description and recording of all inspections, downtime and recovery time (sec) (±10%), and checkout data.

g. Log of SUT baseline checks from the Operator manuals (i.e., self-test), and as needed, diagnostic tests with descriptions of discrepancies.

h. Tank physical and operating configuration during each subtest illumination.

i. Log sheets of test illumination to include induced upsets or failures.

j. Description and calibration of current / voltage measuring probes and the DAS. In addition, description of all probe locations is required. Results of all facility environment measurements expressed in the same units as listed in Paragraph 2.6.4.b.

k. Results of all current and voltage measurements, and Fast Fourier Transforms data obtained from the DAS.

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

l. Results of previous HEMP and CI tests performed by the contractor or another government agency on the M1A2 V2 or Abrams series tank.

m. Detailed description of all deliberate HEMP hardening devices and/or techniques employed on the M1A2 V2.

n. Percent error incorporated into the DAS.

o. Calibration dates for all test equipments.

p. TIRs, if applicable.

G.2.3 Data Analysis / Procedure.

G.2.3.1 Data.

The pre-test analysis will consist of evaluating results from the previous M1A2 V2 HEMP program and reviewing the test system's configuration. Pertinent data, results and information will be incorporated into the test planning of the SVAD HEMP program for the M1A2 V2. The incorporation of all test data will be used to enhance and reduce the scope of testing. Pertinent data will be included in the SVAD failure diagnostics, post-test analysis / assessment, and documented in the detailed test report to support the test results.

G.2.3.2 Criteria Compliance.

The HEMP environmental data from the AFEMP facility will be corrected to account for the percent error associated with the DAS:

a. A mean and standard deviation will be established from the error corrected AFEMP E-Field parameters.

b. The H-Field parameter will be derived by dividing this mean error corrected AFEMP E-Field parameter by 377 ohms.

c. A mean and standard deviation rise time will be established from the AFEMP E-Field data.

d. The AFEMP E-Field data, test point current data, and test point FFTs will be examined using MATLAB to determine the primary coupling frequency or coupling frequency range, critical damping factor, and energy content.

e. The data in Paragraphs 2.6.5.2a through 2.6.5.2d will be compared and evaluated against the MIL-STD-2169B E1 HEMP criteria to determine criteria compliance.

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

G.2.3.3 System Configuration Compliance.

The test system configuration will be evaluated against the expected production configuration and all differences will be identified and documented. Differences that could impact the results will be discussed in the final report. The existing baseline configuration will be updated and documented.

G.2.3.4 Effects Analysis.

Effects will be scored at the test level of occurrence. Cause(s) and victim(s) will be identified, and impact(s) on the M1A2 V2 mission will be discussed. Failures or operational performance degradation occurring at levels above criteria will be classified as system shortcomings, unless verified by additional data and/or energy coupling analysis to be valid as a result of manufacturing variations or assembly. This information will be used primarily to provide the needed level of confidence in the survivability assessment of the M1A2 V2 to meet its defined HEMP criteria.

G.2.3.5 System Performance.

Comparison of pre- and post-illumination functional checkout data for the M1A2 V2 test system will be used to determine the effects of the HEMP test environment on the M1A2 V2. Degradation resulting in system performance outside specifications, or total failure(s), will be addressed with regards to cause(s), victim(s), test level at which they occurred, allowable downtime, and mission impact.

G.2.3.6 Survivability Assessment.

A HEMP survivability assessment will be performed on the production or baseline configuration against the HEMP criteria using the results of Paragraphs 2.6.5.2 through 2.6.5.5. This assessment may produce results different than obtained during the testing due to corrections for manufacturing variations and/or test environment deficiencies.

G.2.3.7 LCN & ES.

Both, the configuration for the test system and proposed baseline production system will be archived into the M1A2 V2 Life-Cycle program database along with pertinent test data and results, extrapolated results, and information from this HEMP subtest. This database will enable implementation of the Life-Cycle program of Hardness and Sustainment Assurance, and Surveillance tests.

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

G.3. GAMMA DASE RATE (GDR).

G.3.1 Objectives.

a. To assess the survivability of the M1A2 V2 to the effects of the GDR environment as specified in SA-SA00001D and USANCA NHC documents.

b. To update the M1A2 V2 LCN&ES program GDR database by inputting pertinent data, results and information resulting from this subtest.

G.3.2 Criteria.

G.3.2.1 GDR Test Levels.

The M1A2 V2 shall perform all its mission essential operational performance functions following exposure to the GDR environment specified in SA-SA00001D and the USANCA criteria (same level). The M1A2 V2 shall remain combat effective without component replacement. The M1A2 V2 does not have an "operate through" requirement (DECU shall circumvent and regain control of engine) and recovery of mission essential functions by crew within seven minutes is permissible. The M1A2 V2 will be exposed to 100% and 150% of the M1A2 GDR criterion levels. The GDR criterion levels for the M1A2 V2 are:

Peak Gamma Dose Rate = Omitted [cGy(Si)/sec] (SA-SA0001D and USANCA)

G.3.2.2 Omission.

The GDR test levels are extracted from the SA-SA00001D document and USANCA documents. The GDR survivability levels are classified CONFIDENTIAL and the GDR criterion level is, therefore, omitted from this document in order to maintain its UNCLASSIFIED status. The criterion is available by contacting TEDT-WS-V, WSMR. The GDR criterion level will be stated in the classified Detailed Test Report (DTR).

G.3.2.3 LCN & ES.

IAW AR 70-75 and the DODD 3180.9, a life-cycle program shall be established and implemented for mission critical systems such as the M1A2 V2 Abrams Tank. The production, operation, maturity, maintenance, sustainment, upgrades, DMS solutions and technology insertions, storage and ambient environment shall not introduce GDR susceptibilities or unacceptable levels of GDR hardness degradation into the M1A2 V2. In addition, the M1A2 V2 Tank's operational performance requirements shall be met throughout its rated life-cycle.

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

G.3.3 Test Procedures.

G.3.3.1 General.

The survivability of the M1A2 V2 to its GDR criterion level will be assessed by:

a. Establishing the V2 SUT performance baseline before exposure using the system diagnostics tests.

b. Exposing the SUT and all equipments to 100% and then 150% of its GDR criterion levels.

c. Repeating the pre-test baseline checks after each GDR irradiation to establish a functional status.

d. Illuminating the SUT in different test positions and hull-turret orientations, in order to insure that all GDR sensitive areas are adequately exposed and to validate the adequacy of intentional GDR hardening.

e. Documenting upsets, failures, downtime, and corrective actions.

G.3.3.2 Test Facility.

The GDR environment for the M1A2 V2 will be provided at the HERMES-III, the largest flash x ray in the world. This flash x ray facility produces Bremsstrahlung photons, which simulates the GDR environment resulting from the detonation of a nuclear weapon. The average end-point photon energy of the HERMES-III is approximately 19 MeV with a FWHM pulsewidth of approximately 28 nsec.

G.3.3.3 Pre-Test Analysis.

A pre-test analysis will be conducted to:

a. Incorporate pertinent test data and results from previous programs.

b. Identify all GDR RT Critical Items (RTCIs) based upon technology.

c. Identify test system's internal configuration.

d. Define DAS and dosimetry requirements.

e. Analyze hardening and analysis performed by contractor.

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f. Identify the most realistic and severe test setup with respect to the HERMES-III anode.

g. Identify all current limiting, power removal, and/or GDR hardening applications.

h. Identify test levels, orientations, configurations, and operational modes.

G.3.3.4 System Test Setup.

a. A calibrated Compton Diode probe will be utilized to monitor the output of the HERMES-III. This probe will be located at a constant position relative to the exit anode to provide consistent pulsewidth measurements of the GDR test environment.

b. Before testing begins, a calibration GDR irradiation will be performed to map the simulator's output. These values will be used for determining the initial STT distance to achieve 100% and 150% of the GDR criterion. As testing proceeds, the STT distance will be refined using the environmental data acquired from the previous irradiation.

c. The M1A2 V2 and electronics in the incident beam of the HERMES-III output will be monitored with CaF_2(Mn) TLDs, which provide the gamma total dose received at the monitored location.

d. The SUT will be positioned at the first STT distance and a pre-test baseline checkout performed utilizing engine power. Any abnormalities will be documented / corrected prior to start of test if judged detrimental to the GDR test.

G.3.3.5 System Test.

a. After a pulse of photons has irradiated the powered and operating SUT, all operational checks will be initiated within five minutes, unless delayed by test facility constraints.

b. If the SUT is determined to be operational to an acceptable level, the TLDs will be replaced and the M1A2 V2 will be re-positioned and/or re-oriented for the next GDR exposure.

c. These procedures will be repeated for each combination of position and hull-turret orientation until all equipments have been exposed to 100% or as close to this value as can be obtained.

d. The SUT will be positioned at the STT distance to achieve 150% of the GDR criterion level. The procedures described for the 100% test will be repeated for the 150% test.

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e. Since the GTD that is received by the SUT during each GDR exposure is small (2 to 150 Rads(Si)), delivered in a very short pulse duration (~28 nsec) and the annealing time between irradiations lengthy (30 – 60 minutes during the test day and 14 hours between test days), the SUT can be utilized in subsequent configurations. Essentially, no permanent GTD effects are introduced by these test procedures.

G.3.3.6 Response Procedure.

If a response occurs, it will be documented and diagnosed. First attempt to recover operational functionality is to cycle power OFF/ON. If operational capability is restored, testing will be repeated at the same combination of position, hull-turret orientation, and GDR level in an attempt to reproduce the same response (GDR induced effect) or to show an anomaly. Following a successful re-test, testing IAW approved procedures will continue. If operational capability is not restored after two OFF/ON cycles, then diagnostic procedures will be implemented to quantify the response and identify the victim. Testing will not be continued until the response is understood and its effect on the SUT has been assessed. If permanent damage occurs, the affected equipment will be identified then diagnosed to the lowest assembly level achievable on-site. After completion of testing, the failed equipment will be returned to GDLS for complete failure analysis. On site, the failed equipment will be replaced or repaired (replacement of a CCA), and the SUT exposed again at the same level unless it has been determined that the failure was GDR induced. In that case, the replaced equipment will be sufficiently shielded to prevent repeat of damage and testing will be continued. Every effort will be made to complete the GDR assessment of the M1A2 V2.

G.3.3.7 Dosimetry.

a. The gamma dose at selected locations on the M1A2 V2 SUT will be measured using CaF_2(Mn) TLDs. The measured gamma dose values in cGy (CaF_2(Mn)) will be expressed in cGy(Si) and cGy(tissue) by the ratios:

$$cGy(Si)/cGy(CaF_2(Mn)) = 1.02 \text{ and } cGy(tissue)/cGy(CaF_2(Mn)) = 1.108$$

b. Each radiation pulse will be measured using a Compton Diode and digitized on a transient digitizing oscilloscope. The FWHM pulsewidth of each radiation pulse will be obtained from this digitized signal. The GDR for each pulse will then be determined from the converted dose recorded on the TLDs divided by the pulsewidth obtained from the oscilloscope.

G.3.4 Data Required.

a. Visual inspections, logs, test conductor notes, and photographs of any damage.

b. Baseline data from SUT checks.

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

c. Detailed description and photographs of the method of producing the GDR test environment showing SUT position relative to the gamma radiation source.

d. Complete set of pre-test mapping data in radiation absorbed dose in cGy, in silicon (cGy(Si))(±7%), and tissue (cGy(tissue)) (±7%) for each test location.

e. Pulsewidth duration of each gamma radiation pulse (nsec) (±5%).

f. Gamma dose expressed in cGy(Si) and cGy(tissue) (±7%).

g. Location of TLDs on the SUT for each test orientation and test exposure.

h. List of all HCIs used in the M1A2 V2 and SUT Abrams Tank.

i. Detailed description and recording of all inspections, downtime (sec) (±10%), checkout data, and pre- and post-test results.

j. Go/no-go log of tank performance checks from self-tests, and diagnostic tests with descriptions of discrepancies.

k. Tank physical and operating configuration during each subtest irradiation.

l. Detailed description of any deliberate hardening devices and/or techniques.

m. Conversion factors (±3%) used to convert cGy(CaF_2(Mn)) and Roentgens to cGy(Si) and cGy(tissue).

n. Results from previous GDR tests performed at SVAD on the M1A2 production tank.

o. TIRs, if applicable.

G.3.4 Data Analysis / Procedure.

G.2.4.1 Data.

Results from other Abrams GDR tests and analysis performed by SVAD, GDLS, and subcontractors will be evaluated as part of the pre-test evaluation and incorporated into the test planning on the GDRSA for the M1A2 V2. The incorporation of all test data, especially the test performed by SVAD, will be used to enhance and reduce the scope of the M1A2 V2 GDRSA program. Pertinent data will be included in the SVAD post-test analysis, survivability assessment, and the detailed test report. Differences greater than 20% will be analyzed particularly in terms of the effect on the test results.

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APPENDIX G. DETAILED TEST PLAN SUBTEST EXAMPLE.

G.3.4.2 Criteria Compliance.

The GDR criterion as expressed in Paragraph 2.8.2 will be compared to the test levels that will be determined in the following manner.

a. A mean incident surface gamma dose for each target area will be established from the TLDs.

b. A mean non-incident surface gamma dose for each target area will be established from the TLDs.

c. A mean GCVP gamma dose for each target area will be established by adding the incident and non-incident gamma dose values and dividing by 2.

d. The GDR value will be established by taking the GCVP gamma dose value in cGy(Si) for each target area and dividing by the FWHM pulsewidth value (nsec) for each GDR irradiation.

e. If adequate dosimetry placement as specified in Paragraphs 2.8.5.2a and 2.8.5.2b cannot be conducted for a target area because of space or access restrictions, then the test item's GCVP will be approximated using available TLD dose values and dividing by the GDR FWHM pulsewidth.

f. The data in Paragraph 2.8.5.2.d will be compared and evaluated against the M1A2 V2 criterion and criterion compliance determined.

g. Total dose readings at each location will be compared to the sum of the individual readings to check accuracy.

G.3.4.3 System Configuration Compliance.

The M1A2 V2 test configuration will be evaluated against the proposed production configuration. All differences will be identified, discussed and documented. Differences that could impact the results will be discussed in the final report. The existing M1A2 V2 baseline configuration will be updated.

G.3.4.4 Response Analysis.

a. Environment induced response(s) will be scored at the test level of occurrence, cause(s) and victim(s) will be identified, and impact on the M1A2 V2 mission critical performance parameters will be discussed. As necessary, data and analysis will be included from the extensive SVAD HCI INR database, especially, in support of diagnostics.

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b. Failures or degradation occurring at levels above criterion will be classified as system shortcomings, unless shown to be valid based on the data and/or data from HCI tests plus circuit analysis. This information will be used primarily to provide additional confidence in the response of a sample size of one for a very large and complex electronic integrated system.

G.3.4.5 System Performance.

Comparison of pre- and post-irradiation functional checkout data for the SUT will be the primary metric used to determine the effects of the test environment on the M1A2 V2 SUT. Observations and engineering judgment will also be used. Degradation resulting in M1A2 V2 performance outside specifications, or total failure(s), will be addressed with regards to cause(s), test level at which they occurred, allowable downtime, and mission impact.

G.3.4.6 Survivability Assessment.

A GDR survivability assessment will be performed on the production or baseline configuration against the M1A2 V2 GDR criterion level using the results of Paragraphs 2.8.5.2 through 2.8.5.5. This assessment may produce results different than those obtained directly during the testing phase due to compliance corrections.

G.3.4.7 LCN & ES.

Both, the configuration for the test system and proposed baseline production system will be archived into the M1A2 V2 program Life-Cycle database along with pertinent test data and results, extrapolated results, and information from this GDR subtest. This database will enable implementation of the Life-Cycle program of Hardness and Sustainment Assurance, and Surveillance tests.

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APPENDIX H. DATA TABLES AND DOCUMENTATION.

TABLE H-1. TESTING DOCUMENTATION (EXAMPLE)

TEST CONDUCTOR: John Anderson — DATE: 20 March 13
FACILITY: HPD-II Facility — PAGE 1 OF 10

Shot # and Item #	Utilized Equipment Serial #s	Test Level kV/m	Test Orientation	Test Mode	Pre-test and Post-test Results	Test Points and Comments
1 # 5723	M1A2 V2 S/N# 26264 *Test Setup #1*	1st	Tank Parallel to Electric-Field (E-Field). Distance = 15 meters GCVP of Tank	Powered and Operational Hull = Turret - 0°	Pre - OK Post - OK	Eleven 8-Input Multiple Links. See Test Point Information
2 #5724	Same as 1 Above	1st	Same as 1 Above	Same as 1 Above	Pre - OK Post - OK	Same as 1 Above
3 # 5725	Same as 1 Above	1st	Same as 1 Above	Same as 1 Above	Pre - OK Post - OK	Same as 1 Above
4 # 5726	Same as 1 Above	1st	Same as 1 Above	Same as 1 Above	Pre - OK Post - OK	Same as 1 Above
5 # 5727	Same as 1 Above	1st	Same as 1 Above	Same as 1 Above	Pre - OK Post - OK	Same as 1 Above
6 # 5728	Same as 1 Above	1st	Same as 1 Above	Same as 1 Above	Pre - OK Post - OK	Same as 1 Above
7 # 5729	Same as 1 Above	1st	Same as 1 Above	Same as 1 Above	Pre - OK Post - OK	Same as 1 Above

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APPENDIX H. DATA TABLES AND DOCUMENTATION.

TABLE H-2. HEMP TEST POINT – CURRENT PROBE INFORMATION (EXAMPLE)

LINK NAME	INPUT #	TEST POINT ID	TEST POINT DESCRIPTION
Alpha	0	DID1	DID J1
Alpha	1	DID2	DID J2
Alpha	2	DID3	DID J3
Alpha	3	HMP1	HMPU J1
Alpha	4	HMP7	HMPU J7
Alpha	5	HMP8	HMPU J8
Alpha	6	HMP9	HMPU J9
Alpha	7	HMPX	HMPU J10
Bravo	0	HPDE	HPDU J14
Bravo	1	HPDB	HPDU J11
Bravo	2	HPD1	HPDU J1
Bravo	3	HPD8	HPDU J8
Bravo	4	HPD7	HPDU J7
Bravo	5	HPDD	HPDU J13
Bravo	6	HPDC	HPDU J12
Bravo	7	HPDA	HPDU J10
Charlie	0	HPD9	HPDU J9
Charlie	1	HPD6	HPDU J6
Charlie	2	HPD5	HPDU J5
Charlie	3	HPD4	HPDU J4
Charlie	4	FEA2	FEA J2
Charlie	5	TCU1	TCU J1
Charlie	6	DEC3	DECU J3
Charlie	7	DEC5	DECU J5
Delta	0	AIM6	AIM J6
Delta	1	AIM7	AIM J7
Delta	2	AIM1	AIM J1
Delta	3	AIM2	AIM J2
Delta	4	AIM3	AIM J3
Delta	5	AIM5	AIM J5
Delta	6	RS21	RSM2 J1
Delta	7	RS22	RSM2 J2

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APPENDIX H. DATA TABLES AND DOCUMENTATION.



Figure H-1. M1A2 V2 HEMP Test Setup – Hull with Turret 0° (example).

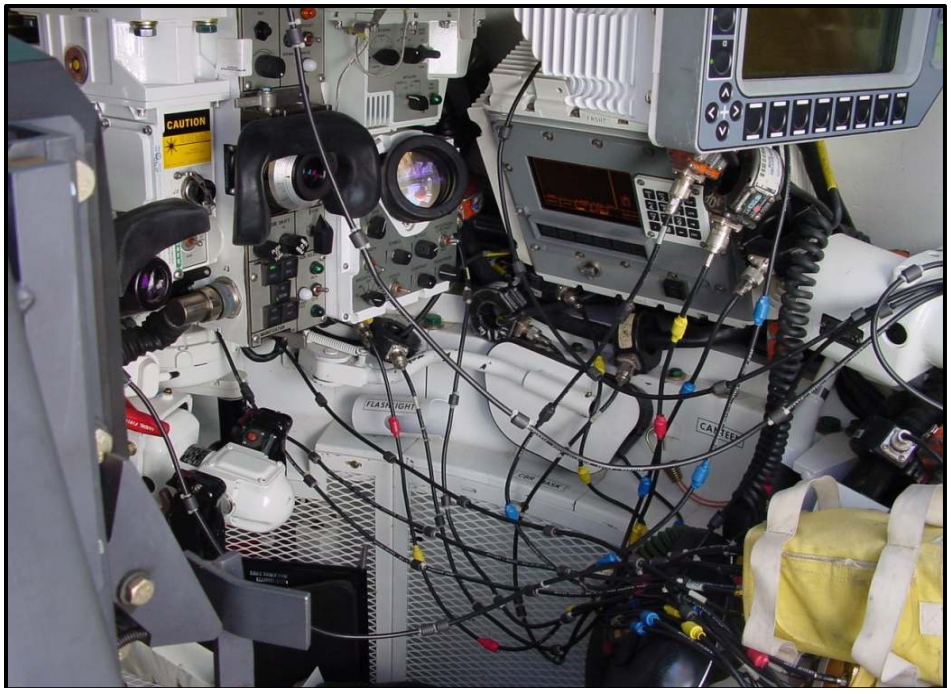


Figure H-2. M1A2 V2 HEMP Test Setup – Instrumentation (example).

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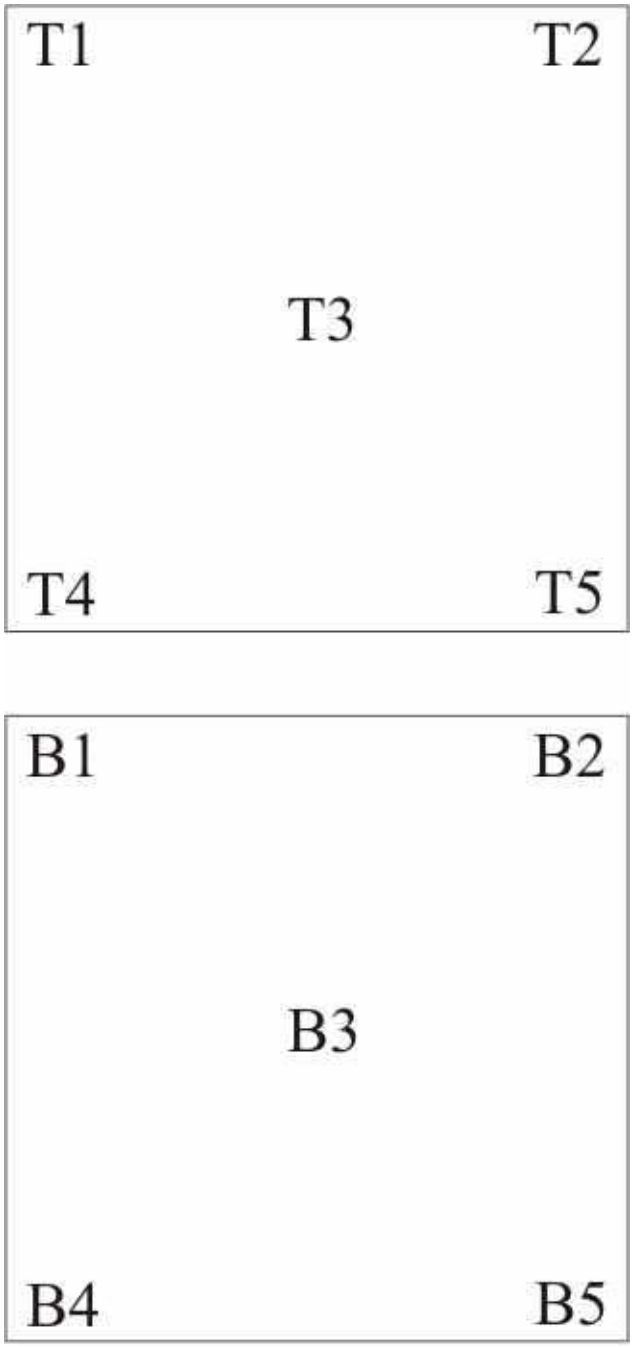
APPENDIX H. DATA TABLES AND DOCUMENTATION.

TABLE H-3. HEMP CURRENT TEST POINT (EXAMPLE)

TEST ID	ORIENTATION	SHOT#	PEAK I	RESFREQ	LOWFREQ	HIGHFREQ	BANDWIDTH	Q
CEU1	parallel-1-turret-front	5723	-0.72343	2.37E+07	2.33E+07	2.59E+07	2.54E+06	9.3128
RSC1	parallel-1-turret-front	5723	-0.78004	2.37E+07	2.34E+07	2.58E+07	2.40E+06	9.86972
AIM6	parallel-1-turret-front	5724	0.254524	1.52E+07	1.40E+07	1.62E+07	2.20E+06	6.90684
CDU1	parallel-1-turret-front	5724	0.854443	2.54E+07	2.17E+07	2.60E+07	4.36E+06	5.84013
CIT1	parallel-1-turret-front	5724	-0.87288	2.32E+07	2.25E+07	2.38E+07	1.29E+06	17.9898
DID1	parallel-1-turret-front	5724	-0.21612	3.15E+07	3.02E+07	3.21E+07	1.93E+06	16.3001
GCD4	parallel-1-turret-front	5724	0.879601	2.30E+07	2.18E+07	2.34E+07	1.58E+06	14.5688
HPD9	parallel-1-turret-front	5724	-0.26658	4.33E+07	4.26E+07	4.51E+07	2.51E+06	17.2656
HPDE	parallel-1-turret-front	5724	0.221622	3.17E+07	3.06E+07	3.22E+07	1.58E+06	20.1163
AIM7	parallel-1-turret-front	5725	-0.22246	1.19E+07	1.14E+07	1.22E+07	863666	13.7974
DID2	parallel-1-turret-front	5725	-0.15849	4.51E+07	4.43E+07	4.62E+07	1.93E+06	23.4194
HPD6	parallel-1-turret-front	5725	-0.43752	5.16E+07	5.02E+07	5.23E+07	2.12E+06	24.2788
HPDB	parallel-1-turret-front	5725	-0.18246	2.27E+07	2.19E+07	2.37E+07	1.84E+06	12.3403
MMU1	parallel-1-turret-front	5725	0.046358	2.37E+07	2.28E+07	2.59E+07	3.11E+06	7.63901
RSC4	parallel-1-turret-front	5725	0.72278	3.38E+07	3.29E+07	3.43E+07	1.42E+06	23.7429
AIM1	parallel-1-turret-front	5726	0.216101	4.89E+07	4.76E+07	4.97E+07	2.10E+06	23.3346
CDU2	parallel-1-turret-front	5726	1.65474	2.38E+07	2.29E+07	2.48E+07	1.96E+06	12.1048
CEU3	parallel-1-turret-front	5726	-7.05391	2.59E+07	2.53E+07	2.69E+07	1.68E+06	15.3943
DID3	parallel-1-turret-front	5726	0.235663	4.56E+07	4.44E+07	4.66E+07	2.25E+06	20.2586
FCE4	parallel-1-turret-front	5726	2.53957	2.39E+07	2.27E+07	2.48E+07	2.15E+06	11.1238

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APPENDIX H. DATA TABLES AND DOCUMENTATION.



Where: T = Top or the Incident Side of Test Item WRT the Radiation Source.
B = Bottom or the Non-Incident Side of Test Item WRT the Radiation Source.

Figure H-3. Dosimetry locations (example).

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APPENDIX H. DATA TABLES AND DOCUMENTATION.



Figure H-4. Gamma total dose test setup (example).

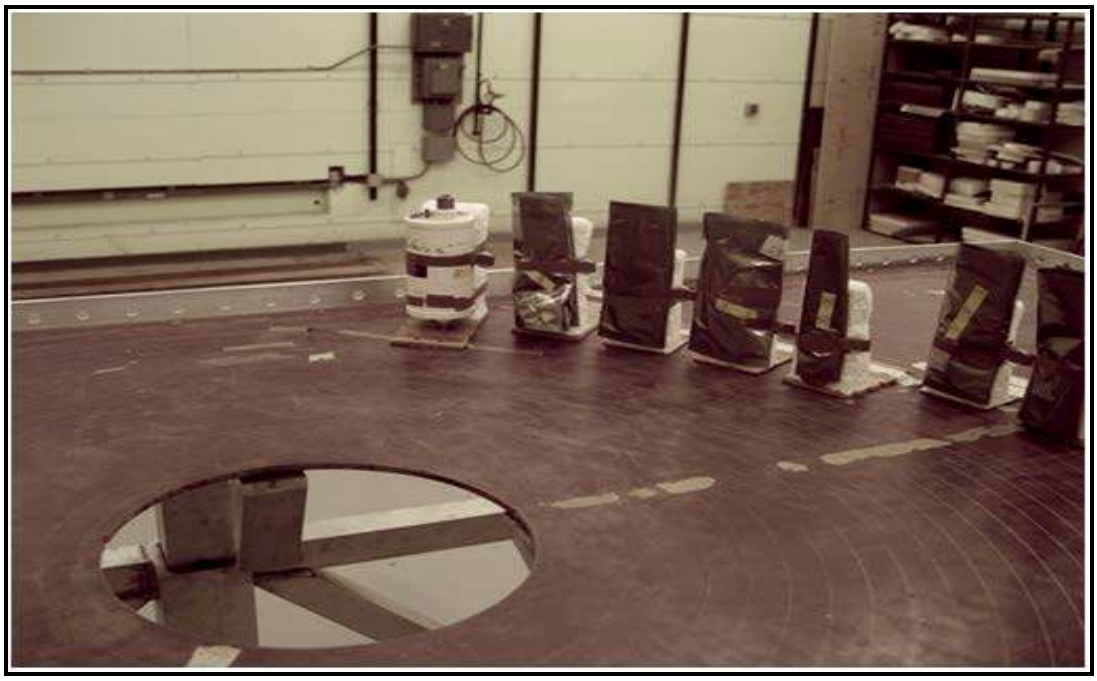


Figure H-5. Neutron fluence test setup (example).

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APPENDIX I. ABBREVIATIONS.

ADSS	ATEC Decision Support System
AEC	U.S. Army Evaluation Center
AEP	Allied Engineering Publication
AFEMP	Advanced Fast Electromagnetic Pulse Simulator
AMC-R	U.S. Army Materiel Command Regulation
Amp	Ampere
APC	armored personnel carrier
AR	Army Regulation
ASAP	as soon as possible
ASTM	American Society for Testing and Materials
ATEC	U.S. Army Test and Evaluation Command
BOB	breakout box
C	Celsius
CAD	computer aided design
$CaF_2(Mn)$	Calcium Fluoride (Manganese)
Cal	calorie
cc	cubic centimeter
CCD	charge coupled device
CDD	Capabilities Development Document
cGy	centiGray
CI	current injection
cm	centimeter
CML	Current Mode Logic
CMOS	Complementary Metal Oxide Semiconductor
CS	conducted susceptibility
DAS	data acquisition system
DOD	Department of Defense
DODD	Department of Defense Directive
DODI	Department of Defense Instruction
DOE	Department of Energy
DPR	Design Parameter Report
DUT	device-under-test
E	total energy emitted
ECL	emitter coupled logic
ECP	Engineering Change Proposal
E-Field	Electric Field
EID	Electrically Initiated Device
EM	electromagnetic

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APPENDIX I. ABBREVIATIONS.

EME	electromagnetic environment
EMP	electromagnetic pulse
FBR	Fast Burst Reactor
FEA	Finite Element Analysis
FFT	Fast Fourier Transform
FPGA	fully programmable gate array
fps	frames per second
Fτ	gain bandwidth product
FWHM	full-width-half-maximum
GDR	gamma dose rate
GHz	GigaHertz
GPF	Gamma Protection Factor
GRF	Gamma Radiation Facility
GTD	gamma total dose
Gy	Gray (100 RADs)
GZ	ground zero
HCC	hardness critical circuit
HCI	hardness critical item
HE	high explosive
HEMP	High-Altitude Electromagnetic Pulse
HERMES	High Energy Radiation Megavolt Electron Source
HF	high frequency
H-Field	magnetic field
HMMWV	High Mobility Multipurpose Wheeled Vehicle
HOB	height of burst
HPD	horizontal polarized dipole
Hz	Hertz
IAP	Independent Assessment Plan
IAW	in accordance with
ICD	Initial Capabilities Document
IGBT	insulated gate bipolar transistors
INR	Initial Nuclear Radiation
I^2L	Current Injection Logic
j-box	junction box
JFET	junction field effect transistor

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APPENDIX I. ABBREVIATIONS.

KAFB	Kirtland Air Force Base
kHz	kilohertz
kJ	kilo-Joule
kPa	kilo Pascal
km	kilometer
kt	kilo-ton
kV/m	kilovolts per meter
kW	kilowatt
lb	pound
LB/TS	Large Blast/Thermal Simulator
LCNS	Life Cycle Nuclear Survivability
LED	light emitting diode
LF	low frequency
LINAC	Linear Accelerator
LRU	line replaceable unit
µs	microsecond
m	meter
m^2	square meter
mA	milliAmpere
MBT	main battle tank
MC	ission critical
MCNP	Monte Carlo N-Particle
MeV	Mega-electron Volt
MF	middle frequency
MHz	MegaHertz
MIL-STD	Military Standard
mm	millimeter
MOS	Metal-Oxide Semiconductor
ms	millisecond
MSDS	Material Safety Data Sheet
msec	millisecond
MT	mega-ton
NATO	North Atlantic Treaty Organization
NF	Neutron Fluence
NHC	Nuclear Hardening Criteria
NIA	neutron induced activity
NIST	National Institute of Standards and Technology
NLT	not later than
NMOS	N-Channel Metal-Oxide-Semiconductor

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APPENDIX I. ABBREVIATIONS.

NPF	Neutron Protection Factor
NRD	neutron rem detector
NRF	Neutron Reduction Factor
ns	nanosecond
NTSA	Nuclear Test and Survivability Analysis
NWE	Nuclear Weapons Effects
P	peak static overpressure
PAM	Pamphlet
PF	Protection Factor
PGTD	prompt gamma total dose
PIN	positive-intrinsic-negative
PM	Program Manager
PMOS	P-Channel Metal-Oxide-Semiconductor
Po	ambient pressure
psi	pounds per square inch
psi-sec	pounds per square inch-second
q	dynamic pressure
Q	thermal fluence
Q_{dot}	thermal flux
QSTAG	Quadripartite Standardization Agreement
R	range (in centimeters)
RAD	radiation absorbed dose
RDM	radiation design margin
RF	radio frequency
RT	radiation tolerance
RTCI	Radiation Tolerant Critical Item
SCR	silicon controlled rectifier
SCT	shielded cable test
SDRAM	synchronous dynamic random access memory
sec	second
SEP	System Evaluation Plan
Si	silicon
SNL	Sandia National Laboratories
SOI	silicon-on-insulator
SR	Safety Release
SREMP	Source Region Electromagnetic Pulse
SSAR	System Safety Assessment Report
STT	source-to-target

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APPENDIX I. ABBREVIATIONS.

SUT	system-under-test
SVAD	Survivability, Vulnerability and Assessment Directorate
τ	transmissivity
TIR	Test Incident Report
TLD	thermoluminescent dosimeter
TO	Test Officer
TOP	Test Operations Procedure
TPD	terminal protection device
TREE	transient radiation effects on electronics
TRIGA	Training, Research, Isotopes, General Atomics
TSARC	Test Schedule and Review Committee
TTL	transistor - transistor logic
UHF	ultra-high frequency
UJT	uni-junction transistor
USA	United States Army
USANCA	United States Army Nuclear and Combating Weapons of Mass Destruction
UV	ultraviolet
V/m	volts per meter
VDLS	VISION Digital Library System
VHF	very high frequency
VISION	Versatile Information Systems Integrated On-Line
VLF	very low frequency
VPC	vertical plane center
VV&A	Validation, Verification and Accreditation
WSMR	White Sands Missile Range

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APPENDIX J. GLOSSARY.

Term	Definition
Absorption	The process by which radiation imparts some or all of its energy to any material through which it passes.
Attenuation	The process by which a beam of radiation is reduced in intensity when passing through some material. It is the combination of absorption and scattering processes and leads to a decrease in flux density of the beam when projected through matter.
Blast Wave	A pulse of air in which the pressure increases sharply at the front, accompanied by winds, propagated continuously from an explosion.
Beta Particle	Charged particle emitted from the nucleus of an atom, with a mass and charge equal in magnitude to that of the electron.
Calorie (gram-calorie)	Amount of heat necessary to raise the temperature of one gram of water l degree Celsius (C) (from 14.5 °C) (abbreviation: cal).
Curie	The unit quantity of any radioactive nuclide in which 3.7 x 10E10 disintegrations occur per second.
Decay, Radioactive	Disintegration of the nucleus of an unstable nuclide by spontaneous emission of charged particles and/or photons.
Dose	A general term denoting the quantity of radiation or energy absorbed per mass of absorbing medium. For special purposes, it shall be appropriately qualified. If unqualified, it refers to absorbed dose.
Dosimeter	Instrument to detect and measure accumulated radiation exposure. A common dosimeter is a pencil-size ionization chamber with a self reading electrometer, used for personnel monitoring.
Dynamic Pressure	The air pressure which results from the mass air flow (or wind) behind the shock front of a blast wave. It is equal to the product of half the density of the air through which the blast wave passes and the square of the particle (or wind) velocity behind the shock front as it impinges on an object, or structure.
Fission, Nuclear	A nuclear transformation characterized by the splitting of a nucleus into at least two other nuclei and the release of a relatively large amount of energy.

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APPENDIX J. GLOSSARY.

Term	Definition
Fusion, Nuclear	Act of combining two or more atomic nuclei into a heavier element, releasing substantial amounts of energy.
Gamma, Prompt	Gamma radiation emitted at the time of fission of a nucleus.
Gamma Ray	Short-wavelength electromagnetic radiation of nuclear origin (range of energy from 10 Kev to 9 Mev) emitted from the nucleus.
Impulse (per unit area)	The integral, with respect to time, of the overpressure (or dynamic pressure), the integration being between the time of arrival of the blast wave and that at which the overpressure (or dynamic pressure) returns to zero at the given point.
Ionization	The process by which a neutral atom or molecule acquires a positive or negative charge.
Isotopes	Nuclides having the same number of protons in their nuclei, and hence the same atomic number, but differing in the number of neutrons, and therefore in the mass number. Isotopes of the same element have almost identical chemical properties.
Mev	One million electron volts (10E6 ev). An electron volt is the amount of energy acquired by an electron when it falls through a potential of 1 volt.
Neutron	A neutral particle of approximately one atomic mass unit, present in all atomic nuclei except those of ordinary hydrogen.
Neutron Flux	The product of the neutron density and the neutron velocity, expressed as neutrons per unit area per unit time.
Nuclear Radiation	Particulate and electromagnetic radiation emitted from atomic nuclei in various nuclear processes. The important nuclear radiations from the weapon detonations are gamma, alpha and beta particles, and neutrons.
Overpressure	The transient pressure that exceeds ambient conditions. It is manifested in the shock or blast wave from an explosion. Usually expressed in pounds per square inch (psi).

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APPENDIX J. GLOSSARY.

Term	Definition
Test Officer	The test officer or project engineer is responsible for developing the cost estimates, initial planning, experimental design/development, ADSS development (if required), project/program safety, environmental, security information, scheduling resources, testing documentation , test execution and test recovery.
Thermal Radiation	Electromagnetic radiation emitted (in two pulses from an air burst) from the fireball as a consequence of its high temperature; it consists essentially of ultraviolet, visible, and infrared radiations. In the early stages (first pulse of an air burst), when the temperature of the fireball is extremely high, the ultraviolet radiation predominates; in the second pulse, the temperatures are lower and most of the thermal radiation lies in the visible and infrared regions of the spectrum. From a high-altitude burst, the thermal radiation is emitted in a single short pulse.
X-rays	Penetrating electromagnetic radiations whose wavelengths are shorter than those of visible light. They are usually produced by bombarding a metallic target with fast electrons in high vacuum. In nuclear reactions, it is customary to refer to photons originating in the nucleus as gamma rays and those originating in the extra-nuclear part of the atom as x-rays. These rays are sometimes called roentgen rays after their discoverer, W. C. Roentgen.

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Check the source to verify that this is the current version before use.

APPENDIX K. REFERENCES.

1. AR 70-75, Survivability of Army Personnel and Materiel, 2 May 2005.

2. DODD 5000.01, The Defense Acquisition System, 20 November 2007.

3. A Complete Guide to Nuclear Weapons Effects (NWE) Simulator Facilities and Applications, April 2013.

4. QSTAG 244, Edition 4, Nuclear Survivability Criteria for Military Equipment, 19 January 1993.

5. MIL-STD-2169B Notice 1, High Altitude Electromagnetic Pulse (HEMP) Environment, 19 January 2012.

6. ASTM E666-14, Standard Practice for Calculating Absorbed Dose from Gamma or X Radiation, 2014.

7. ASTM E668-13, Standard Practice for Application of Thermoluminescence-Dosimetry (TLD) Systems for Determining Absorbed Dose in Radiation-Hardness Testing of Electronic Devices, 2013.

8. Annual Book of ASTM Standards, Section12, 2014.

9. ASTM E720, Standard Guide for Selection and Use of Neutron Sensors for Determining Neutron Spectra Employed in Radiation-Hardness Testing of Electronics.

10. ASTM E721, Standard Guide for Determining Neutron Energy Spectra from Neutron Sensors for Radiation-Hardness Testing of Electronics.

11. ASTM E722, Standard Practice for Characterizing Neutron Fluence Spectra in Terms of an Equivalent Monoenergetic Neutron Fluence for Radiation-Hardness Testing of Electronics.

12. AEP-14, Allied Engineering Publication, 1 March 2014.

13. ATEC PAM 73-1, Volume I, System Test and Evaluation Procedures, 9 July 2013.

14. AR 200-1, Environmental Protection and Enhancement, 31 Dec 2007

15. AR 200-2, Environmental Effects of Army Actions, 15 Jan 2006.

16. AMC-R 385-100, Safety Manual, 26 September 1995.

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Check the source to verify that this is the current version before use.

APPENDIX K. REFERENCES.

17. DODI 3150.09 change 1, The Chemical, Biological, Radiological, and Nuclear (CBRN) Survivability Policy, 17 August 2009.

18. MIL-STD-461F, Department of Defense Interface Standard, Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment, 10 December 2007.

19. MIL-STD-464C, Department of Defense Interface Standard, Electromagnetic Environmental Effects Requirements for Systems, 1 December 2010.

20. MIL-STD-883J with change 2, Department of Defense Test Method Standard: Microcircuits, 14 March 2014.

21. QSTAG 620, Edition 2, Nuclear Survivability Criteria for Communications-Electronics Equipment, 29 January 1993.

22. TOP 01-2-620, High-Altitude Electromagnetic Pulse (HEMP) Testing, 10 November 2011.

For information only (related publications).

a. G.C. Messenger and M.S Ash, The Effects of Radiation on Electronic Systems, 1986.

b. S. Glasstone and P.J. Dolan, The Effects of Nuclear Weapons, published by United States Department of Defense and United States Department of Energy, 1977.

c. L.W. McNaught, Nuclear Weapons and Their Effects, published by Brassey's Defence Publishers, 1977.

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APPENDIX L. APPROVAL AUTHORITY.

CSTE-TM 28 October 2014

MEMORANDUM FOR

Commanders, All Test Centers
Technical Directors, All Test Centers
Directors, U.S. Army Evaluation Center
Commander, U.S. Army Operational Test Command

SUBJECT: Test Operations Procedure (TOP) 01-2-612A, Nuclear Environment Survivability, Approved for Publication

1. TOP 01-2-612A, Nuclear Environment Survivability, has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency. The scope of the document is as follows:

This TOP provides a general outline and guidelines of the test and analysis procedures required to determine the effects of specified tactical nuclear weapons effects environments on Army material. The purpose of these test and analysis procedures is to ascertain the degree to which the Initial Capabilities Document, Capabilities Development Document, System Evaluation Plan, and Army Nuclear Hardening Criteria are met.

2. This document is approved for publication and has been posted to the Reference Library of the ATEC Vision Digital Library System (VDLS). The VDLS website can be accessed at https://vdls.atc.army.mil/.

3. Comments, suggestions, or questions on this document should be addressed to U.S. Army Test and Evaluation Command (CSTE-TM), 2202 Aberdeen Boulevard-Third Floor, Aberdeen Proving Ground, MD 21005-5001; or e-mailed to usarmy.apg.atec.mbx.atec-standards@mail.mil.

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FOR

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Forward comments, recommended changes, or any pertinent data which may be of use in improving this publication to the following address: Range Infrastructure Division (CSTE-TM), U.S. Army Test and Evaluation Command, 2202 Aberdeen Boulevard, Aberdeen Proving Ground, Maryland 21005-5001. Technical information may be obtained from the preparing activity: Commander, U.S. Army White Sands Missile Range, ATTN: TEDT-WS-V, Survivability, Vulnerability and Assessment Directorate, WSMR NM 88002-5002. Additional copies can be requested through the following website: http://www.atec.army.mil/publications/topsindex.aspx, or through the Defense Technical Information Center, 8725 John J. Kingman Rd., STE 0944, Fort Belvoir, VA 22060-6218. This document is identified by the accession number (AD No.) printed on the first page.

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